# A HAND BOOK OF
# P.G.I. BATTERY OF BRAIN DYSFUNCTION
# ( PGI - BBD )

NATIONAL PSYCHOLOGICAL CORPORATION AGRA

DWARKA PERSHAD
SANTOSH K. VERMA

*Handbook*

*of*

# P.G.I. Battery of Brain Dysfunction

# (PGI-BBD)

*Dwarka Pershad*
*Santosh K. Verma*

PERSHAD AND VERMA'S REVISED

# Handbook

## of

# P.G.I. Battery

## of

# BRAIN DYSFUNCTION

# (PGI-BBD)

**Dwarka Persad**

*and*

**Santosh K. Verma**



Estd. : 1971 ✆ : (0562) 2464926

**NATIONAL PSYCHOLOGICAL CORPORATION**

**4/230, KACHERI GHAT, AGRA-282 004**

Revised Second Edition : 2009

*Lasser Typesetting :*

MAHESHWARI GRAPHICS
Motilal Nehru Road, AGRA-282 003

*Printed at :*

RASHTRA BHASHA OFFSET PRESS
Raja Mandi, AGRA-282 002

# PREFACE

This research monograph cum handbook of Cognitive Dysfunction Battery embodies the work of the authors carried out at the Department of Psychiatry, Postgraduate Institute of Medical Education and Research, Chandigarh during the period of last two decades. From the very beginning, based on clinical experiences it was realised that the concept of culture-free as well as culture-fair was a farce. No psychometric test could be of universal applicability because people differ in their experiences, customs, attitudes, beliefs, socio-economic conditions, education and cultural heritage. Based on these assumptions, a crash programme was undertaken to refine the psychological tests one by one. The choice of the tests for refinement / adaptation / construction was guided by the needs and priorities of the clinicians.

Area of measurement of cognitive functioning in psychiatric and neurological patients was found to be deficient in more than one way. Only one memory test was in use in India and that too with western norms and philosophy. Except Bhatia's and Raven's tests which were measure of non-verbal ability, no verbal test of intelligence could be used for clinical purposes in our adult clinic population with any degree of confidence. Even tests that we had, were developed long time back, without any further revisions of norms and refinement or modification. In India, education and sex of the subjects were found to be highly significant in influencing their responses on any psychological test. The findings of the researches carried out were, however, not utilized in the tests that we had been using in the clinic.

Findings of cognitive dysfunctions in adult neuro-psychiatric cases are important in planning the total treatment strategy and finding a preliminary clue to a neurological deficit before the costly and sometimes painful radio and laboratory diagnostic procedures are recommended. Of the two approaches of neuro-psychological test battery, the psychometric approach was considered to be more useful than the approach claiming to assess the lobe functions separately. The reasons being, the former approach is based on the tests and concepts that have been accepted as valid measures and are more useful in total care, follow-up and evaluation of the patients, supplementing the knowledge gained through the sensitive procedures based on latest electro-magnetic waves (MRI – Magnetic Resonance Imagery) and X-Rays Computer Techniques (CAT Scan).

The work that has been included in the "PGI Battery of Brain Dysfunction" was carried out in the Department of Psychiatry, Postgraduate Institute of Medical Education and Research, Chandigarh under different projects, like those from the departmental resources, from personal resources, from PGI Research grants and from ICMR's research grants sanctioned from time to time. Authors gratefully acknowledge the assistance rendered by them and are thankful to the Head, Department of Psychiatry, for his active encouragement, to the Dean and Director of the PGI for permitting to get the present monograph published and to the faculty colleagues and friends for their active participation in different projects and encouragement. Prof. V. K. Varma, Dr. Anil Malhotra, Dr. Savita Malhotra, Dr. S. Prabhakar, Dr. P. Kulhara, Dr. Awasthi, Dr. Adarsh Kohli, Mr. T. B. Singh, Mr. H. A. Khan, Mrs. Karobi Das and Dr. Arunima need special thanks for their very active help in varied capacity, but for which it would not have been possible to accumulate material and use in a consolidated manner in this manual. Further work on the Battery is going on, particularly by Dr. Adarsh Kohli at Postgraduate Institute of Medical Education and Research, Chandigarh for which the authors are grateful.

**Dr. Dwarka Pershad** *and* **Dr. S. K. Verma**

Both are Ex-Additional Professors of Clinical Psychology,
PGIMER, Chandigarh
*and*
Ex-Consultants of Clinical Psychology
Govt. Med. College and Hospital
Sector 32, Chandigarh

also, Ex-Assistant Editor and Ex-Editor of
Indian Journal of Clinical Psychology
respectively

Revised Manual 2009

Chandigarh

# CONTENTS

# PART I

# STANDARDIZATION

Handbook
of

# PGI Battery of Brain Dysfunction (PGI-BBD)

## INTRODUCTION

Interest in the Neuro psychological testing has been developed nearly half a century earlier. In the beginning psychological investigations were considered to be very useful in arriving at a decision whether a particular patient is suffering from a brain pathology or a functional pathology. With the advent of the sophisticated Radio-diagnostic procedures such as computerized axial tomography (C.A.T.) Scan, Neuro-psychological testing has now become an important ancillary procedure in neurological diagnostic evaluations (Heaton and Pendleton, 1981). The results of such psychological testing are rarely considered sufficient by themselves to make diagnoses, but their role has increased in the understanding of the patients for total management and to provide rehabilitatory measures.

Need for laboratory investigations, whether biophysical, biochemical or psychological, is much more in the 'difficult to diagnose' cases, where clinicians on the basis of presenting complaints, history, and the clinical examination are not sure about the nature of patients' illness. In the medical speciality of psychiatry and neurology, sensitive laboratory investigations are often painful, time consuming and also costly in terms of their exorbitant, recurring expenses. Such investigations moreover, impart informations about the struotural unity of the brain. Psychological tests are less costly, impart information about functional unity of the brain and about the deterioration in psychological functioning, if any. The profile of various psychological parameters, therefore, might be of help in the formulation of diagnosis, management and prediction of outcome of such cases. If the profile is made on the cases, which are easy to classify as psychiatric or neurological, then it usually does not give satisfactory results, specially when a decision has to be taken on 'difficult to diagnose' cases. The real utility of an investigation lies only in helping to decide about difficult cases.

So far, literature is scanty on this topic. A couple of psychological test batteries for the assessment of organic brain dysfunction are available (Kapur, 1978 ; Golden et al., 1980 ; Hemmeke et al, 1978 ; Yates, 1954 ; Halstead, 1947 ; Reitan, 1966 ; Goldstein and Scheerer, 1941). The researchers in this area, however, have not been able to provide unequivocal results with regard to their screening capacity, especially in "difficult to diagnose" cases. A disturbingly large number of false positives and false negatives have been reported in many such studies (Kapur, 1978 ; Chawla, 1973 ; Filskow and Goldstein, 1974).

The battery of Psychological tests initially developed by Halstead (1947) and later on modified by Reitan (1966) for the purpose of localization of a possible brain lesion is the most popular and said to be the most sensitive. It, however, also cannot differentiate between functional and organic cases with promising results (Lacks, Colbert, Harrow and Levine, 1970 ; Watson, Thomas, Anderson, and Felling 1968 ; Heaton, Vogt, Hoehn, Lewis, Crowley and Stallings, 1979). Basing on this battery and adopting multivariate approach, Kapur (1978) has also not demonstrated its utility in 'difficult to diagnose' cases. The Luria-Nebraska Neuro-psychological Battery, a comparatively new tool to measure the neuro-psychological functioning (Golden et al., 1978) of the individual has also failed to measure upto the standards that must be required of any new instrument purporting to offer a comprehensive assessment of neuro-psychological functioning (Spiers, 1982).

Recently, a few researches have demonstrated the limitations of the Luria-Nebraska Battery of Neuro-psychological tests. Performance of chronic brain damage or, chronic schizophrenic or schizoaffective disorder was compared on the Luria-Nebraska Neuro-psychological Battery (LNNB). The psychotics were discriminated from the brain-damaged patients at high levels by a relatively few LNNB variables. Level of performance discrimination was compared with sampling criteria as bases of the high group discriminability. Hit rates appeared to be a function of the strictness of neurological inclusion criterion. It is, therefore, felt that hit rate differences among studies may be artifacts of sampling bias. The importance of analyses of demographic variable is stressed. The task of differential diagnosis of chronic psychosis from neurological disorder appears to be more difficult when the subjects have less than a high school educational level but not when comparable subjects have completed one or more year of college work (Moses, Thompson and Cardelline, 1983).

Stanley and Howe (1983) attempted to cross validate double discrimination scales of the LNNB for identification of multiple sclerosis. This group was selected randomly but with special regard to diagnosis, current symptomatology and incurred deficit to increase the homogeneity of the sample. Neither of the two scales was effective in discriminating multiple sclerosis from brain damaged subjects and no multiple sclerosis subject was correctly diagnosed by either scale using the recommended cut off points.

Delis and Kaplan (1983) have commented that the LNNB generated a profile of scaled scores presumed to measure specific neuropsychological functions (e.g., expressive speech, visual functions). The scales of the battery have, however, been validated only for predicting "brain damage" *per se* and not for assessing the neuropsychological functions. Thus, achievement on a given scale may be erroneously associated with the integrity of the designated cognition of motor function. The low content validity of the scales of the battery precludes a valid assessment of the nature of a patient's neuropsychological functions, the most critical part of the evaluation for the neurological patient. These problems in the battery promote clinical misinterpretation. Whenever an assessing instrument contains unvalidated scales with low content validity, then, it is indeed the test that invites clinical misuse. In the validation of the battery, elevation on a scale was analysed only in terms of differentiating a group of brain damaged patients from normals. The construct validity of whether elevation on a particular scale represented an impairment on the ability, was not even considered.

Seidenberg, Giordani, Berent and Boll (1983) reported that on the Halstead-Reitan Neuro-psychological Battery there was a significant influence of I.Q. on the level of performance for 6 of the 14 test variables. Tests of problem solving abilities, language skills, and auditory perceptual analysis were most affected whereas little influence was observed over tests of basic and simple motor functions. Rt-Lt. hand differences were unaffected by I.Q. level,

Apart from the use of comprehensive test battery there is, another approach unitary to assess the individual's cognitive functioning such as intelligence, memory, perceptuo-motor skills, visual organization etc. which can give useful cues for the diagnosis of brain damage.

Drawing performance is popularly used as a psychological test procedure to assess organic brain pathology. There is, however, no agreement among researchers (Alexander, 1970 ; Chawla, 1973 ; Pershad and Wig, 1974 ; Turland, Steinhard, 1969) who have used drawing tests like Bender-Visuo Motor-Gestalt Test and Benton's Visual Retention Test, about the capacities of these tests to screen out organic cases from amongst the suspected cases. Memory impairment (Talland, 1965 ; Baddeley and Warrington, 1970 ; Lewis, 1973 ; Bachrach and Mintz, 1974 ; Pershad and Wig 1976) and deterioration in intelligence (International classification of diseases, 1967 ; Matarazzo, 1972 ; Ramalingaswami, 1975 ; Pershad, 1977 ; Wechsler, 1958 ; Pershad and Verma, 1978) are well known to occur in various organic conditions.

In a recent study conducted with financial assistance of Indian Council of Medical Research, it was found that severely head injured patients showed decline in their memory function (On PGI Memory Scale) and perceptual acuity (On Bender Visual Motor Gestalt Test) compared to that of mildly head injured cases. The Loss in memory functioning persisted even after 18 months of head injury. Perceptual acuity, however, continued to decline gradually on follow-ups. Severity of head injury was found to be associated with dysfunctioning in social, vocational, personal, familial and cognitive areas of the Dysfunction Analysis Questionnaire (Natarajan, 1987).

Above discussion clearly indicates that there are at least two different approaches for the diagnosis of brain dysfunction. These are (a) multifactorial and (b) unitary. Halstead Reitan Neuro-psychological Battery, Luria-Nebraska Neuro-psychological Battery, NIMHANS Battery of Brain Dysfunction, Goldstein-

Scheerer Test Battery are some famous tools of the former category. Whereas, Figure drawing test, Bender Visuo-motor Gestalt Test, Nahor and Benson Test of organicity, Benton's Visual Retention Test, Hooper's Visual Organization Test, Trail Making Test, McGill Picture Anomalies Test. Wechsler's Adult intelligence Scale, Wechsler's Memory Scale, Perceptual Maze Test, Graham Kendall Memory for Design Test, Wisconsin card sorting test, Minnesota Percepto Diagnostic Test, William's Memory Scale. PGI-Memory Scale, etc., are some of the tests fall under the latter category.

The former (multifactorial) approach is based on neurological facts that the brain has differential functions and different structures of the brain are responsible for different functions whereas the latter (unitary) approach is based on functional unity of the brain. Different parts of the brain have adaptability, equipotentiality, compensatory function and are governed by the mass action rather than

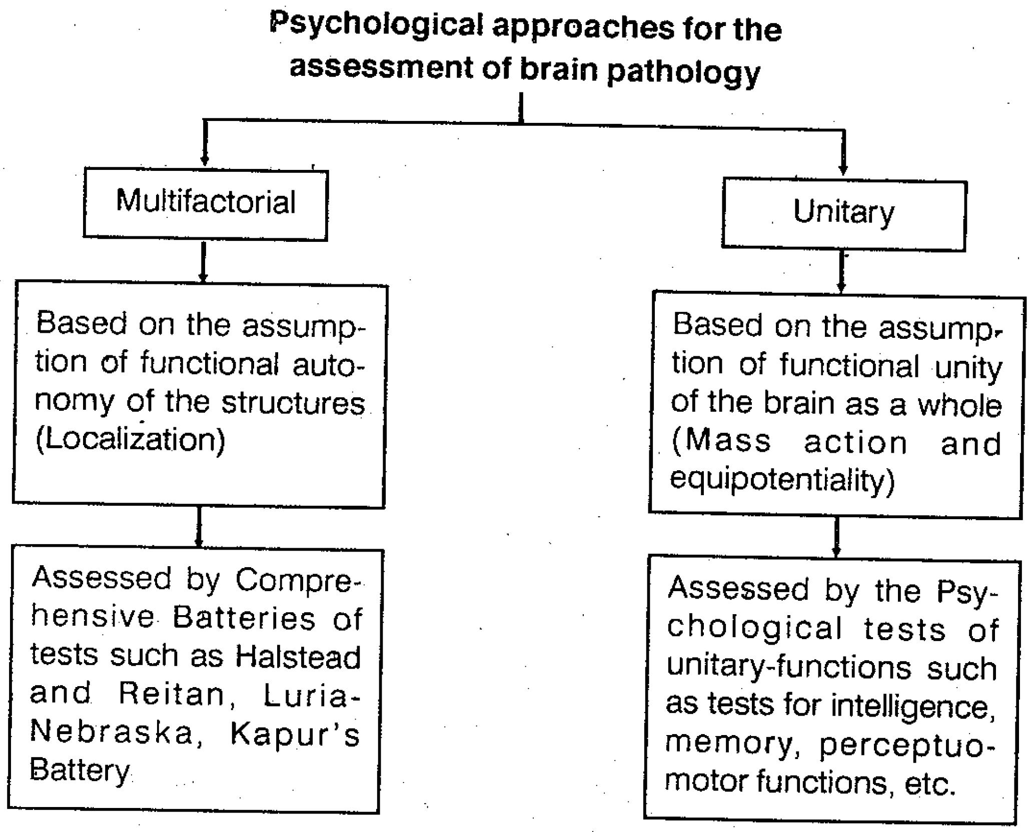



**Figure 1**

localization. Stimulation of the brain cells and motivation of the subject are the main variables in the differential functioning.

Review as cited above indicates that none of these two approaches is satisfactory though the multifactorial approach came into vogue because of certain limitations of traditional unitary approach. This new approach was also not found to be satisfactory, at least when viewed from two different angles, i.e., effectiveness in the screening and practicability. The batteries are either very time consuming (requiring upto 6 hours for administration) or the gadgets being used are very costly and involve heavy recurring expenses. Hence their practical use in the clinical sphere is restricted. On the theoretical side their content validity has never been established. The older approach, i.e., the unitary method is also not free from limitations. Single function verbal or non-verbal cannot adequately indicate functional unity of both the hemispheres of brain, well known for their entirely different functions, i.e., in right handed people, left hemisphere is related with verbal and right hemisphere is related with non-verbal behaviour.

These batteries though claim that they can localize the brain lesion yet their capacity to do so is doubtful. The reason being, functioning of the various parts of the body is based upon acquired capacities whereas structure alone is innate. Even the neurologists do agree that differentiation of the various lobes of the brain on the basis of functions is not clearly demarcated. This approach may still be useful only in those small number of cases belonging to Neurology Department where lesion is clearly demonstrated only in one lobe of the brain without intruding into the other for experimental purposes alone and not for the routine clinical screening.

Morgan (1966) has rightly stated, "We find it natural to say that people are different in the measurements of personality, intelligence, or some other aspect of behaviour but we often seem to assume that brains are standard products tinned out on an assembly line so that they look as much alike as new cars. The fact is that brains vary a lot in their size, shape and function".

Education, age and sex of the subject are some of the obvious factors which might influence the performance on psychological tests and if some of these variables are given due weightage in the interpretation of the test score, it might increase the screening capacity of the tests. Apart from this, combination of test variables of known functions and of established validity might increase the sensitivity and specificity of the psycho-diagnostic exercise and may help the clinician to quantify the extent of psychological functions/dysfunctions. It is with this aim that a number of suitable psychological measures of intellignece, memory and perceptual acuity have been developed and adopted. Their reliabilities and validities have been determined in the subjects of different educational levels. The tests of these functions are 'Verbal Adult Intelligence Scale, PGI Memory Scale, Bender-Visuo-Motor Gestalt Test, Nahor and Benson Test of Organicity'. These tests are being used regularly with difficult to diagnose cases since 1976.

The present study attempts to establish the utility of psychological assessment in brain dysfunction cases with a view to evolve a psychometric profile which can differentiate patients of organic brain dysfunction from those having no organic pathology, to validate the psychometric profile and assess its sensitivity and utility.

Prevailing tests of mental functions are useful as research tools but their clinical utility is always a suspect because of their inherent limitations. One of the limitations is that these tests provide norms without considering the social and educational background of the subject. The need is to develop simple, cheap and brief tests appropriate even for less educated patients (Wig, Pershad and Verma, 1974). This seems to be especially relevant in case of underdeveloped and developing countries where the level of education varies greatly. Lishman (1978) empasizes that "we need clearer delineation of core areas of disability and careful selection of the parameters and methods most suited for measuring change. Conventional psychological tests are clearly unsatisfactory for the purpose".

There have been many attempts to develop and standardize psychological tests to measure brain pathology. In one of the studies carried out in India

(Chawla, 1973), localization of brain pathology on the psychological tests has given unsatisfactory results.

Inspite of dissatisfaction with both the approaches theoretical and practical problems associated with them, the work on development of test battery has been continuing at premier centres of psychiatry in India.

Central Institute of Psychiatry, Ranchi, is carrying out the work of adaptation and translation of Luria's Battery. National Institute of Mental Health and Neuro Sciences, Bangalore is developing a battery of neuro psychological tests, including wide variety of psychological tests, neurological examination and computerised brain diagnostic procedures (Rao, Subbakrishna and Gopukumar, 2004 ; Kar, Rao, Chandramouli and Thennarasu, 2004). All India Institute of Medical Sciences, New Delhi, is working on an Indian Council of Medical Research's funded project for the development of a battery, using latest concepts of brain behaviour research, but restricting to paper-pencil tests of different lobes of the brain.

**Psychological Test Variables**

A number of psychological test variables like personality, sexual inhibition and interest, ego strength, cognitive functions, speech, and motor functions are likely to be disturbed in the individuals with brain pathology. One time evaluation and measurement of personality, sexual and emotional factors may not give an idea as to how much deterioration or change has occurred. Whereas cognitive factors such as intelligence, memory, percepto-motor functions can be evaluated in a single sitting and may give a sufficiently reliable and valid index of deterioration, if any. Probably based on this philosophy a number of psychological tests of these functions have been used singularly or in combination of more than one test depending upon the resources available. Some of well-known tests tried in India, are given below :

*(A) Perceptuo Motor Organization Tests*

1. Benton's Visual Retention Test (Benton, 1963 ; Shukla, Jha and Misra, 1971).

2. Graham and Kendall's Memory for Design Test (Graham and Kendall, 1960).
3. Bender Visual Motor Gestalt Test (Bender, 1938 ; Verma 1974 ; Verma, Wig and Shah, 1972 ; Singh and Virmani, 1970 ; Mendhiratta, Wig & Verma, 1978 ; Bhargava and Sandhu, 1987 ; etc.).
4. A Screening Test for Organic Brain Disease (Nahor and Benson 1970 ; Pershad and Verma, 1978).
5. Hooper Visual Organization Test (Hooper, 1966 ; Pershad and Verma, 1980).
6. Stroop Colour and Word Test (Golden, 1975).
7. Trail Making Test (Reitan, 1958).
8. Set Test (Isaacs and Akhtar, 1972).
9. Smell Matching Test (Abraham, 1981; Abraham and Mathai, 1983).
10. Minnesota Percepto-diagnostic Test (Faller and Lair, 1963).

*(B) Memory Tests*

1. Boston Memory Scale (An adaptation of Wells and Martin, 1922 ; Pershed and Verma, 1975).
2. Wechsler's Memory Scale (Wechsler, 1948).
3. Cronholm and Molandors' (1957) Memory Test Battery.
4. Williams scale for the measurement of Memory (Williams, 1968).
5. Clinical Tests of Sensorium (Withers and Hinton, 1971).
6. PGI-Memory Scale (Pershad and Wig, 1988, 1976).

*(C) Intelligence Tests*

1. Bhatia's Battery of Performance Tests of Intelligence (Bhatia, 1955 ; Murthy, 1966).

2. Ravens Progressive Matrices (Raven, 1958).
3. Wechsler's Adult Intelligance Scale (Wechsler, 1958, 1981).
4. Cattell's Culture Free Intelligence Test (Cattell and Cattell, 1960).
5. Stanford Binet Intelligence Scale (Kulshreshtha, 1971).
6. Verbal Adult Intelligence Scale (Pershad and Verma, 1988).
7. Revised Bhatia's Short Battery of Performance Tests of Intelligence for Adults (Verma, Pershad, Malhotra and Arunima, 1988).

## STANDARDIZATION OF BATTERY OF BRAIN DYSFUNCTION

On the basis of the philosophy that none of the two methods of screening of the brain dysfunction cases is completely satisfactory specially amongst the psychiatric patients 'difficult to diagnose', a simplest possible approach of testing was examined critically that had (a) established reliability validity, (b) required lesser training of testing personnel, (c) greater objectivity of scoring and interpretation, (d) required less psychological sophistication, and (e) could be applied on larger population of the clinics in india, a majority of whom belonged to low educational level, poor socio-economic status and rural and semi urban areas.

On the basis of experience it was decided that intelligence tests both verbal and performance, comprehensive battery of memory tests and two drawing tests could satisfy the above cited criteria. Therefore, the selection of tests of these functions amongst available tools was preferred. Having found that the available tests as such can not be put together because a majority of them were either not properly standardised on the type of population seen in the clinics in india, or had not provided national norms. It was thought necessary to modify them or to develop them afresh. Finally, the following tests were chosen after doing considerable research work separately with each of them to constitute a battery of brain dysfunction tests :–

### 1. PGI-Memory Scale (PGIMS)

It defines memory as the ability to retain and reproduce impressions once perceived intentionally. It includes verbal and non verbal material and measures remote, recent and immediate, short term, very short term, intermediate term and long term memories. There are 10 subtests, standardized on adult subjects in the age range of 20 to 45 years. Its test-retest reliability over a period of one week ranges from .69 to .85 for 10 subtests (N = 40) and for the total test about .90 (test-retest and splithalf). Correlation of PGIMS with Boston's Memory Scale and Wechsler's Memory Scale were found to be .71 and .85 respectively. Elderly subjects obtained significantly lower score than the younger subjects. Cases suffering from organic brain pathology and functional psychotic conditions obtained significantly less score than normals and neurotics. It had significantly high correlation with education and low correlation with I.Q. It has satisfactory cross-validity and provides quintile norms and a profile, Scores of the subjects suffering from organic brain pathology, functional psychosis and neurosis, fall in the lowest, 2nd and middle quintiles respectively. Separate norms are provided for three education levels, i.e., '0 to 5', '6 to 9' and '10 and above' years of schooling (Pershad, 1977 ; Pershad and Wig, 1976, 1988).

### 2. Revised Bhatia's Short Battery of Performance Tests of Intelligence (SB-R)

This is an adaptation of Bhatia's Battery of Performance Tests of Intelligence (Bhatia, 1955) Short Scale (Murthy, 1966) consisting of Kohs' Block Design and Alexander's Pass-A-Long tests. This short scale was standardized on adult, healthy subjects in the age range of 20- 59 years. Scoring was modified, norms were developed for 4 age groups, 3 educational levels separately for male and female to increase sensitivity of the scores. It gives Test Quotients (T.Q.) for two sub-tests which allow comparability of functions measured by Kohs' Block Design Test and Pass-A-Long test. T.Qs of Kohs' Block and Pass-a-long tests were found to be significantly correlated with the IQs on Verbal Adult Intelligence Scale.

Wechsler's Adult Performance Intelligence Scale (WAPIS by Ramalingaswamy) and Standard Progressive Matrices. Distribution of I.Q. formed a normal probability curve. Elderly subjects and subjects with low education obtained significantly low scores (Pershad, Mahajan and Verma, 1988 ; Mahajan, Pershad and Verma, 1988 ; Verma, Pershad, Malhotra and Arunima, 1988 etc.)

### 3. Verbal Adult Intelligence Scale (VAIS)

This battery of Verbal Tests of Intelligence includes four sub-tests (1) Information, (2) Digits span (3) Arithmetic, and (4) Comprehension. To develop the items for these sub-tests help was sought from the Wechsler's Adult Intelligence Scale. Pass percentages of the items were determined for different educational groups of adult subjects. The final standardization sample included 3600 subjects following 2 × 3 × 5 factorial design (i.e., *Sex* : Male-female ; *Education* : 0-5, 6-9 and 10 and above years of schooling and *Age* groups five each of 10 years range from 20 to 69) from Hindi speaking subjects of different states.

Test retest reliability over a period of 1-2 weeks was found to be in the range of ·87 to ·98 (N = 99) and split-half reliability in the range of .59 to .85 for four subtests, separately for male-female of different educational levels.

Correlations of scores on Information, Digit span, Arithmetic, Comprehension and total score with Bhatias Short Scale (N = 39) were ·56, ·56, ·73, ·67 and ·68 ; with Standard Progressive Matrices (N = 34) they were ·66, ·41, ·75, ·71 and ·71 ; with Stanford Binet (N = 18) they were ·84, ·79, ·95, ·68 and ·87, with Wechsler's Performance Scale (N = 30), ·75, ·64, ·82, ·77 and ·76 and with WAIS- R (N = 23), ·81, 1·0, ·91, ·87 and ·91, respectively. Scores on all the four sub-tests tend to increases with increase in education. Subjects using mass media like Radio, T.V. and newspapers obtained significantly higher scores than non mass media users on all the four sub-tests, (Verma, Pershad, Singh and Singh, 1982).

This test battery also provides norms for three educational levels, and 5 age groups separately for males and females. It uses concept of Test Quotient for determining level of performance on each of the 4 sub-tests and mean of 4 T.Q.s is Verbal Quotient (V.Q.). It does not take into account the speed or, quickness, rather it emphasizes the power or, the capacity of the subjects (Pershad and Verma, 1988).

**4. Nahor-Benson Test**

Nahor and Benson (1970) had developed a quick and simple screening test for organic brain pathology. A retrospective study was conducted at this centre to find out the screening capacity and to ascertain the claims of utility of the authors about this test. It was found that the cases who were diagnosed as brain dysfunction on follow-up had reproduced and made the drawings incorrect in sufficiently large number of cases compared to not brain dysfunction cases. Nearly ninety per cent of the patients in non brain dysfunction made errors on two or less drawings, whereas in brain dysfunction group, 75% of the subjects made more than 2 errors. Age of the adult subjects was not found to be associated with the number of errors on this test (r = ·08) but education had a little significant but negative correlation (—·47). Correlation of errors on this test with that of PQ and VQ were ·27 and ·45 respectively (Pershad and Verma, 1978).

**5. Bender Visual Motor Gestalt Test (Bender Gestalt Test)**

Based on Wertheimer's classical work in the field of perception, the Bender Gestalt Test was introduced by Bender in 1938. Various surveys conducted in India and elsewhere to find out the 10 most frequently used tests, revealed that Bender Gestalt Test occupied between Ist to Vth position at the top (Dubey, Pershad and Verma, 1981). The nine drawings to be copied therein, are simple and non threatening to the subjects. Various errors in the copying of the drawings were more frequently seen in the subjects with brain dysfunction than in non brain dysfunction cases (Verma, Wig and Shah 1972 ; Singh and Virmani, 1970 ; etc.). Orientation of the presented paper for copying the design had a little influence on the performance of the subjects (Verma 1974).

## A RETROSPECTIVE STUDY (DETERMINATION OF DYSFUNCTION RATING)

The tests, as mentioned earlier, though published subsequently had been in use in the 'Department of Psychiatry of the Postgraduate Institute of Medical Education and Research, Chandigath (popularly known as PGI)' for routine use. Files of such cases, that were referred earlier (during 1976 and 1980) for the diagnosis of Brain dysfunction (BD) and were in the age range of 20-50 years, were retrieved.

Psychological tests as mentioned earlier were re-scored wherever needed. The cases where clinical diagnosis could not be ascertained were asked to come for follow-up. Out of 180 such files where diagnosis was established, 55 belonged to brain dysfunction and 125 were functional psychiatric conditions. All the psychological test scores were not available in all 180 subjects. Various test scores were available in 37 to 48 cases of B.D. and 73 to 116 cases of non brain dysfunction (NBD). Frequency distribution of each of the psychological test variables were prepared separately for BD and NBD cases and Ogives were plotted. Points of maximum differentiations were located. These points/scores were then assigned arbitrary ratings indicating 'no deviation' deviation and great deviation. Different models (weights) were tried for this. For example, for great deviation on each test variable a rating of 3, for slight deviation a rating of 2 and for no deviation a rating of 1 or ratings of 3, 2, 0, or 2, 1, 0, respectively were tried. Rating of 3, 2, 0 was found to be more useful sum of ratings on all the variables thus was called total Dysfunction Rating Score (DRS).

### 1. P.G.I. Memory Scale

This scale contained ten test variables and all of them were included here on the basis of past experience (Pershad, 1977 ; Pershad and Wig, 1988). The distribution of scores on 155 BD and NBD cases is shown in the following table: –

**TABLE 1**

*Percentage frequency distribution of scores of PGI Memory Scale*

| Sr. No. | Sub-tests of PGI Memory Scale | BD (N = 47) | NBD (N = 108) | p |
|---|---|---|---|---|
| I. | **Remote Memory** | | | |
| | 3rd quintile and above | 57·4 | 58·3 | $\chi^2$ = 17·58 |
| | 2nd quintile | 6·4 | 28·7 | p = ·01 |
| | 1st quintile | 36·2 | 13·0 | |
| II. | **Recent Memory** | | | |
| | 3rd quintile and above | 63·8 | 69·4 | $\chi^2$ = 6·14 |
| | 2nd quintile | 12·8 | 21·3 | p = ·05 |
| | 1st quintile | 23·4 | 9·3 | |
| III. | **Mental Balance** | | | |
| | 3rd quintile and above | 34·0 | 50·9 | $\chi^2$ = 5·66 |
| | 2nd quintile | 17·0 | 19·4 | p = NS |
| | 1st quintile | 49·0 | 29·7 | |
| IV. | **Attention-Concentration** | | | |
| | 3rd quintile and above | 34·0 | 56·5 | $\chi^2$ = 5·77 |
| | 2nd quintile | 27·7 | 20·4 | p = NS |
| | 1st quintile | 38·3 | 23·1 | |
| V. | **Delayed Recall** | | | |
| | 3rd quintile and above | 27·7 | 45·4 | $\chi^2$ = 8·59 |
| | 2nd quintile | 19·1 | 25·9 | p = ·05 |
| | 1st quintile | 53·2 | 28·7 | |
| VI. | **Immediate Recall** | | | |
| | 3rd quintile and above | 40·4 | 57·4 | $\chi^2$ = 4·80 |
| | 2nd quintile | 19·2 | 18·5 | p = NS |
| | 1st quintile | 40·4 | 24·1 | |
| VII. | **Similar Pairs** | | | |
| | 3rd quintile and above | 40·4 | 77·8 | $\chi^2$ = 20·48 |
| | 2nd quintile | 21·3 | 8·3 | p = ·01 |
| | 1st quintile | 38·3 | 13·9 | |

*Contd....*

| Sr. No. | Sub-tests of PGI Memory Scale | BD (N = 47) | NBD (N = 108) | p |
|---|---|---|---|---|
| VIII. | **Dissimilar Pairs**<br>3rd quintile and above<br>2nd quintile<br>1st quintile | <br>31·9<br>21·3<br>46·8 | <br>44·4<br>18·5<br>37·1 | $\chi^2 = 2{\cdot}14$<br>p = NS |
| IX. | **Visual Retention**<br>3rd quintile and above<br>2nd quintile<br>1st quintile | <br>29·8<br>27·7<br>42·5 | <br>52·8<br>13·9<br>33·3 | $\chi^2 = 6{\cdot}95$<br>p = ·05 |
| X. | **Recognition**<br>3rd quintile and above<br>2nd quintile<br>1st quintile | <br>36·2<br>10·6<br>53·2 | <br>62·0<br>14·8<br>23·2 | $\chi^2 = 15{\cdot}59$<br>p = ·01 |

## 2. Battery of Performance Tests of Intelligence

It was analysed as having three variables, i.e., scores on 'Kohs' Block', 'Pass-A-Long' and Performance I.Q. It was found that scores on the Kohs' Block and the Pass-A-Long tests did not discriminate groups of B.D. (organic brain dysfunction) and NBD (Non organic brain dysfunction). The ratio of Pass-along and Kohs'

$$= \left[P : K = \frac{\text{Scores on Pass–A–Long}}{\text{Scores on Kohs'}} \times 100\right]$$

however was found to differentiate between the two groups. Performance IQ was found to differentiate the groups. Performance I.Q. was found to be significantly different in two groups of subjects. Thus, on this test two variables were considered.

The complete record of this test was available in 110 BD and NBD cases excluding epilepsy.

**TABLE 2**

*Percentage Distribution and Mean scores on the variables of Battery of Performance Tests of Intelligence*

| Variables | BD (N = 37) | NBD (N = 73) | p |
|---|---|---|---|
| **P / K × 100** | | | |
| Upto 199 | 48·7% | 74·0% | $\chi^2 = 8{\cdot}04$ |
| 200 – 250 | 27·0% | 9·6% | p = ·02 |
| 251 and above | 24·3% | 16·4% | |
| Mean | 236·00 | 184·75 | t = 1·76 |
| S.D. | 153·78 | 116·03 | p = N.S. |
| **Performance IQ** | | | |
| Upto 60 | 35·1 | 5·5 | $\chi^2 = 17{\cdot}12$ |
| 61 – 80 | 29·7 | 28·8 | df = 2 |
| 81 – 100 | 21·6 | 46·6 | p = ·01 |
| 101 and above | 13·5 | 19·1 | |
| Mean | 64·44 | 86·69 | t = 2·98 |
| S.D. | 21·60 | 16·68 | p = ·01 |

### 3. Verbal Adult Intelligence Scale (VAIS)

It contained five test scores—four sub-tests (information, Digit span, Arithmetic and Comprehension) and a V.Q. (Verbal Quotient). Though the sub-test scores did not differ significantly (except for Digit span) yet they were retained as important indices on the basis of reported significance of these items (Matarazzo, 1972). Mean scores and distribution of V.Q. did not differ in two groups of subjects (BD *vs.* NBD) but the difference between V.Q. and P.Q. was found to be highly discriminatory and hence was retained. Thus this test had five indices as given in Table 3 on page 20. Complete scores on four sub-tests were available in 120 cases only.

**TABLE 3**

*Percentage Frequency distribution and mean scores on V.A.I.S. variables*

| Variables | BD (N = 38) | NBD (N = 82) | p |
|---|---|---|---|
| **Information Test Quotient** | | | |
| Upto 60 | 13·2 | 7·3 | $\chi^2 = 4{\cdot}27$ |
| 61 – 80 | 42·1 | 28·0 | df = 2 |
| 81 – 100 | 23·7 | 46·3 | p = N.S. |
| 101 and above | 21·0 | 18·3 | |
| Mean | 82·47 | 86·09 | t = 0·95 |
| S.D. | 20·80 | 15·68 | p = N.S. |
| **Digit Span T. Q.** | | | |
| Upto 60 | 0·0 | 1·2 | $\chi^2 = 3{\cdot}94$ |
| 61 – 80 | 28·9 | 13·4 | df = 1 |
| 81 – 100 | 63·2 | 72·0 | p = ·05 |
| 101 and above | 7·9 | 13·4 | |
| Mean | 85·78 | 90·28 | t = 2·10 |
| S.D. | 10·95 | 10·81 | p = ·05 |
| **Arithmetic T. Q.** | | | |
| Upto 60 | 18·4 | 7·3 | $\chi^2 = 3{\cdot}30$ |
| 61 – 80 | 18·4 | 20·7 | df = 2 |
| 81 – 100 | 34·2 | 31·7 | p = N.S. |
| 101 and above | 29·0 | 40·2 | |
| Mean | 86·34 | 92·13 | t = 1·20 |
| S.D. | 25·54 | 22·14 | p = N.S. |
| **Comprehension T. Q.** | | | |
| Upto 60 | 13·1 | 8·5 | $\chi^2 = 4{\cdot}35$ |
| 61 – 80 | 47·4 | 31·7 | df = 2 |
| 81 – 100 | 23·7 | 46·3 | p = N.S. |
| 101 and above | 15·8 | 13·4 | |
| Mean | 79·18 | 83·91 | t = 1·13 |
| S.D. | 22·87 | 17·35 | p = N.S. |

*Contd....*

| Variables | BD (N = 38) | NBD (N = 82) | p |
|---|---|---|---|
| **Difference of VQ and PQ** | | | |
| Upto 14 | 40·5 | 61·6 | $\chi^2 = 9{\cdot}97$ |
| 15–23 | 21·6 | 26·0 | df = 2 |
| 24–31 | 21·6 | 6·8 | p = ·01 |
| 32 and above | 16·2 | 5·5 | |
| N | 37 | 73 | |
| Mean | 20·52 | 14·34 | t = 2·19 |
| S.D. | 14·74 | 12·30 | p = ·05 |

## 4. Percepto-motor Task

This included two variables viz scores on Bender Visuomotor Gestalt Test and on Nahar and Benson Test. Both the variables were found to differ significantly in BD and NBD cases (Table 4).

**TABLE 4**

*Percentage frequency distribution and mean scores on Bender Visuo-motor Gestalt Test (BVMG) and Nahar and Benson Test (NBT)*

| Test Variables | BD | NBD | p |
|---|---|---|---|
| **BVMG Score** | | | |
| 0–4 | 39·6 | 81·3 | $\chi^2 = 25{\cdot}74$ |
| 5–8 | 27·1 | 8·0 | p = ·01 |
| 9–12 | 20·8 | 9·8 | |
| 13 and above | 12·5 | 0·9 | |
| N | 48 | 112 | |
| Mean | 6·31 | 2·26 | t = 4·17 |
| S.D. | 4·91 | 3·51 | p = ·01 |
| **NBT Error Score** | | | |
| 0–3 | 56·5 | 89·7 | $\chi^2 = 23{\cdot}00$ |
| 4–5 | 26·1 | 6·9 | p = ·01 |
| 6 and above | 17·4 | 3·4 | |
| N | 46 | 112 | |
| Mean | 2·82 | 1·12 | t = 4·47 |
| S.D. | 2·37 | 1·74 | p = ·01 |

The various cut off points mentioned in the frequency distribution tables given earlier, were assigned arbitrary points. Different models were tried for this, for example, extremely low score on each variable a score of 3, middle group-a-score of 2, normal or near normal-a-score of 1, or scores of 3/2/0 or score of 2/1/0 respectively. A trial was also conducted dividing normal category into two divisions, i.e., within average range and above average and assigning scores to these four categories as 3, 2, 1 and 0. The rating scores are called 'Dysfunction Rating Score'. Ogives were prepared and cut off points for maximum differentiation between BD and NBD were worked out. It was found that a combined score of pooled rating had a maximum differentiation at cut off point of 20 when a three point rating for each variable as 3, 2, and 0 is adopted. The table given on page 23 indicates precisely the test scores on each of the tests and their ratings for the BD scoring.

All the test variable scores were available only in 27 BD (excluding Epilepsy) and 60 NBD patients. The 'brain' dysfunction Rating scores on memory, intelligence and perception separately and as a whole are given below in percentage frequency tables (Table 5 to 8).

From the frequency distribution of the brain dysfunction rating score on memory, intelligence and perception a cut off point with maximum differentiation was taken for each of the functions and are shown below in Table 9.

*Inter-correlations* of three cognitive functions and of total Dysfunction Rating Scores are given in Table 10 separately for BD and NBD subjects.

Linkage Analysis (McQuitty, 1957) was performed and it was found that the four scores (3 cognitive functions and sum of the DRS) formed one cluster in both groups of subjects. The nature of Linkage, however, was different in two groups. In the B.D. group, brain damage scores of three variables were found to revolve around total DRS whereas in NBD cases total DRS were found to be related with memory and intelligence, and perception scores had a distant relation with it via intelligence.

**TABLE 5**

*Percentage frequency distribution of 'Dysfunction Rating Score' on the variables of memory, in BD and NBD subjects*

| Dysfunction Rating Score on Memory | BD (N = 27) | NBD (N = 60) |
|---|---|---|
| 25–29 | 14·8 | 5·0 |
| 20–24 | 18·5 | 6·7 |
| 15–19 | 22·2 | 18·3 |
| 10–14 | 14·8 | 15·0 |
| 5–9 | 25·9 | 30·0 |
| 0–4 | 3·7 | 25·0 |
| Mean | 15·51 | 10·34 |
| S.D. | 7·43 | 7·24 |
| t | 3·00 | |
| | p = 0·1 | |

Minimum and Maximum DRS on memory ranges from 0 to 30

**TABLE 6**

*Percentage frequency distribution of 'Dysfunction Rating Score' on the variables of Intelligence in BD and NBD subjects*

| Dysfunction Rating Score on Intelligence | BD (N = 27) | NBD (N = 60) |
|---|---|---|
| 15–17 | 3·7 | 1·7 |
| 12–14 | 18·5 | 3·3 |
| 9–11 | 7·4 | 20·0 |
| 6–8 | 40·8 | 21·7 |
| 3–5 | 18·5 | 21·7 |
| 0–2 | 11·1 | 31·7 |
| Mean | 7·46 | 5·41 |
| S.D. | 4·05 | 3·87 |
| t | 2·20 | |
| | p = ·05 | |

Minimum and maximum DRS on Intelligence ranges from 0 to 21

**TABLE 7**

*Percentage frequency distribution of 'Dysfuction Rating Score' on the Variable of Perception in BD and NBD Subjects*

| Dysfunction Rating Score on Perception | BD (N = 27) | NBD (N = 60) |
|---|---|---|
| 6 | 11·1 | 3·3 |
| 5 | 18·5 | 0·0 |
| 4 | 7·4 | 1·7 |
| 3 | 0·0 | 1·7 |
| 2 | 22·2 | 13·3 |
| 1 | 00·0 | 0·0 |
| 0 | 40·7 | 80·0 |
| Mean | 2·42 | 0·54 |
| S.D. | 2·29 | 1·34 |
| t | 3·68 $p = \cdot 01$ | |

Minimum and maximum DRS on perception ranges from 0 to 6.

**TABLE 8**

*Percentage frequency distribution of 'Total Dysfunction Rating Score' on Psychological Variable in BD and NBD Subjects*

| Total Dysfunction Rating Score | BD (N = 27) | NBD (N = 60) |
|---|---|---|
| 40 – 49 | 14·8 | 1·7 |
| 30 – 39 | 18·5 | 6·7 |
| 20 – 29 | 33·3 | 26·7 |
| 10 – 19 | 29·6 | 33·3 |
| 0 – 9 | 3·7 | 31·7 |
| Mean | 25·46 | 15·34 |
| S.D. | 11·54 | 9·57 |
| t | 3·98 $p = \cdot 01$ | |

Minimum and maximum total Dysfunction Rating Score on Psychological test variables ranges from 0 to 57.

**TABLE 9**

*Percentage frequency distribution of the subjects above and below cut off points with maximum differentiation on three cognitive functions*

| Cognitive Functions with Cut off Points on DRS | BD (N = 27) | NBD (N = 60) | $\chi^2$ |
|---|---|---|---|
| **Memory** | | | |
| 10 and above | 70·4 (19) | 45·0 (27) | 4·76* |
| Upto 9 | 29·6 ( 8) | 55·0 (33) | |
| **Intelligence** | | | |
| 6 and above | 70·4 (19) | 46·6 (28) | 4·19* |
| Upto 5 | 29·6 ( 8) | 53·4 (32) | |
| **Perception** | | | |
| 2 and above | 59·3 (16) | 20·0 (12) | 13·11** |
| Upto 1 | 40·11 (11) | 80·0 (48) | |
| **Total DRS** | | | |
| 20 and above | 66·7 (18) | 35·0 (21) | 7·56** |
| Upto 19 | 33·3 ( 9) | 65·0 (39) | |

* p = ·05, ** p = ·01

Sensitivity = 18 + 39 / 87 × 100 = 65·5%

Specificity = 18 / 27 × 100 = 66·7%

Number of subjects is given in parenthesis

**TABLE 10**

*Inter correlation Matrix among 'Brain Dysfunction Rating Scores' on Different Cognitive Functions*

| Sr. No. | Brain Damage Score | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| 1. | Intelligence | | ·48 | ·35 | ·72 |
| 2. | Memory | ·48 | | ·46 | ·54 |
| 3. | Perception | ·17 | ·01 | | ·58 **BD** |
| 4. | Pooled Score | ·17 | ·92 | ·15 | **Subjects** |
| | | **NBD Subjects** | | | |

This strengthens the initial impression with which intelligence, memory and perceptuo-motor variables were selected as indicators of B.D. The linkage also confirmed that total score is a valid composition of three variables but more so in B.D. cases rather than in NBD cases. If this is true then the screening of BD subjects based on total cut off point should have been significantly better than on the basis of its individual component. This justification comes from the following table where subejcts were categorised under various heads depending upon on how many variables they had above cut off point score.

**TABLE 11**

*Percentage of subjects obtaining scores beyond cut-off point*

| Group | Beyond cut-off point on | | | | Total BDS |
|---|---|---|---|---|---|
| | None | Any 1 | Any 2 | All the 3 | |
| B.D. | 11·1 | 26·0 | 18·5 | 44·4 | 66·7 |
| N.B.D. | 35·0 | 26·7 | 33·3 | 5·0 | 35·0 |

The above results showed that the presented approach and developed dysfunction rating score could identify more than 65% of the cases of brain dysfunction ahead of clinical diagnosis of brain pathology amongst the difficult to diagnose cases. Total dysfunction rating score is a better predictor than the dysfunction rating score on individual cognitive functions. This sensitivity of the dysfunction rating score is likely to increase in case of prospective study of similar cases (for detail refer to Pershad, Verma, Malhotra and Prabhakar, 1984a).

## CONFIRMATION OF RATING (A PROSPECTIVE STUDY)

Twentyeight adult subjects in whom diagnosis of brain pathology was arrived at subsequently by the clinicians included 20 patients of non psychotic mental disorder following brain damage and 8 patients of organic psychatric conditions. It also included 45 cases of NBD where various psychiatric conditions (neurosis in 12, and psychosis in 33) emerged on follow-up. However all the 73 cases initially had doubtful impression of organic brain damage and were referred for psychometric assessment for the evidence of organic illness. All the cases were administered PGI Battery of Brain Dysfunction as described earlier and their scores were given dysfunction rating as discussed in the retrospective study (Pershad, Verma, Malhotra and Prabhakar, 1984 b and c).

**TABLE 12**

*Percentage Frequency of the Patients scoring Above and Below cut-off point on total Dysfunction Rating Score*

| DRS | BD (N = 28) | NBD (N = 45) |
|---|---|---|
| 35 and above | 39.5% (11) | 2.2% ( 1) |
| 20 – 34 | 21.0% ( 6) | 31.1% (14) |
| 0 – 19 | 39.5% (11) | 66.7% (30) |
| **TOTAL** | 100.0% (28) | 100.0% (45) |

Number of cases are given in the parenthesis.

*Sensitivity* cut off point 20 and above

$$\frac{17 + 30}{73} \times 100 = 64.4$$

*Specificity* $$\frac{17}{28} \times 100 = 60.7$$

Sensitivity and specificity rates in two studies (retrospective and prospective) were not found to be significantly different.

## CROSS VALIDITY

The two studies retrospective and prospective reported earlier were published in 1984 (Pershad, Verma, Malhotra and Prabhakar,1984a, b, c). The cross validity reported in 1987 (Pershad and Verma 1987) was conducted on 47 patients (16 BD and 31 NBD). The correlations of dysfunction rating score with age, education and duration of illness and sensitivity and specificity rates cut point are given below in Tables 13 and 14.

**TABLE 13**

*Correlations of Age, Education and Duration of illness with Dysfunction Rating Score (DRS)*

| Correlation of DRS with | BD cases (N = 16) | NBD cases (N = 31) |
|---|---|---|
| Age | ·51* | –·02 |
| Education | –·30 | –·11 |
| Duration of illness | –·01 | –·21 |

* p = ·05 except for one all correlations (r) were not significant

**TABLE 14**

*Sensitivity and specificity of cut off point of Dysfunction Rating Score*

| Cut off point | BD cases | NBD cases | Total |
|---|---|---|---|
| 20 and above | 12 | 4 | 16 |
| below 20 | 4 | 27 | 31 |
| **Total** | 16 | 31 | 47 |

Sensitivity $\frac{12 + 27}{47} \times 100 = 83\%$

Specificity $\frac{12}{16} \times 100 = 75\%$

False Negative $\frac{4}{16} \times 100 = 25\%$

False Positive $\frac{4}{31} \times 100 = 13\%$

These results further confirm the screening capacity of the unitary procedure of organic brain dysfunction battery. The rates of correct hit would definitely be much higher if the present battery is administered on clear cut cases where detection of brain pathology does not have any problem. The Dysfunction Rating Score on present battery as anticipated did not correlate significantly with the education of the patients because while developing norms proper weightage had already been introduced. Thus PGI Battery of Brain Dysfunction can be used with equal confidence with the patients of all grades of education.

PGI battery of Brain Dysfunction has also been used in number of diversified studies where theoretically cognitive dysfunctioning is assumed to occur. For example longer use of alcohol, chronic hypertension and generalised anaesthetic agents are known to cause some disturbances in cognitive area.

Arora (1988) administered a battery of psychological tests similar to that of PGI Battery of Brain Dysfunction on the hypertensive subjects three times, i.e., once before surgery and twice after surgery on 4th and 7th day post operatively to find out influence of hypertension and of general anaesthesia on cognitive functions. It was observed that memory (on PGI Memory Scale) and verbal intelligence (on Verbal Adult Intelligence Scale – VAIS) deteriorated after operation under general anaesthesia but gained preoperative level on 7th day post operatively. The hypertensives were found to have Verbal Quotient 8 points lower than their non hypertensive counterparts on pre-operative evaluation (on VAIS). Performances on the Bender Visual Motor Gestalt Test and the Hooper Visual Organization Test were consistently poor in hypertensives compared to controls.

Garg (1989) used PGI Battery of Brain Dysfunction to find out extent of cognitive impairment in chronic alcoholics. Subjects were kept alcohol free and when withdrawal symptoms subsided they were administered the PGI-BBD. It was found that 50% of the alcoholics obtained a dysfunction rating score of 20 and above. The comparable control group (non-alcoholic) obtained dysfunction rating score of 10 and below.

These two studies further demonstrate the use of PGI Battery of Brain Dysfunction in scaling the cognitive impairment and confirms its validity.

# PART II

# ADMINISTRATION, SCORING AND INTERPRETATION

# PGI BATTERY OF BRAIN DYSFUNCTION

It consists of five subtests and nineteen variables as under :

| | | |
|---|---|---|
| 1. | PGI-Memory Scale | 10 variables |
| 2. | Battery of Performance Tests of Intelligence | 2 variables |
| 3. | Verbal Adult Intelligence Scale (VAIS) | 5 variables |
| 4. | Nahor-Benson Test (NBT) | 1 variable |
| 5. | Bender Visual Motor Gestalt Test (BVMGT) | 1 variable |

PGI-BBD was standardized on adult males and females, both for literate and illiterate in the age range of 20 to 45, this however can be used on subjects up to 50 years of age.

It requires nearly 90 to 120 minutes for its complete administration. Scoring and rating may require about 30 minutes in the beginning but with experience this may be reduced considerably and should not take more than 5-7 minutes.

This test battery should be used as a whole, however, if situation demands only the few tests can be used depending on the requirement because separate norms are available for each of the five tests of the battery.

This test battery measures the well known cognitive functions of the brain-behaviour, namely intelligence (both verbal and performance), memory, perceptual acuity and transference from one hemisphere to another. Thus it is based on reception, consolidation (search) and execution of the stimulations.

**Objectives of the Use**

PGI-BBD is proposed to be used to quantify cognitive dysfunction / impairment / decline / deficit [1] for clinical purposes, [2] to evaluate extent of decline or loss in cognitive area as a result of illness/accident/injury/natural calamities etc., [3] to plan rehabilitatory strategy, [4] helping judiciary in taking decision about extent of decline for grant of compensation, [5] to plan training/ treatment, [6] for the evaluation of out come of the treatment / management.

**Theoretical Assumptions**

PGI-BBD is based on the assumption of functional unity of the brain as a whole (mass action and equi-potentiality of the brain) not on the assumption of functional autonomy of the structures (localization). Thus PGI-BBD, evaluates functioning of the brain not the structure of the brain.

PGI-BBD is not purported to be used for brain pathology/lateralization or localization of the hemispheric functions. Now-a-days more sophisticated laboratory and radio diagnostic tests/investigations are available to meet this challenge. Decline in cognitive function, however, may give some clue about the structural health of the brain.

**Variety of Areas Where PGI-BBD is Used**

This test battery has been in use at all the training institutes of clinical psychology as part of the requisite training and practice. This is also used in research works leading to M Phil. or Ph D degrees of the universities.

This test battery in full or part has been used on variety of subjects for research purposes including-

ALCOHOLIC/DRUG ABUSER (Narang, R.L. ; Pershad, D. ; Gupta, R. and Garg, D. 1991 ; Safia, A. et al., 1989 ; Singh, S. 1995).

SCHIZOPHRENICS (Siddiqui, R.S. ; Verma, S.K. and Pershad, D. 1992 ; Kumar, R. and Sharma, R.G. 1991 ; Srivastava, A.K., Kumar, S., Kumar, R. and Singh, B. 2004).

EPILEPTICS (Siddiqui, R.S., Pershad, D., Verma, S.K. and Prabhakar, S. 1992).

EPILEPTICS with or without CT lesions (Siddiqui, R.S., Verma, S.K., Pershad, D. ; Prabhakar, S. ; Suri, S. and Avasthi, A. 1994).

MANICS (Kar, B.C. 2006 ; Sharma, M.K. and Varghese, S.T. 2005).

HEAD INJURY (Kak, V.K., Pershad, D. and Kumar, H. 1995 ; Dhand, U.K., Chandrashekhar, M. ; Chopra, J.S. and Pershad, D. 1990).

(THIS IS NOT A COMPREHENSIVE LIST OF PUBLICATIONS. Those desirous to know in detail may consult Indian Psychological Abstract.

# 1. PGI MEMORY SCALE

Memory is a highly complex phenomenon and a subject matter of many biological and behavioural sciences. Its importance lies in the fact that mere imagination of loss of memory may threaten the very existence of human being. Here memory is considered from psychological points of view as required for the behavioural adequacy. Memory is generally believed to be a function of temporal lobe of the cerebral cortex – Right Temporal lobe is responsible for non-verbal and temporal lobe towards the left hemisphere is responsible for verbal part of the memory. Here 'PGI-Memory Scale (Pershad, 1977 and Pershad and Wig, 1988) is included in the Battery of Brain Dysfunction. The PGI-Memory Scale (PGIMS) provides a comprehensive and simple scale to measure verbal and non-verbal memories on the basis of neurological theory; very short term, short term and long term memories on the basis of experimental evidences and remote, recent and immediate memories on the basis of clinical practice of evaluation of memory.

**Material**

(1) Memory test proforma containing items for the 10 subtests i.e. Remote Memory, Recent Memory, Mental Balance, Attention-concentration, Delayed Recall, Immediate Recall (sentence reproduction) Verbal retention for similar pairs, Verbal retention for dissimilar pairs, Visual retention and Recognition.

(2) A set of 5 cards for Visual Retnetion subtest IX.

(3) A set of 2 cards – one having pictures of 10 common objects and second having pictures of 20 common objects for Recognition-subtest X.

(4) Stop watch / stop clock with push back device / cell phone.

(5) A pencil.

(6) An erasure.

# पी. जी. आई. स्मृति परीक्षण (PGI MEMORY SCALE)

## I. पूर्वकालीन स्मृति (Remote Memory)

1. आपकी उम्र कितनी है ?
   How old are you ?
2. आपका जन्म कहाँ हुआ ?
   Where were you born ?
3. (a) आपकी शादी कब हुई ?
   When were you married ?
   (b) आपने नौकरी या व्यवसाय करना कब से चालू किया ?
   When did you start earning ?
   (c) आपने पढ़ना कब छोड़ा या हाई स्कूल कब पास किया ?
   When did you left study/pass high school ?
4. आपके सबसे छोटे बच्चे या भाई बहिन की उम्र कितनी है ?
   How old is your youngest child/brother/sister ?
5. आप इस विभाग (अस्पताल) में पहली बार अपने इस इलाज के लिये कब आये ?
   When did you come first time in this clinic/department (Hospital) ?
6. पिछली बार आप इस विभाग (अस्पताल) में कब आये थे ?
   When did you come here last time ?

## II. वर्तमान स्मृति (Recent Memory)

1. कल आपने रात के खाने में क्या खाया ?
   What did you eat in your last dinner ?
2. आज सुबह आपने नाश्ते में क्या खाया ?
   What did you eat this morning ?
3. इस महीने का क्या नाम है ?
   What is the name of this month ?
4. आज कौन-सा दिन है ?
   What day is today ?
5. कल आपसे कौन-कौन मिलने आया या कल आप किस-किस से मिलने गये ?
   Who came to visit you or to whom you visited yesterday ?

### III. मानिसक संतुलन (Mental Balance)

1. A, B, C, – Z तक बोलें । (Alphabet of any Language)
   Recite A to Z (Alphabet of any language).
2. 20 से 1 तक उल्टी गिनती बोलें ।
   Count backward from 20 to 1.
3. 40 में से तीन घटाते हुए उल्टी गिनती बोलें ।
   Count backward by 3s starting from 40.

### IV. ध्यान एवं एकाग्रता (Attention and Concentration)

1. मैं कुछ अंक पढ़कर सुनाने वाला हूँ ध्यान से सुनें । जब मैं अंक पढ़ चुकूँ तो आप उन अंकों को दुहरायेंगे ।
   I am going to say some numbers. Listen them carefully, when I read them, you will repeat them in the same order.

| Item | Set I | Set II |
|---|---|---|
| 1. | 5–7–3 | 4–1–7 |
| 2. | 5–3–8–7 | 6–1–5–8 |
| 3. | 1–6–4–9–5 | 2–9–7–6–3 |
| 4. | 3–4–1–7–9–6 | 6–1–5–8–3–9 |
| 5. | 7–2–5–9–4–8–3 | 4–7–1–5–3–8–6 |
| 6. | 4–7–2–9–1–6–8–5 | 9–2–5–8–3–1–7–4 |

2. अब मैं कुछ और अंक पढ़कर सुनाने वाला हूँ, परन्तु ये अंक आपको विरुद्ध क्रम में (अन्त से प्रारम्भ करके) सुनाने होंगे जैसे—यदि मैं कहूँ '2-5' तो आप कहें '5-2' ।
   I am going to read some numbers but you will be required to repeat them backward. For example, I say 2, 5 you will say 5,2.

| Item | Set I | Set II |
|---|---|---|
| 1. | 8–5 | 2–8 |
| 2. | 4–3–7 | 8–5–1 |
| 3. | 8–5–6–3 | 3–7–5–9 |
| 4. | 4–7–2–9–1 | 9–2–5–8–4 |
| 5. | 2–5–9–4–8–3 | 7–1–5–3–9–6 |
| 6. | 3–5–8–6–1–9–2 | 6–3–7–1–4–8–5 |
| 7. | 8–5–2–3–6–1–9–4 | 2–8–4–5–9–7–1–3 |

### V. समयान्तरित स्मरण (Delayed Recall)

मैं कुछ चीजों के नाम पढ़ूँगा, आप ध्यान से सुनें और जब मैं आपसे उन चीजों के नाम दुहराने के लिये कहूँ तो आप उन चीजों के नाम बतलावें जिनको मैंने पढ़ा था। (एक नाम प्रति सैकण्ड की रफ्तार से पढ़ें और एक मिनट बाद सब नामों को दुहराने के लिये कहें।)

I am going to read the name of some objects, listen carefully and when I ask you to repeat, you will do so. (Read at the rate of one word per second and ask the subject to repeat it after an interval of one minute.)

| Sr. No. | List I | List II |
|---|---|---|
| 1. | छाता (Umbrella) | मछली (Fish) |
| 2. | फूल (Flower) | लैम्प (Lamp) |
| 3. | घड़ी (Clock) | रुपया (Rupee) |
| 4. | तस्वीर (Picture) | ताज (Taj) |
| 5. | पैंसिल (Pencil) | खिलौना (Toy) |

### VI. तत्कालीन स्मरण (Immediate Recall)

मैं एक-एक करके कुछ वाक्य पढ़ूँगा, आप ध्यान से सुनें, जब मैं बोल चुकूँ तो आपको वही वाक्य दोहराना होगा।

I am going to read a few small sentences one by one. Listen them carefully because when I am through I would like you to tell me the whole sentence as precisely as you can.

1. राम कुर्सी से उठा, दरवाजा खोला और बाजार चला गया।
   Ram got up from the chair, opened the door and went to market.
2. रोगी को मेज पर लिटाया, उसको देखा, दवा लिखी और कल आने के लिए कहा।
   Patient was asked to lie down on the table, he was seen, medicine was prescribed and was told to come next day.
3. मोहन के घर पानी नहीं था, उसने बाल्टी उठाई, बाजार के नल पर गया, पानी भरा और वापिस लौट आया।
   Mohan did not have water in his house, he picked-up the bucket, went to street well, filled it up and returned back.

## VII. समान युग्मों का शाब्दिक धारण (Verbal Retention for Similar Pairs)

मैं कुछ चीजों के जोड़े बताऊँगा । आप ध्यान से सुनें । इन जोड़ों में से जब एक चीज का नाम लिया जाये तो आपको जोड़े की दूसरी चीज का नाम बतलाना होगा । (दो सैकण्ड प्रति युग्म की रफ्तार से पढ़ें । प्रत्येक जोड़े के बाद लगभग 5 सैकण्ड का अन्तर रखें, अन्तिम युग्म के बाद लगभग 10 सैकण्ड का अन्तर रखकर जोड़े की प्रथम वस्तु का नाम पढ़ें और इसके साथ की दूसरी चीज का नाम बतलाने के लिये कहें ।)

Now, I am going to read to you a list of pairs, i.e., two words at a time. Listen carefully, when I name one word of the pair you will tell the second word of the pair (read the pairs at the rate of 2 seconds per pair, with a pause of 5 seconds between pairs ; give an interval of 10 seconds after reading last pair and then present a stimulus word of the pair and ask the subject to recall the 2nd word of the pair).

| Sr. No. | | |
|---|---|---|
| 1. | पेड़ (Tree) | फूल (Flower) |
| 2. | मीठा (Sweet) | नमकीन (Sour) |
| 3. | आदमी (Man) | औरत (Woman) |
| 4. | दिन (Day) | रात (Night) |
| 5. | काला (Black) | सफेद (White) |

## VIII. असमान युग्मों का शाब्दिक धारण (Verbal Retention for Dissimilar Pairs)

निर्देशन एवं परीक्षण प्रशासन उपर्युक्त भाँति रहेगा, लेकिन युग्म उनके सामने लिखे क्रम में पूछें । यदि प्रयोज्य सन्तोषजनक उत्तर न दे पाये तो उत्तर बतलाकर पुनः दूसरे युग्म का उत्तेजक शब्द बोलें, इस प्रकार तीन प्रयास करवाएँ । अगर प्रयोज्य प्रथम प्रयास में ही सब सही उत्तर दे तब भी दो प्रयास और करवाने अनिवार्य हैं ।

Instruction and administration procedure is the same as above for subtest VII, with a difference that stimulus words are to be presented in the order as mentioned for each of the trial. If subject fails to give correct answer, correct it and proceed to next stimulus word. If all his answers are correct in first trial even then other two trials are to be completed but in no case, pairs be repeated, only incorrect answers are to be corrected.

| Sr. No. | | | | | |
|---|---|---|---|---|---|
| 1. | मेज (Table) | काला (Black) | 4 | 2 | 1 |
| 2. | पेड़ (Tree) | ऊँचा (High) | 2 | 1 | 5 |
| 3. | लैम्प (Lamp) | खुरदरा (Uneven) | 1 | 5 | 3 |
| 4. | बच्चा (Child) | कड़वा (Bitter) | 3 | 4 | 2 |
| 5. | सपना (Dream) | गहरा (Deep) | 5 | 3 | 4 |

**IX. दृष्टिक धारणा (Visual Retention)**

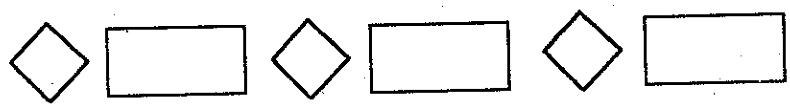
मैं आपको एक-एक करके कुछ कार्ड दिखाऊँगा, आप अच्छी तरह से देखें। कुछ देर (15 सैकण्ड) बाद मैं कार्ड उठा लूँगा और फिर जब मैं आपसे कहूँ (30 सैकण्ड बाद) तब आप वैसा ही चित्र कागज पर बनाएँगे जैसा कि आपने देखा था (एक कागज, एक पेन्सिल तथा एक रबड़ प्रयोज्य को दें लेकिन यह न कहें कि रबड़ का प्रयोग कर सकता है या नहीं।)

I am going to show you a card, see it carefully. After some time (15 seconds) I will take it away and when I ask you to draw (after 30 seconds) it, draw the things you saw in the card from your memory on this paper (give a paper, a pencil and an erasure to the subject but do not instruct whether he can use the erasure or not).

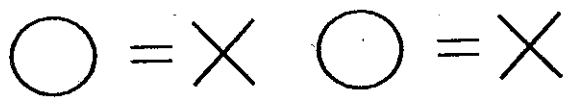
**Card 1 :**

**Card 2 :**

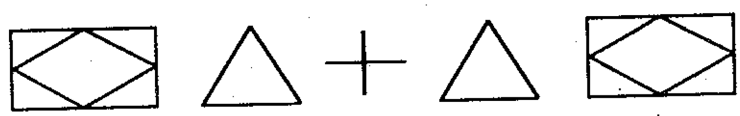
**Card 3 :**



**Card 4 :**



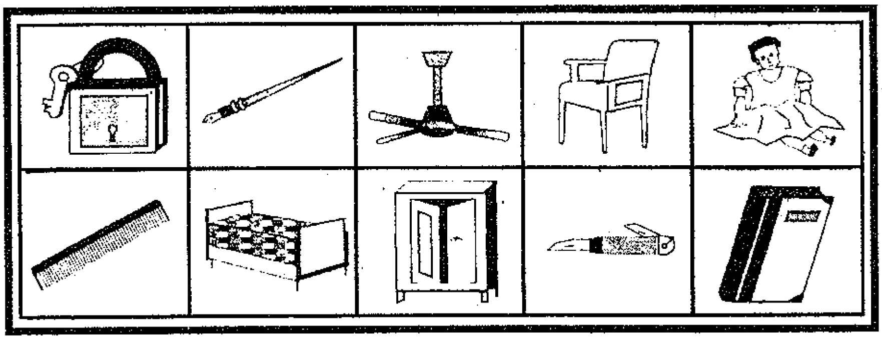
**Card 5 :**

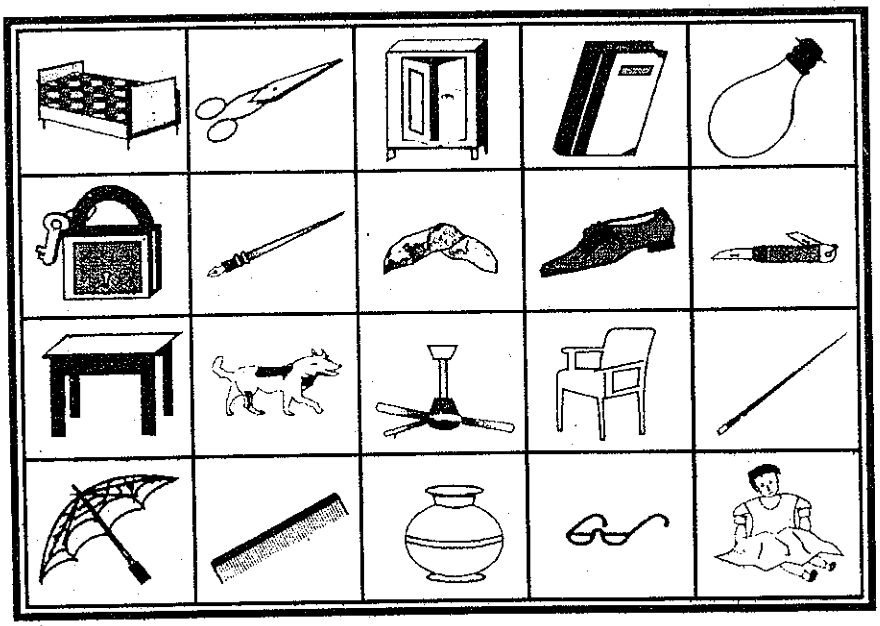


## X. प्रत्यभिज्ञान (Recognition)

मैं आपको एक कार्ड दिखलाऊँगा, जिसमें कुछ चीजें बनी हुई हैं। इन्हें आप ध्यान से (30 सैकण्ड तक) देखें, कुछ देर (2 मिनट) बाद मैं आपको दूसरा कार्ड दिखलाऊँगा जिसमें से आपको पहले देखी हुई चीजें पहचान कर उनके नाम बतलाने होंगे। (प्रयोज्य को यह न बतलावें कि पहले वाले कार्ड में कितनी चीजें थीं तथा अभी कितनी चीजें उसे और बतलानी हैं, बतलाये हुये नामों को नीचे लिखें)

I will show you a card containing pictures of many objects, see the whole card attentively (expose for 30 seconds), After some time (120 seconds interval), I will place before you another card. From this card you will be required to identify and name the objects you saw in earlier card (Do not tell the subject, the exact number of objects seen in first card and how many things he is yet to identify).





## SCORING

**I. Remote Memory.....** **Maximum Score 6**

Each correct answer (supported by attendants or from the file or from other reliable source) was given a score of one. In case of normals, where items 5 and 6 could not be administered (because of non-applicability of these items) an arbitrary scoring system was as follows : –

| | |
|---|---|
| If all the 4 items were correct then score | 6 |
| If only 3 items were correct then score | 5 |
| If only 2 items were correct then score | 3 |
| If only 1 item was correct then score | 1 |
| If none of the four items was correct then score | 0 |

**II. Recent Memory .....** **Maximum Score 5**

Each correct answer was given a score of one as above.

**III. Mental Balance ....** **Maximum Score 9**

(1) Alphabet –

| | |
|---|---|
| All correct within 15 seconds | 3 |
| All correct beyond 15 seconds | 2 |
| With one mistake | 1 |
| More than one mistake | 0 |

(2) Counting backward from 20 to 1

Scoring was same as mentioned for alphabet.

(3) Counting backward by 3's

| | |
|---|---|
| Correct within 30 seconds | 3 |
| Correct beyond 30 seconds | 2 |
| With one mistake | 1 |
| More than one mistake | 0 |

**IV. Attention Concentration.....** **Maximum Score 15**

(1) Digit forward – number of digits in the longest series correctly reproduced.

(2) Digit backward – number of digits in the longest series correctly reporoduced in reversed order. Summation of scores earned on Digit Forward and Backward was the score for this subtest.

**V. Delayed Recall...** **Maximum Score 10**

One score for each word correctly recalled. Summation of scores earned on both the trials were the score for this subtest.

**VI. Immediate Recall....** **Maximum Score 12**

One mark for each clause correctly reproduced. Thus the maximum possible range of scores on each sentence were as follows :

| | | |
|---|---|---|
| Sentence 1 | – | 0–3 |
| Sentence 2 | – | 0–4 |
| Sentence 3 | – | 0–5 |

**VII. Verbal Retention for Similar Pairs ....** **Maximum Score 5**

One mark for each correct reproduction of the associated word of the pair.

**VIII. Verbal Retention for Dissimilar Pairs ....** **Maximum Score 15**

One mark each for the correctly reproduced word of the pair, separately for each trial. Summation of marks on three trials was the score for this test.

**IX. Visual Retention .....** **Maximum Score 13**

One mark was given for each type of geometrical figure correctly reproduced in sequence and two marks were given to the one and the only compound figure on card 5. Thus the possible range of score for each design was as follows : –

| | | |
|---|---|---|
| Card 1 | – | 0–2 |
| Card 2 | – | 0–2 |
| Card 3 | – | 0–2 |
| Card 4 | – | 0–3 |
| Card 5 | – | 0–4 |

*Partial Scoring* for each card is allowed as illustrated below :

**Card 1 :** reproduction *(a)* is to be scored as zero because none of the designs is in its correct number and, *(b)* as one, because only 'minus' sign is equal in number.

(a) + – + – + –

(b) – + – + – +

**Card 2 :** reproduction as shown in illustration is to be given a score of one because only circle is correct in number and order.

○ + ○ + ○ +

**Card 3 :** Is scored similar to cards 1 and 2.

**Card 4 :** reproduction *(a)* is to be scored as zero because none of the design is in their original position, *(b)* as one because only parallel straight lines are in their correct order and number, *(c)* is to be scored as 2 because circles and parallel straight lines are both correct in their number and order.

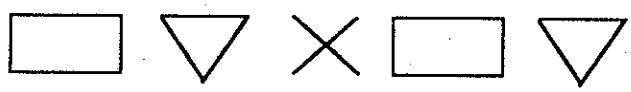
(a) × ○ = × ○ = **(Score 0)**

(b) × = ○ × = ○ **(Score 1)**

(c) ○ = + ○ = + **(Score 2)**

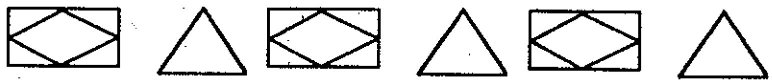
**Card 5 :** reproduction *(a)* is to be scored as zero because none of the designs is correct ; *(b)* is to be scored one because only inside figure is correct ; *(c)* is scored two (one for plus and one for inside designs being correct) ; *(d)* is scored two because only extreme compound design is correct ; *(e)* is scored three because only one type of design, i.e., 'plus' is replaced by cross and remaining whole design in correct.

(a)



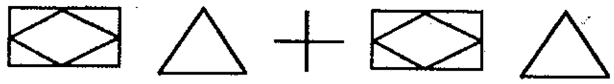
(Scored as zero because none of the design was correct)

(b)



(Scored as one because only inside figure was correct)

(c)

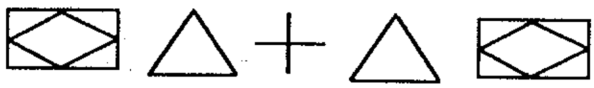


(Scored as two, one score for '+' and one for drawing inside figure of the rectangular).

(d)

(Scored as 2 because extreme compound figures were correct).

(e)



(Scored as three because only + was replaced by x).

**X. Recognition.....** **Maximum Score 10**

Each object correctly identified and named was given a score of one. An object correctly identified but either not named (even on asking) or wrongly named or showing inability to name was given a score of 1/2.

To minimize the effect of guessing number of wrongly identified objects were deducted from the earned score. No minus scoring was done.

### Scoring Criteria for Various items of Recognition Sub-test X

Give a score of one for correct identification and correct naming of an item.

Give half score for correct identification but wrong naming of an item.

Few subjects sometimes use different names for a particular item. Correct names for each item of this sub-test are given below. Any name other than those given here and only the description of the item, should be regarded as incorrect. However, if the examiner feels that the response given be the subject is not markedly wrong, he may give credit for that.

1. Lock = Lock/Lock-key/Close/Sed
2. Pen = Holder/ Pencil/Ink
3. Fan = Fan/Ceiling fan
4. Chair = Chair/Stool/Seat
5. Child = Boy/Girl/Lad/Doll/Baby
6. Comb = Disentangle, dress
7. Cot = Sofa/Charpai/Bed
8. Almirah = Door/Godrej Shelf/Wooden Case
9. Knife = Cutter/Razer sword/Whittle
10. Book = Register/Volume/Monograph

### Administration

Administration of the tests is simple and similar to other memory tests and clinical evaluation of memory. Broad hints for administration are provided in the test blank itself. It is, however, advised to consult research monograph by Pershad (1977) for more details. Administration takes nearly 15-20 minutes. Special care, however, needs to be taken with regard to the reliability of the responses on first two sub-tests, rate of presentation of digits, words and pairs on subtest IV, V, VII and VIII and interval between presentation of material and asking the subjects to recall.

**Subtest I : Remote Memory**

Out of the items at S.N. 3a, 3b, and 3c, only one item is to be enquired but these three items are not alternate of one another. If item at 3a is not applicable then ask 3b, if this is also not applicable then ask item at S.N. 3c. Reliability of the answer can be checked from the attendant and if there is no attendant, items may be repeated again after completing all the 10 sub test. Any discripancy in the answer may be settled or answer may be marked wrong.

**Sub-test II : Recent Momory**

It consists of five items. Answers on these items should be verified from the attendant who stay with patient or may be checked from nurse or hospital record if patient is admitted. If these efforts are not fruitful, then the items may be repeated after completing full battery of memory tests. Any discrepancy in the answer needs to be settled or answer may be marked as wrong.

**Sub-test III : Mental Balance**

It consists of three items. Time required to complete the recitation should be noted down precisely with the help of stop watch/clock/mobile phone and write down any omission or error made on each item.

**Subtest IV : Attention Concentration**

It consists of digits which are to be read by the tester and immediately after he is through, subject needs to repeat it either in same order or in reverse order as he is instructed. While reading the digits, subject needs to be asked to be attentive. Digits need to be read out at a steady rate of one digit per second. While reading the digits raise your right arm, keep elbow at table and open palm towards the subjects when you are through with an item-down your arm and palm indicating now he has to repeat the digits heard. Both the sets of each item need to be administered even when subject is able to repeat correctly on the first set. Put a tick mark on each correctly reproduced set on all the items administered. Start from the lowest length, i.e., from item No. 1 and continue till subject fails on both the sets of the same length of digits. Always when you read the digit, raise your arm and lower it when you are through. If his reproduction is not in a desired sequence on any item remind again saying listen me carefully.

**Sub-test V : Delayed Recall**

There are two lists of five names each of common objects. First give the instruction as written in the memory scale blank. Then read the name of the common objects from list – I, at a steady rate of one second each. After the expiry of one minute (60 seconds) post presentation period observed silently ask the subject to recall the names he has heard. Name may be recalled in any order. Simply put a tick on each name correctly recalled and write down any name he has recalled but was not read to him. In the same manner administer the 2nd list also, but before that ask him again, 'listen to me carefully'. If he asks, should he repeat the words in the same sequence. Answer him, 'as you wish' If he indulges in any conversation during the rest pause, discourage him, saying 'Let L's, complete this work, then we will discuss". If he rehearses loudly during interval period ask him 'gently' please rehearse it yourself silently.'

**Sub-test VI : Immediate Racall**

There are three sentences of increasing length, first sentence has 3 clauses, 2nd has 4 clauses and the third sentence has 5 clauses, Read out the instruction as written on the memory scale blank. Read first sentence slowly, distinctly and at an uniform rate of presentation. Immediately after presentation ask the subject to recall it. Note down the recalled sentence verbatim or each of the correctly recalled clauses may be ticked leaving incorrect or omitted clauses as such. Like this present sentences 2 and 3 also. If subject made any error or omission or substituted a work from other languages then while presenting the subsequent sentence, remind him to be attentive, careful and to reproduce the sentence, exactly as it was presented / read out.

**Sub-test VII : Verbal Retention for Similar Pairs**

There are five noun-noun pairs. Read out the instructions from the memory Scale blank Each pair is to be read distinctly and slowly at the rate of 2 seconds per pair. Give 5 seconds pause (intervals) between any two pairs. After last pair give 10 seconds interval and read first word (noun) of the pair and ask what was the 2nd word (noun) of the pair, proceed exactly in the same manner till all the 5 items are exhausted. Put a tick on each pair if response is correct.

**Sub-test VIII : Retention of Dissimilar Pairs**

There are five noun-adjective pairs. Three trials are given. In each trial stimulus word (noun) is presented in random order as written against each pair. Give the instruction to the subject as written for sub test VII. Read the noun adjective pairs at the rate of 2 seconds per pair. Keep pause of 5 seconds between any two pairs. On last pair give an interval (pause) of 10 seconds and then read the stimulus word 'lamp' (under first column one is written against lamp). If the response is correct put a tick in the cell where 1 is written in first column. If answer is wrong or no answer is given then tell the correct response and put a cross in the cell. And proceed to stimulus word 'Tree' (2 under first column), if response is wrong correct it before moving to other stimulus word.

Like this complete the first trial in the order mentioned in first column. Before starting second trial there is no need of presenting pairs again because where ever response was wrong, it was corrected with a request to remember. For 2nd trial stimulus words are to be presented in the sequence as mentioned in column two, wrong responses are to be corrected putting a cross in the cell and if answer is correct put a tick, finishing the 2nd trial, proceed with third trial in a sequence as given in column three.

### Sub-test IX : Visual Retention

There are five cards, Each card is to be presented for *fifteen* seconds and *thirty* seconds post exposure period be observed silently. Thereafter subject is asked to draw the same design from his memory. If his reproduction is incorrect in any way, then ask him to be attentive before presenting the next card-say ध्यान से देखिए इसमें जैसा बना है वैसा ही आपको अपनी याद से बनाना होगा । If he seems to have forgotten an element (sub part) of the design, ask him to put a point (dot) for that and complete the design. If he asks "*can I use the eraser*", say – "as you please".

### Sub-test X : Recognition

There are two cards of similar size. First card contains 10 pictures of common objects and second card contains 20 pictures (10 pictures of the first card are mixed with other 10 pictures). First card (with 10 picture) is to be presented (exposed to the subjects for 30 seconds with a request to see and remember the objects therein. On expiry of 30 seconds, card is removed gently and after an interval of 120 seconds observed silently, second card (card with 20 pictures) is presented and subject is requested to identify the objects that were present in earlier exposure. Name of all the objects as identified by the subjects are to be noted down at the space provided in the test proforma. Even the wrongly identified objects are to be noted. If subject identifies an object correctly but either names it wrongly or does not name it correctly the fact is to be noted down. He is not to be told at any stage about the number of objects present in card one and how many objects yet he has to identify from card two.

**Interpretation**

For obtaining dysfunction rating the following table needs to be used. First of all, the raw score on each subtest is to be noted on the proforma and then according to the education of the subjects his scores need to be rated.

Suppose there is an individual with an education of VIII class who has secured the following score column (2)

| 1<br>Sub-test | 2<br>Raw Score | 3<br>Converted Score | 4<br>Dysfunction Rating |
|---|---|---|---|
| I | 4 | 0–2 | 3 |
| II | 5 | 5+ | 0 |
| III | 8 | 5+ | 0 |
| IV | 8 | 5+ | 0 |
| V | 7 | 3–4 | 2 |
| VI | 9 | 5+ | 0 |
| VII | 4 | 3–4 | 2 |
| VIII | 12 | 5+ | 0 |
| IX | 9 | 5+ | 0 |
| X | 8 | 3–4 | 2 |

His converted score from the Table 15 are given in column 3 and dysfunction rating is given in column 4. Score in column 3 will help in completing the first page of the test blank form (proforma).

**TABLE 15**

*Conversion of Raw Score of Memory Subtests in Quintile for 0-V, VI-IX and X and above Educational Levels*

| Converted Score | Sub-tests and Raw Scores | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | I | II | III | IV | V | VI | VII | VII | IX | X |
| **Subjects having Education upto V Class** | | | | | | | | | | |
| 5+ | 6 | 5 | 3–9 | 7–15 | 7–10 | 6–12 | 4–5 | 8–15 | 5–13 | 8–10 |
| 3–4 | 5 | 4 | 2 | 6 | 6 | 5 | 3 | 5–7 | 4 | 7 |
| 0–2 | 0–4 | 0–3 | 0–1 | 0–5 | 0–5 | 0–4 | 0–2 | 0–4 | 0–3 | 0–6 |
| **Subjects having Education upto VI-IX Class** | | | | | | | | | | |
| 5+ | 6 | 5 | 7–9 | 8–15 | 8–10 | 7–12 | 5 | 12–15 | 8–13 | 9–10 |
| 3–4 | 5 | 4 | 5–6 | 7 | 7 | 6 | 4 | 10–11 | 6–7 | 8 |
| 0–2 | 0–4 | 0–3 | 0–4 | 0–6 | 0–6 | 0–5 | 0–3 | 0–9 | 0–5 | 0–7 |
| **Subjects having Education of X and Above** | | | | | | | | | | |
| 5+ | 6 | 5 | 8–9 | 10–15 | 9–10 | 9–12 | 5 | 13–14 | 10–13 | 9–10 |
| 3–4 | 5 | 4 | 7 | 9 | 8 | 7–8 | 4 | 10–11 | 8–9 | 8 |
| 0–2 | 0–4 | 0–3 | 0–6 | 0–8 | 0–7 | 0–6 | 0–3 | 0–9 | 0–7 | 0–7 |

Adapted from Pershad (1977) and Pershad and Wig (1988)

**Rating of Converted Score for Dysfunction Score**

| Converted Score of | be given a Dysfunction Rating |
|---|---|
| 0–2 | 3 |
| 3–4 | 2 |
| 5+ | 0 |

Maximum Dysfunction Rating Score for each subtest is 3 and there are 10 sub-tests, so total Dysfunction Rating Score on PGIMS would be $3 \times 10 = 30$.

## 2. BATTERY OF PERFORMANCE TESTS OF INTELLIGENCE

There are a number of ways of classifying types of intelligence, i.e., Cattell's (1976) *Fluid* and crystallized ; Hebb's intelligence A and B (quoted from Kulshrestha, 1979), Jensen's (1973 and 1974) *Level I* and Level II intelligence. In each of these three classifications, the former (in italics) is generally considered to be more related with genetic component of intelligence and the latter is presumed to be more related with environmental development of intelligence. The former can be measured by so called 'culture fair' tests. To some extent non verbal and performance tests are said to be the measure of inborn aspect.

On the basis of neurological theory of 'brain behaviour' by and large intellectual functioning is determined by the frontal lobe. Non-verbal/Perfomance ability resides in the right hemisphere in right handed individuals. Therefore deterioration in performance I. Q. may be considered as indication of lesion in the right hemisphere.

Revised Bhatia's Short Battery of Performance Tests of Intelligence for adults (Verma, Pershad, Malhotra, and Arunima, 1988) included in the 'PGI Battery of Brain Dysfunction' consists of (1) Kohs' Block Design Test and (2) Pass-A-Long Test. By and large the administration procedure of these tests is similar to that of Bhatia's Battery as given in its manual.

**Material**

1. Kohs' Block Design Test as used in Bhatia's Battery of Performance Tests of Intelligence consists of 10 cards of design and 16 cubes (6 sides of a cube coloured as Blue, White, Red, Green, Half red-half white and Half blue half yellow).
2. Pass-A-Long Test as used in Bhatia's Battery of Performance Tests of Intelligence consists of 8 cards of design, 4 Boxes and rectangular Blocks (6 blue small, 2 blue long, 1 blue big and 2 red small, 1 red big, 1 red long).
3. Stop watch / clock / mobile phone.
4. Paper to record the time.

**Administration**

Administer the tests in a natural relaxed manner. Establish workable rapport with the subject. Talk in the subject's own dialect. The seating arrangment should be comfortable and test administered in a distraction free, noise free atmosphere to the extent possible. The subject should be encouraged to do as well as he can under the circumstnces. Note the observation very carefully and faithfully. Record the time required to complete each item, as carefully as you can, with the help of a stop-watch. Give only as much aid or hint to the subjects as is permissible. When the subject fails, do no put on a serious look.

**For Kohs' Block Design Test**

1. Place four cubes before the subject. Explain how they are all alike and coloured in a particular way, Let him handle and examine the cubes at leisure, confidently, and let him feel at home.
2. Show him card No. 1, Tell him that a design like this has to be prepared with the cubes. Even if he attempts to prepare the design himself, you should demonstrate this design to every subject.
3. Mix up the blocks. Now ask him to prepare Design No. 1 as you have already shown him. Note the time.
4. If the subject succeeds in the above within the time limit, namely 2 minutes, proceed to Design No. 2 and ask him to make it similar to the design on the card (put Card 2).
5. Proceed in this manner with successive designs.
6. When the subject fails in a particular design within the time limit, demonstrate the design after he has failed. Do not discuss ? do not let him again try the design he has failed ? pass on to the next which, of course, he must again try independently.

7. When the subject comes to Design No. 6, give him five cubes more, making the total nine ; when he comes to design No. 8, give him the remaining seven, making the total sixteen.
8. Stop the test when failure has been recorded twice in succession.
9. The maximum time limit for Design Nos 1-5 is 2 minutes each and for the Designs Nos. 6-10 is 3 Minutes each.
10. In the Remarks column you may note anything particular or peculiar you find about the subject in making the designs (e. g., his actions, movements, expressions. spontaneous verbalizations to indicate his ways to meet new tasks, challenges, frustration.)

**For Pass-A-Long Test**

1. Take the first and the smallest box, and the Card No. 1. The red side should be in the upward direction (towards the tester) and the blue side facing down (towards the subject). Point out to the subject that the red block has been placed near the blue end (blue line) and the blue blocks near the red end (red line). Explain that the red block must be brought to the red side and the blue blocks to the blue side, as shown in the card. Emphasize that blocks should not be lifted, but may only be moved. Demonstrate the solution of the first box to every subject (Smallest Box and Card 1).
2. Again place the card No. 1 and the box and ask the subject to do it (as he has already seen). Record the time required to complete it successfully.
3. Proceed to Designs No. 2, 3, etc. with the appropriate boxes, and after having placed the blocks properly in the initial position as required in the test. The initial position is obtained by putting the blocks wrongly Red blocks be put towards blue end of the box and blue blocks towards red end of the box. Put the cards with Red at top and blue at bottom.
4. When the subject fails in a particular design within the time limit, demonstrate the design after he has failed. Do not discuss. Do not let him again try this

design. Pass on to the next which, of course, he must again try independently.

5. Stop the test when failure has been recorded twice in succession.
6. The maximum time limit for Designs (cards/items) 1 to 4 is 2 minutes each and for Designs 5-8 is 3 minutes each.
7. In the remarks column you may note anything particular or peculiar you find about the subject in solving the designs (e.g., verbal remarks, facial expression, restlessness, irritability, giving up before time, asking for cues as help, showing signs of distress, repeating same errors, etc.)

**Scoring Procedure**

| For Kohs' Block Design | | For Pass-A-Long Test | |
|---|---|---|---|
| **Designs** | | **Designs** | |
| Designs I–V | | Designs I–IV | |
| upto 30 secs | 4 | upto 30 secs | 4 |
| 31 – 60 secs | 3 | 31 – 60 secs | 3 |
| 61 – 90 secs | 2 | 61 – 90 secs | 2 |
| 91 – 120 secs | 1 | 91 – 120 secs | 1 |
| 121 + secs | 0 | 121 + secs | 0 |
| Designs VI–X | | Designs V–VIII | |
| upto 30 secs | 6 | upto 30 secs | 6 |
| 31 – 60 secs | 5 | 31 – 60 secs | 5 |
| 61 – 90 secs | 4 | 61 – 90 secs | 4 |
| 91 – 120 secs | 3 | 91 – 120 secs | 3 |
| 121 – 150 secs | 2 | 121 – 150 secs | 2 |
| 151 – 180 secs | 1 | 151 – 180 secs | 1 |
| 181 + secs | 0 | 181 + secs | 0 |
| Max. Possible Score | 50 | Max. Possible Score | 40 |
| | | **Grand Total** | **90** |

### Recording the Performance

Record the time required by the subject to complete each item of Kohs' Block and Pass-A-Long tests. Demonstrate the incomplete or wrongly completed items and write (D) demonstrated against the number of such items. Stop after two consecutive failures. The second failure should not be followed by any demonstration. The last design should not be demonstrated as there is no other design to follow it. This is because the purpose of demonstration is to see if the subject can profit from it in solving newer problems (i.e., capacity to learn from past experience).

| Kohs' Time taken to complete successfully | | | Pass-A-Long Time taken to complete successfully | | | Remarks |
|---|---|---|---|---|---|---|
| **Sr. No.** | **Time** | **Score** | **Sr. No.** | **Time** | **Score** | |
| B. 1 | | | B. 1 | | | |
| B. 2 | | | B. 2 | | | |
| B. 3 | | | B. 3 | | | |
| B. 4 | | | B. 4 | | | |
| B. 5 | | | B. 5 | | | |
| B. 6 | | | B. 6 | | | |
| B. 7 | | | B. 7 | | | |
| B. 8 | | | B. 8 | | | |
| B. 9 | | | | | | |
| B. 10 | | | | | | |
| Total Score | | | Total Score | | | |
| T. Q. = | | | T. Q. = | | | |
| Performance Quotient = | | | | | | |

**How to determine T. Q.'s and P. Q. (Performance Quotient)** – T.Q. can be perceived as similar to P. Q. except for the difference, that T. Q. (Test Quotient) is a P. Q. (Performance Quotient) determined on the basis of performance on a single subtest and P. Q. is an arithmetical mean of the two T. Q.'s Thus the concept of P. Q. is a little broader than T. Q. The advantage of T. Q. is that the performance on the two subtests of the battery might be of some significance in the screening, guidance, counselling and rehabilitation purposes.

In the subsequent tables you will find that T. Q.'s are determined for each test separately – for each of the four age levels, for the three educational levels and for the two sex categories. Thus there are 24 sets of norms given in 12 tables. Each table is meant for a particular age group along with one of the three categories of educational level. The norms for male and female are separated by a bold line in each table.

The following *example* will make it clear as to how to determine T. Q. and P. Q. of a particular case: – Suppose there is a male, 32 years of age, educated upto graduate level, who obtains raw scores on Short Battery of Performance Tests as follows :

On Kohs' Block Design test – 18

On Pass-A-Long test – 23

For this gentleman left half of the table P. Q. 6 (which is for males) has to be consulted which will give the following standard scores in terms of Test Quotients :

| Bhatia's Short Scale | Obtained Score | Test Quotients from Table |
|---|---|---|
| Kohs' Block Design | 18 | 91 |
| Pass-A-Long Test | 23 | 110 |
| | Total of T.Q.'s = | 201 |
| | P. Q. (Mean of T.Q.'s) = 100.5 | |

How to determine $\frac{P}{K} \times 100$

P stands for the raw score on Pass-A-Long test and K for the raw Score on Kohs' Block Design test. Divide the obtained raw score of Pass-A-Long by the obtained raw score of Kohs's Block Design. This product is then multiplied by 100.

**Interpretation**

For obtaining Dysfunction ratings the following table is to be used.

| Variable | Dysfunction Rating |
|---|---|
| P. Q. (Mean of two T. Q.'s) | |
| 81 and above | 0 |
| 61 to 80 | 2 |
| Less than 61 | 3 |
| $\frac{P}{K} \times 100$ | |
| Upto 199 | 0 |
| 200 to 250 | 2 |
| above 250 | 3 |

## TABLES OF NORMS FOR KOHS' BLOCK AND PASS-A-LONG

### TABLE : PQ 1 ; AGE : 20–29 ; EDU : 0–5

| Raw Score | MALE | | FEMALE | |
|---|---|---|---|---|
| | Kohs' Block | Pass-A-Long | Kohs' Block | Pass-A-Long |
| 1 | 62 | 50 | 62 | 40 |
| 2 | 67 | 55 | 69 | 47 |
| 3 | 72 | 59 | 75 | 54 |
| 4 | 77 | 64 | 82 | 61 |
| 5 | 82 | 68 | 89 | 68 |
| 6 | 87 | 73 | 96 | 75 |
| 7 | 92 | 77 | 103 | 83 |
| 8 | 97 | 82 | 109 | 90 |
| 9 | 102 | 86 | 116 | 97 |
| 10 | 107 | 91 | 123 | 104 |
| 11 | 112 | 95 | 130 | 111 |
| 12 | 117 | 100 | 136 | 118 |
| 13 | 122 | 104 | 143 | 125 |
| 14 | 127 | 109 | 150 | 132 |
| 15 | 132 | 113 | 157 | 139 |
| 16 | 137 | 118 | 164 | 146 |
| 17 | 142 | 122 | 170 | 153 |
| 18 | 147 | 127 | 177 | 160 |
| 19 | 152 | 131 | 184 | 167 |
| 20 | 157 | 136 | 191 | 175 |
| 21 | 162 | 140 | 198 | 182 |
| 22 | 168 | 145 | 204 | 189 |
| 23 | 173 | 149 | 211 | 196 |
| 24 | 178 | 154 | 218 | 203 |
| 25 | 183 | 158 | 225 | 210 |
| 26 | 188 | 163 | | |
| 27 | 193 | 167 | | |
| 28 | 198 | 172 | | |
| 29 | 203 | 176 | | |
| 30 | 208 | 181 | | |
| 31 | | 185 | | |
| 32 | | 190 | | |
| 33 | | 194 | | |
| 34 | | 198 | | |
| 35 | | 203 | | |
| 36 | | | | |
| 37 | | | | |
| 38 | | | | |
| 39 | | | | |
| 40 | | | | |
| Mean | 8·59 | 12·07 | 6·63 | 9·47 |
| S.D. | 2·98 | 3·34 | 2·21 | 2·12 |
| N | 30 | 30 | 30 | 30 |

**TABLE : PQ 2 ; AGE : 20–29 ; EDU : 6–9**

| Raw Score | MALE | | FEMALE | |
|---|---|---|---|---|
| | Kohs' Block | Pass-A-Long | Kohs' Block | Pass-A-Long |
| 1 | 58 | 49 | 66 | 52 |
| 2 | 61 | 52 | 69 | 56 |
| 3 | 65 | 56 | 72 | 59 |
| 4 | 68 | 59 | 75 | 63 |
| 5 | 72 | 63 | 78 | 67 |
| 6 | 75 | 66 | 81 | 70 |
| 7 | 78 | 70 | 84 | 74 |
| 8 | 82 | 73 | 87 | 78 |
| 9 | 85 | 77 | 89 | 81 |
| 10 | 89 | 80 | 92 | 85 |
| 11 | 92 | 84 | 95 | 89 |
| 12 | 95 | 87 | 98 | 93 |
| 13 | 99 | 91 | 101 | 96 |
| 14 | 102 | 94 | 104 | 100 |
| 15 | 105 | 98 | 107 | 104 |
| 16 | 109 | 102 | 110 | 107 |
| 17 | 112 | 105 | 113 | 111 |
| 18 | 116 | 109 | 116 | 115 |
| 19 | 119 | 112 | 119 | 118 |
| 20 | 122 | 116 | 122 | 122 |
| 21 | 126 | 119 | 125 | 126 |
| 22 | 129 | 123 | 128 | 129 |
| 23 | 133 | 126 | 130 | 133 |
| 24 | 136 | 130 | 133 | 137 |
| 25 | 139 | 133 | 136 | 140 |
| 26 | 143 | 137 | 139 | 144 |
| 27 | 146 | 140 | 142 | 148 |
| 28 | 149 | 144 | 145 | 151 |
| 29 | 153 | 147 | 148 | 155 |
| 30 | 156 | 151 | 151 | 159 |
| 31 | 160 | 154 | 154 | 163 |
| 32 | 163 | 158 | 157 | 166 |
| 33 | 166 | 162 | 160 | 170 |
| 34 | 170 | 165 | 163 | 174 |
| 35 | 173 | 169 | 166 | 177 |
| 36 | 177 | 172 | 169 | 181 |
| 37 | 180 | 176 | 171 | 185 |
| 38 | 183 | 179 | 174 | 188 |
| 39 | 187 | 183 | 177 | 192 |
| 40 | 190 | 186 | 180 | 196 |
| Mean | 13.39 | 15.57 | 12.6 | 14.03 |
| S.D. | 4.43 | 4.25 | 5.12 | 4.07 |
| N | 30 | 30 | 30 | 30 |

## TABLE : PQ 3 ; AGE : 20–29 ; EDU : 10+

| Raw Score | MALE | | FEMALE | |
|---|---|---|---|---|
| | Kohs' Block | Pass-A-Long | Kohs' Block | Pass-A-Long |
| 1 | 61 | 53 | 29 | 55 |
| 2 | 63 | 55 | 34 | 57 |
| 3 | 65 | 58 | 38 | 60 |
| 4 | 67 | 60 | 42 | 63 |
| 5 | 69 | 62 | 47 | 65 |
| 6 | 71 | 65 | 51 | 68 |
| 7 | 73 | 67 | 56 | 70 |
| 8 | 74 | 69 | 60 | 73 |
| 9 | 76 | 72 | 65 | 76 |
| 10 | 78 | 74 | 69 | 78 |
| 11 | 80 | 76 | 74 | 81 |
| 12 | 82 | 79 | 78 | 83 |
| 13 | 84 | 81 | 83 | 86 |
| 14 | 86 | 83 | 87 | 89 |
| 15 | 87 | 86 | 91 | 91 |
| 16 | 89 | 88 | 96 | 94 |
| 17 | 91 | 90 | 100 | 96 |
| 18 | 93 | 93 | 105 | 99 |
| 19 | 95 | 95 | 109 | 101 |
| 20 | 97 | 98 | 114 | 104 |
| 21 | 99 | 100 | 118 | 107 |
| 22 | 101 | 102 | 123 | 109 |
| 23 | 102 | 105 | 127 | 112 |
| 24 | 104 | 107 | 131 | 114 |
| 25 | 106 | 109 | 136 | 117 |
| 26 | 108 | 112 | 140 | 120 |
| 27 | 110 | 114 | 145 | 122 |
| 28 | 112 | 116 | 149 | 125 |
| 29 | 114 | 119 | 154 | 127 |
| 30 | 115 | 121 | 158 | 130 |
| 31 | 117 | 123 | 163 | 133 |
| 32 | 119 | 126 | 167 | 135 |
| 33 | 121 | 128 | 172 | 138 |
| 34 | 123 | 130 | 176 | 140 |
| 35 | 125 | 133 | 180 | 143 |
| 36 | 127 | 135 | 185 | 146 |
| 37 | 128 | 137 | 189 | 148 |
| 38 | 130 | 140 | 194 | 151 |
| 39 | 132 | 142 | 198 | 153 |
| 40 | 134 | 144 | 203 | 156 |
| Mean | 21.73 | 21.07 | 16.93 | 18.43 |
| S.D. | 8.07 | 6.42 | 3.37 | 5.78 |
| N | 30 | 30 | 30 | 30 |

**TABLE : PQ 4 ; AGE : 30–39 ; EDU : 0–5**

| Raw Score | MALE | | FEMALE | |
|---|---|---|---|---|
| | Kohs' Block | Pass-A-Long | Kohs' Block | Pass-A-Long |
| 1 | 45 | 29 | 66 | 48 |
| 2 | 50 | 35 | 70 | 52 |
| 3 | 56 | 41 | 75 | 57 |
| 4 | 61 | 47 | 79 | 61 |
| 5 | 67 | 53 | 83 | 66 |
| 6 | 73 | 59 | 88 | 71 |
| 7 | 78 | 65 | 92 | 75 |
| 8 | 84 | 71 | 96 | 80 |
| 9 | 90 | 77 | 101 | 85 |
| 10 | 95 | 83 | 105 | 89 |
| 11 | 101 | 89 | 109 | 94 |
| 12 | 107 | 95 | 113 | 99 |
| 13 | 112 | 101 | 118 | 103 |
| 14 | 118 | 107 | 122 | 108 |
| 15 | 124 | 113 | 126 | 113 |
| 16 | 129 | 119 | 131 | 117 |
| 17 | 135 | 125 | 135 | 122 |
| 18 | 140 | 131 | 139 | 127 |
| 19 | 146 | 137 | 144 | 131 |
| 20 | 152 | 143 | 148 | 136 |
| 21 | 157 | 149 | 152 | 141 |
| 22 | 164 | 155 | 156 | 145 |
| 23 | 169 | 161 | 161 | 150 |
| 24 | 174 | 167 | 165 | 155 |
| 25 | 180 | 173 | 169 | 159 |
| 26 | 186 | 179 | 174 | 164 |
| 27 | 191 | 185 | 178 | 169 |
| 28 | 197 | 192 | 182 | 173 |
| 29 | 202 | 198 | 187 | 178 |
| 30 | 208 | 204 | 191 | 183 |
| 31 | | | 195 | 187 |
| 32 | | | 199 | 192 |
| 33 | | | 204 | 197 |
| 34 | | | 208 | 201 |
| 35 | | | 212 | 206 |
| 36 | | | | |
| 37 | | | | |
| 38 | | | | |
| 39 | | | | |
| 40 | | | | |
| Mean | 10.83 | 12.8 | 8.87 | 11.8 |
| S.D. | 2.66 | 2.49 | 3.49 | 3.62 |
| N | 30 | 30 | 30 | 30 |

**TABLE : PQ 5 ; AGE : 30–39 ; EDU : 6–9**

| Raw Score | MALE | | FEMALE | |
|---|---|---|---|---|
| | Kohs' Block | Pass-A-Long | Kohs' Block | Pass-A-Long |
| 1 | 53 | 34 | 64 | 48 |
| 2 | 57 | 38 | 68 | 52 |
| 3 | 61 | 43 | 71 | 57 |
| 4 | 64 | 47 | 74 | 61 |
| 5 | 68 | 51 | 77 | 66 |
| 6 | 72 | 55 | 81 | 71 |
| 7 | 76 | 60 | 84 | 75 |
| 8 | 80 | 64 | 87 | 80 |
| 9 | 83 | 68 | 91 | 85 |
| 10 | 87 | 73 | 94 | 89 |
| 11 | 91 | 77 | 97 | 94 |
| 12 | 95 | 81 | 101 | 99 |
| 13 | 99 | 86 | 104 | 103 |
| 14 | 102 | 90 | 107 | 108 |
| 15 | 106 | 94 | 112 | 113 |
| 16 | 110 | 101 | 114 | 117 |
| 17 | 114 | 103 | 117 | 122 |
| 18 | 118 | 107 | 121 | 127 |
| 19 | 121 | 112 | 124 | 131 |
| 20 | 125 | 116 | 127 | 136 |
| 21 | 129 | 120 | 130 | 141 |
| 22 | 133 | 124 | 134 | 145 |
| 23 | 137 | 129 | 137 | 150 |
| 24 | 140 | 133 | 140 | 155 |
| 25 | 144 | 137 | 144 | 159 |
| 26 | 148 | 142 | 147 | 164 |
| 27 | 152 | 146 | 150 | 169 |
| 28 | 156 | 150 | 154 | 173 |
| 29 | 160 | 155 | 157 | 178 |
| 30 | 163 | 160 | 160 | 183 |
| 31 | 167 | 163 | 164 | 187 |
| 32 | 171 | 168 | 167 | 192 |
| 33 | 175 | 172 | 170 | 197 |
| 34 | 179 | 176 | 174 | 201 |
| 35 | 182 | 180 | 177 | 206 |
| 36 | 186 | 185 | 180 | |
| 37 | 190 | 189 | 183 | |
| 38 | 194 | 193 | 187 | |
| 39 | 198 | 198 | 190 | |
| 40 | 201 | 202 | 193 | |
| Mean | 13.37 | 16.33 | 11.8 | 12.27 |
| S.D. | 3.94 | 3.48 | 4.53 | 3.22 |
| N | 30 | 30 | 30 | 30 |

**TABLE : PQ 6 ; AGE : 30–39 ; EDU : 10+**

| Raw Score | MALE | | FEMALE | |
|---|---|---|---|---|
| | Kohs' Block | Pass-A-Long | Kohs' Block | Pass-A-Long |
| 1 | 29 | 45 | 59 | 32 |
| 2 | 32 | 48 | 62 | 37 |
| 3 | 36 | 51 | 65 | 41 |
| 4 | 40 | 54 | 67 | 45 |
| 5 | 43 | 57 | 70 | 49 |
| 6 | 47 | 60 | 73 | 53 |
| 7 | 51 | 63 | 75 | 57 |
| 8 | 54 | 66 | 78 | 61 |
| 9 | 58 | 69 | 81 | 66 |
| 10 | 62 | 72 | 83 | 70 |
| 11 | 66 | 75 | 86 | 74 |
| 12 | 69 | 78 | 89 | 78 |
| 13 | 73 | 81 | 91 | 82 |
| 14 | 77 | 84 | 94 | 86 |
| 15 | 80 | 87 | 97 | 90 |
| 16 | 84 | 90 | 99 | 95 |
| 17 | 88 | 93 | 102 | 99 |
| 18 | 91 | 96 | 105 | 103 |
| 19 | 95 | 99 | 107 | 107 |
| 20 | 99 | 101 | 110 | 111 |
| 21 | 102 | 104 | 113 | 115 |
| 22 | 106 | 107 | 116 | 119 |
| 23 | 110 | 110 | 118 | 124 |
| 24 | 114 | 113 | 121 | 128 |
| 25 | 117 | 116 | 124 | 132 |
| 26 | 121 | 119 | 126 | 136 |
| 27 | 125 | 122 | 129 | 140 |
| 28 | 128 | 125 | 132 | 144 |
| 29 | 132 | 128 | 134 | 148 |
| 30 | 136 | 131 | 137 | 153 |
| 31 | 139 | 134 | 140 | 157 |
| 32 | 143 | 137 | 142 | 161 |
| 33 | 147 | 140 | 145 | 165 |
| 34 | 151 | 143 | 148 | 169 |
| 35 | 154 | 146 | 150 | 173 |
| 36 | 158 | 149 | 153 | 177 |
| 37 | 162 | 152 | 156 | 182 |
| 38 | 165 | 155 | 158 | 186 |
| 39 | 169 | 158 | 161 | 190 |
| 40 | 173 | 161 | 164 | 194 |
| Mean | 20.33 | 19.5 | 16.2 | 17.3 |
| S.D. | 4.06 | 5.02 | 5.61 | 3.62 |
| N | 30 | 30 | 30 | 30 |

**TABLE : PQ 7 ; AGE : 40–49 ; EDU : 0–5**

| Raw Score | MALE | | FEMALE | |
|---|---|---|---|---|
| | Kohs' Block | Pass-A-Long | Kohs' Block | Pass-A-Long |
| 1 | 71 | 45 | 66 | 36 |
| 2 | 74 | 50 | 71 | 43 |
| 3 | 77 | 55 | 76 | 50 |
| 4 | 80 | 60 | 80 | 57 |
| 5 | 83 | 64 | 85 | 64 |
| 6 | 86 | 69 | 90 | 71 |
| 7 | 89 | 74 | 95 | 78 |
| 8 | 92 | 78 | 100 | 86 |
| 9 | 95 | 83 | 104 | 93 |
| 10 | 98 | 88 | 109 | 100 |
| 11 | 102 | 93 | 114 | 107 |
| 12 | 105 | 97 | 119 | 114 |
| 13 | 108 | 102 | 123 | 121 |
| 14 | 111 | 107 | 128 | 128 |
| 15 | 114 | 112 | 133 | 135 |
| 16 | 117 | 116 | 138 | 142 |
| 17 | 120 | 121 | 143 | 150 |
| 18 | 123 | 126 | 147 | 157 |
| 19 | 126 | 131 | 152 | 164 |
| 20 | 129 | 135 | 157 | 171 |
| 21 | 133 | 140 | 162 | 178 |
| 22 | 136 | 145 | 167 | 185 |
| 23 | 139 | 150 | 171 | 192 |
| 24 | 142 | 154 | 176 | 199 |
| 25 | 145 | 159 | 181 | 206 |
| 26 | 148 | 164 | 186 | |
| 27 | 151 | 168 | 191 | |
| 28 | 154 | 173 | 195 | |
| 29 | 157 | 178 | 200 | |
| 30 | 160 | 183 | 205 | |
| 31 | 164 | 187 | | |
| 32 | 167 | 192 | | |
| 33 | 170 | 197 | | |
| 34 | 173 | 202 | | |
| 35 | 176 | 206 | | |
| 36 | 179 | | | |
| 37 | 182 | | | |
| 38 | 185 | | | |
| 39 | 188 | | | |
| 40 | 191 | | | |
| Mean | 10.5 | 12.53 | 8.1 | 10.30 |
| S.D. | 4.84 | 3.17 | 3.13 | 2.11 |
| N | 30 | 30 | 30 | 30 |

**TABLE : PQ 8 ; AGE : 40–49 ; EDU : 6–9**

| Raw Score | MALE | | FEMALE | |
|---|---|---|---|---|
| | Kohs' Block | Pass-A-Long | Kohs' Block | Pass-A-Long |
| 1 | 55 | 35 | 54 | 43 |
| 2 | 59 | 39 | 58 | 48 |
| 3 | 62 | 43 | 62 | 52 |
| 4 | 66 | 48 | 66 | 57 |
| 5 | 70 | 52 | 70 | 62 |
| 6 | 73 | 56 | 75 | 67 |
| 7 | 77 | 60 | 79 | 71 |
| 8 | 81 | 65 | 83 | 76 |
| 9 | 84 | 69 | 87 | 81 |
| 10 | 88 | 73 | 92 | 85 |
| 11 | 92 | 78 | 96 | 90 |
| 12 | 95 | 82 | 100 | 95 |
| 13 | 99 | 86 | 104 | 100 |
| 14 | 103 | 90 | 108 | 104 |
| 15 | 106 | 95 | 113 | 109 |
| 16 | 110 | 99 | 117 | 114 |
| 17 | 114 | 103 | 121 | 118 |
| 18 | 118 | 108 | 125 | 123 |
| 19 | 121 | 112 | 130 | 128 |
| 20 | 125 | 116 | 134 | 133 |
| 21 | 129 | 121 | 138 | 137 |
| 22 | 132 | 125 | 142 | 142 |
| 23 | 136 | 129 | 146 | 147 |
| 24 | 140 | 134 | 151 | 151 |
| 25 | 143 | 138 | 155 | 156 |
| 26 | 147 | 142 | 159 | 161 |
| 27 | 151 | 146 | 163 | 166 |
| 28 | 154 | 151 | 168 | 170 |
| 29 | 158 | 155 | 172 | 175 |
| 30 | 162 | 159 | 176 | 180 |
| 31 | 165 | 164 | 180 | 184 |
| 32 | 169 | 168 | 185 | 189 |
| 33 | 173 | 172 | 189 | 194 |
| 34 | 177 | 177 | 193 | 199 |
| 35 | 180 | 181 | 197 | 203 |
| 36 | 184 | 185 | 201 | |
| 37 | 188 | 189 | 206 | |
| 38 | 191 | 194 | 210 | |
| 39 | 195 | 198 | 214 | |
| 40 | 199 | 202 | 218 | |
| Mean | 13.23 | 16.2 | 12.0 | 13.1 |
| S.D. | 4.07 | 3.49 | 3.55 | 3.18 |
| N | 30 | 30 | 30 | 30 |

TABLE : PQ 9 ; AGE : 40–49 ; EDU : 10+

| Raw Score | MALE | | FEMALE | |
|---|---|---|---|---|
| | Kohs' Block | Pass-A-Long | Kohs' Block | Pass-A-Long |
| 1 | 51 | 42 | 30 | 46 |
| 2 | 54 | 45 | 35 | 49 |
| 3 | 57 | 48 | 40 | 52 |
| 4 | 60 | 52 | 45 | 55 |
| 5 | 62 | 55 | 50 | 59 |
| 6 | 65 | 58 | 55 | 62 |
| 7 | 68 | 61 | 60 | 65 |
| 8 | 71 | 64 | 65 | 68 |
| 9 | 73 | 67 | 70 | 71 |
| 10 | 76 | 70 | 75 | 74 |
| 11 | 79 | 73 | 80 | 77 |
| 12 | 82 | 76 | 85 | 80 |
| 13 | 85 | 79 | 90 | 83 |
| 14 | 87 | 82 | 95 | 86 |
| 15 | 90 | 85 | 100 | 89 |
| 16 | 93 | 88 | 105 | 93 |
| 17 | 96 | 91 | 110 | 96 |
| 18 | 98 | 94 | 115 | 99 |
| 19 | 101 | 98 | 120 | 102 |
| 20 | 104 | 101 | 125 | 105 |
| 21 | 107 | 104 | 130 | 108 |
| 22 | 109 | 107 | 135 | 111 |
| 23 | 112 | 110 | 140 | 114 |
| 24 | 115 | 113 | 145 | 117 |
| 25 | 118 | 116 | 150 | 120 |
| 26 | 120 | 119 | 155 | 124 |
| 27 | 123 | 122 | 160 | 127 |
| 28 | 126 | 125 | 165 | 130 |
| 29 | 129 | 128 | 170 | 133 |
| 30 | 131 | 131 | 175 | 136 |
| 31 | 134 | 134 | 180 | 139 |
| 32 | 137 | 137 | 185 | 142 |
| 33 | 140 | 140 | 190 | 145 |
| 34 | 142 | 144 | 195 | 148 |
| 35 | 145 | 147 | 200 | 151 |
| 36 | 148 | 150 | | 154 |
| 37 | 151 | 153 | | 158 |
| 38 | 153 | 156 | | 161 |
| 39 | 156 | 159 | | 164 |
| 40 | 159 | 162 | | 167 |
| Mean | 18.63 | 19.8 | 15.03 | 18.04 |
| S.D. | 5.45 | 4.89 | 3.01 | 4.85 |
| N | 30 | 30 | 30 | 30 |

**TABLE : PQ 10 ; AGE : 50–59 ; EDU : 0–5**

| Raw Score | MALE | | FEMALE | |
|---|---|---|---|---|
| | Kohs' Block | Pass-A-Long | Kohs' Block | Pass-A-Long |
| 1 | 62 | 44 | 57 | 55 |
| 2 | 67 | 50 | 64 | 60 |
| 3 | 72 | 55 | 71 | 66 |
| 4 | 77 | 61 | 78 | 71 |
| 5 | 82 | 67 | 85 | 77 |
| 6 | 87 | 72 | 92 | 82 |
| 7 | 92 | 78 | 99 | 88 |
| 8 | 97 | 83 | 106 | 93 |
| 9 | 102 | 89 | 113 | 98 |
| 10 | 107 | 94 | 120 | 104 |
| 11 | 112 | 100 | 127 | 109 |
| 12 | 117 | 105 | 133 | 115 |
| 13 | 122 | 111 | 140 | 120 |
| 14 | 127 | 116 | 147 | 125 |
| 15 | 132 | 122 | 154 | 131 |
| 16 | 137 | 127 | 161 | 136 |
| 17 | 142 | 133 | 168 | 142 |
| 18 | 147 | 138 | 175 | 147 |
| 19 | 152 | 144 | 182 | 153 |
| 20 | 157 | 149 | 189 | 158 |
| 21 | 162 | 155 | 196 | 163 |
| 22 | 167 | 160 | 203 | 169 |
| 23 | 172 | 166 | 210 | 174 |
| 24 | 177 | 171 | 217 | 180 |
| 25 | 182 | 177 | 224 | 185 |
| 26 | 187 | 182 | | 190 |
| 27 | 192 | 188 | | 196 |
| 28 | 197 | 193 | | 201 |
| 29 | 202 | 199 | | 207 |
| 30 | 207 | 204 | | 212 |
| 31 | | | | |
| 32 | | | | |
| 33 | | | | |
| 34 | | | | |
| 35 | | | | |
| 36 | | | | |
| 37 | | | | |
| 38 | | | | |
| 39 | | | | |
| 40 | | | | |
| Mean | 8.6 | 11.07 | 7.2 | 9.3 |
| S.D. | 3.01 | 2.72 | 2.15 | 2.77 |
| N | 30 | 30 | 30 | 30 |

**TABLE : PQ 11 ; AGE : 50–59 ; EDU : 6–9**

| Raw Score | MALE | | FEMALE | |
|---|---|---|---|---|
| | Kohs' Block | Pass-A-Long | Kohs' Block | Pass-A-Long |
| 1 | 52 | 48 | 58 | 28 |
| 2 | 56 | 51 | 62 | 33 |
| 3 | 60 | 55 | 67 | 39 |
| 4 | 64 | 59 | 71 | 45 |
| 5 | 68 | 63 | 75 | 51 |
| 6 | 72 | 66 | 79 | 57 |
| 7 | 76 | 70 | 84 | 63 |
| 8 | 80 | 74 | 88 | 69 |
| 9 | 84 | 78 | 92 | 75 |
| 10 | 88 | 81 | 96 | 81 |
| 11 | 92 | 85 | 101 | 87 |
| 12 | 96 | 89 | 105 | 93 |
| 13 | 100 | 93 | 109 | 99 |
| 14 | 104 | 96 | 113 | 105 |
| 15 | 108 | 100 | 117 | 111 |
| 16 | 112 | 104 | 122 | 116 |
| 17 | 116 | 107 | 126 | 122 |
| 18 | 120 | 111 | 130 | 128 |
| 19 | 124 | 115 | 134 | 134 |
| 20 | 128 | 119 | 139 | 140 |
| 21 | 132 | 122 | 143 | 146 |
| 22 | 136 | 126 | 147 | 152 |
| 23 | 140 | 130 | 151 | 158 |
| 24 | 144 | 134 | 156 | 164 |
| 25 | 148 | 137 | 160 | 170 |
| 26 | 152 | 141 | 164 | 176 |
| 27 | 156 | 145 | 168 | 182 |
| 28 | 160 | 149 | 173 | 188 |
| 29 | 164 | 152 | 177 | 194 |
| 30 | 168 | 156 | 181 | 199 |
| 31 | 172 | 160 | | |
| 32 | 176 | 163 | | |
| 33 | 180 | 167 | | |
| 34 | 184 | 171 | | |
| 35 | 188 | 175 | | |
| 36 | 192 | 178 | | |
| 37 | 196 | 182 | | |
| 38 | 200 | 186 | | |
| 39 | 204 | 190 | | |
| 40 | 208 | 193 | | |
| Mean | 13.0 | 15.0 | 10.89 | 13.22 |
| S.D. | 3.75 | 4.02 | 3.54 | 2.53 |
| N | 30 | 30 | 30 | 30 |

**TABLE : PQ 12 ; AGE : 50–59 ; EDU : 10+**

| Raw Score | MALE | | FEMALE | |
|---|---|---|---|---|
| | Kohs' Block | Pass-A-Long | Kohs' Block | Pass-A-Long |
| 1 | 56 | 30 | 39 | |
| 2 | 59 | 34 | 43 | |
| 3 | 62 | 38 | 47 | |
| 4 | 65 | 42 | 51 | |
| 5 | 67 | 46 | 55 | 22 |
| 6 | 70 | 50 | 59 | 28 |
| 7 | 73 | 54 | 63 | 34 |
| 8 | 76 | 59 | 67 | 40 |
| 9 | 78 | 63 | 71 | 46 |
| 10 | 81 | 67 | 75 | 52 |
| 11 | 84 | 71 | 79 | 58 |
| 12 | 87 | 75 | 83 | 65 |
| 13 | 89 | 79 | 87 | 71 |
| 14 | 92 | 83 | 91 | 77 |
| 15 | 95 | 87 | 94 | 83 |
| 16 | 98 | 91 | 98 | 89 |
| 17 | 100 | 95 | 102 | 95 |
| 18 | 103 | 99 | 106 | 102 |
| 19 | 106 | 103 | 110 | 108 |
| 20 | 109 | 107 | 114 | 114 |
| 21 | 111 | 111 | 118 | 120 |
| 22 | 114 | 115 | 122 | 126 |
| 23 | 117 | 119 | 126 | 133 |
| 24 | 120 | 123 | 130 | 139 |
| 25 | 122 | 127 | 134 | 145 |
| 26 | 125 | 132 | 138 | 151 |
| 27 | 128 | 136 | 142 | 157 |
| 28 | 131 | 140 | 146 | 163 |
| 29 | 133 | 144 | 150 | 170 |
| 30 | 136 | 148 | 154 | 176 |
| 31 | 139 | 152 | 157 | 182 |
| 32 | 142 | 156 | 161 | 188 |
| 33 | 145 | 160 | 165 | 194 |
| 34 | 147 | 164 | 169 | 200 |
| 35 | 150 | 168 | 173 | |
| 36 | 153 | 172 | 177 | |
| 37 | 156 | 176 | 181 | |
| 38 | 158 | 180 | 185 | |
| 39 | 161 | 184 | 189 | |
| 40 | 164 | 188 | 193 | |
| Mean | 16.83 | 18.23 | 16.4 | 16.73 |
| S.D. | 5.45 | 3.7 | 3.81 | 2.43 |
| N | 30 | 30 | 30 | 30 |

## 3. VERBAL ADULT INTELLIGENCE SCALE (VAIS)

Verbal intelligence is crystallized by social interaction, experiences, schooling etc., and hence it continues to develop beyond the age of physiological maturity. Aging and brain injury, generally, do not reduce the verbal ability (Cattell, 1970) until domineering hemisphere (left side of the brain in right handed individuals) is involved. In a battery of intelligence tests some of the tests are considered as 'hold' and others 'don't hold' and a deterioration quotient is worked out to indicate as to how much deterioration has occurred in functioning intelligence. In a number of studies, however, schizophrenics and organic psychiatric cases were not found to be significantly different in verbal intelligence (Verma, Pershed, Ojha, Misra, Khorana, Gupta, Shukla and Singh, 1985). The reasons for such findings could be traced in the literature where impairement in learning (Campbell, 1957), over inclusion (Payne, 1968 and Payne and Hewlett 1960), scanning deficits (Shokow, 1946) and preoccupation in schizophrenic are held responsible for their poor functioning. Nevertheless, significance of different sub scales in clinical use can not be ignored.

Verbal Adult Intelligence Scale (VAIS) consisting of four subtests, i. e., information, Digit span, Arithmetic and Comprehension is used here. The scale is meant for adult subjects in the age range of 20 and above. It gives Test Quotients separately for four sub tests and a V. Q. (Verbal Quotient) that is mean of T. Qs (Tests Quotients), separately for male and female of different age and educational levels (Pershad and Verma, 1988). It stresses on power or capacity rather than on speed or efficiency.

### Material

Items of four subtests and tables of norms for the evaluation of the answers.

### Administration

Administration of the test is simple and takes about 20-30 minutes. Procedure for the administration of Information, Arithmetic and Comprehension tests alongwith the items, standard answers, and scoring is given here below separately for each test. Since the 'digit span' test has already been administered earlier under memory scale, it is not repeated here. Only the scoring procedure of this test will be described. It is important to remember that the complete structure and the complexity of the items to be answered should not be changed to make them readily understandable. If need be, item as a whole can be repeated. Under compelling circumstance (when administrator feels that a word has different meaning or is not prevalent in a geographical region) only a word can be changed not the configuration of the sentence. If a subject does not normally understand conversation in Hindi, the test should not be applied on him.

## A. INFORMATION-SUBTEST

This subtest consists of 33 items. Read each question exactly as stated. If the response to a question is incomplete or not clear. You may say, "Explain what you mean or tell me more about it," but do not ask leading questions or spell the words. Do not alter the wording of question.

Record verbatim the subject's response to each item in the appropriate space on the Record Form.

Discontinue : After 7 consecutive failures.

### Scoring

1 point for each correct response. Essentials of acceptable answers are noted below, where several acceptable answers are listed, the subject needs to give only one to receive the credit.

*Maximum score* : 33 points.

| Sr. No. | TEST QUESTIONS | ACCEPTABLE ANSWERS |
| --- | --- | --- |
| 1. | गेंद किस आकार की होती है ?<br>What is the shape of the Ball. | गोल ।<br>Circle |
| 2. | एक साल में कितने महीने होते हैं ?<br>How many months in a year ? | 12 महीने ।<br>12 Months. |
| 3. | थर्मामीटर क्या होता है ?<br>What is Thermometer ? | बुखार नापने या ताप देखने की चीज/यंत्र ।<br>The instrument to measure temperature or fever. |
| 4. | सूरज किधर से निकलता है ?<br>Where does the Sun rise ? | पूर्व में (दिशा का नाम बतलाना जरूरी है) ।<br>East. |
| 5. | ऐसे तीन व्यक्तियों के नाम बताइये जो भारत के प्रधानमंत्री रहे हों ?<br>Tell the names of three persons who have been the prime-minister of India ? | जवाहर लाल नेहरू, लाल बहादुर शास्त्री, गुलजारी लाल नंदा, मोरारजी देसाई, चरण सिंह, इन्दिरा गाँधी, राजीव गाँधी, देवगोड़ा, अटल बिहारी, मनमोहन सिंह ।<br>Jawahar Lal Nehru, Lal Bahadur Shastri, Gulzari Lal Nanda, Morarji Desai, Charan Singh, Indira Gandhi, Rajiv Gandhi, Devagora, Atal Bihari, Manmohan Singh. |
| 6. | गुरु नानक (मिर्जा गालिब) एक प्रसिद्ध व्यक्ति थे, वह क्या थे ?<br>Guru Nanak (Mirza Ghalib) was a famous person. What was he ? | सिक्खों के पहले गुरु, सिक्खों के गुरु, सिक्ख धर्म के प्रवर्तक, चलाने वाले (उर्दू भाषा के कवि, शायर) ।<br>He was the first Guru of Sikhs, the precussor of Sikh religion. Who originated it. (The poet of Urdu language). |
| 7. | भारत के झण्डे में कौन-कौन से रंग हैं ?<br>What are the colours in India's national flag ? | लाल (केसरी, पीला), हरा, सफेद ।<br>3 Colours – Red (Saffron, Yellow), Green, White. |
| 8. | लोढ़ी (मकर संक्रान्ति) कब मनाते हैं ?<br>When is Lohri celebrated ? | 13-14 जनवरी को, जनवरी में, साल के पहिले महीने में, माघ में ।<br>13-14th January, in the first month of the year, in Magh. |
| 9. | रामायण किसने लिखी ?<br>Who wrote the Ramayan ? | तुलसीदास, वाल्मीकि, रामानन्द सागर ।<br>Tulsi Das, Balmiki, Ramanand Sagar. |

Contd.....

| Sr. No. | TEST QUESTIONS | ACCEPTABLE ANSWERS |
|---|---|---|
| 10. | कुरान (बाईवल) क्या है ?<br>What is Quran (Bible) ? | मुसलमानों की धार्मिक पुस्तक, इस्लाम धर्म की पाक/पवित्र पुस्तक (ईसाइयों का धार्मिक ग्रंथ/पुस्तक)।<br>The Religious book of Muslims, the holy book of Islam religon (The religious book of Christians). |
| 11. | औरतों का औसत कद कितना होता है।<br>What is the average height of women ? | 5 फुट। 4 फुट 9 इंच से 5 फुट 3 इंच तक के बीच का कोई भी उत्तर।<br>5 feet. Any answer between 4 feet 9 inches and 5 feet 3 inches. |
| 12. | हिन्दुस्तान कब आजाद हुआ ?<br>When did India become free ? | 15 अगस्त 1947 या अगस्त 1947।<br>15th August 1947 or August 1947. |
| 13. | बीरबल कौन था ?<br>What was Birbal ? | अकबर का मंत्री/बजीर। नवरत्नों में से एक अकबर का दरबारी।<br>Akbar's minister. One courtier among nine jewels of Akbar. |
| 14. | गाँधी जयन्ती कब मनाते हैं ?<br>When Gandhi jayanti is celebrated ? | 2 अक्टूबर को।<br>On 2nd October. |
| 15. | रबड़ कहाँ से निकलती (पैदा होती) है ?<br>From where rubber is produced ? | पेड़ से, पेट्रोलियम से।<br>From tree, From patroliam. |
| 16. | ताजमहल किसने बनवाया ?<br>Who built Taj Mahal ? | शाहजहाँ, मुगल बादशाह ने, मुमताज के शौहर ने।<br>Shahjahan, A Mughal emperor, Mumtaj's husband. |
| 17. | भारत की जनसंख्या कितनी है ?<br>What is the population of India ? | 100 और 110 करोड़ के बीच की कोई भी संख्या।<br>Any number between 100 crore and 110 crore. |
| 18. | त्रिवेणी (संगम) कहाँ है ?<br>Where is Triveni (Sangam) ? | इलाहाबाद, प्रयाग में।<br>In Allahabad (Prayag). |
| 19. | बम्बई कौन से राज्य में है ?<br>In which state Bombay is situated ? | महाराष्ट्र।<br>Maharastra |

*Contd.....*

| Sr. No. | TEST QUESTIONS | ACCEPTABLE ANSWERS |
|---|---|---|
| 20. | अगर आप दिल्ली से मद्रास गए तो आपने किस दिशा में सफर किया ?<br>If you go to Medras from Delhi, in which direction did you travel ? | दक्षिण/दक्षिण पूर्व ।<br>South / South East. |
| 21. | दही कैसे जमता है ?<br>How the curd is prepared ? | किटाणुओं से / Fermentation से, जामन से खमीर पड़ जाता है, Baterial action से ।<br>With the help of Bacteria / उबाल (उफान) से, Yeast is made from the runnet used in coagulating milk, कीटाणुओं की क्रिया से । |
| 22. | गहरे रंग के कपड़े हल्के रंग के कपड़ों से ज्यादा गर्म क्यों लगते हैं ।<br>Why the dark coloured clothes seem hot than light coloured clothes ? | सूर्य की गर्मी को सोख लेते हैं/ आकर्षित कर लेते हैं । सूर्य की किरणें खींच लेते हैं ।<br>They absorve the heat from the Sun / attract the heat. Attract the rays of Sun. |
| 23. | एक साल में कितने सप्ताह होते हैं ?<br>How many weeks are there in one year ? | 52 सप्ताह ।<br>52 weeks. |
| 24. | पानी कितने ताप पर उबलता है ?<br>What is the temperature boiling of water ? | 100°C, 212°F, 373°K (यदि पैमाना विशिष्टीकृत नहीं हो तो बताइये, कौन से पैमाने पर)<br>100°C, 212°F, 373°K (If Scale is not specified say, what scale) । |
| 25. | श्री लंका (सीलोन) की राजधानी का नाम बताओ ?<br>Tell the name of the capital of Sri Lanka (Sylon) ? | कोलम्बो ।<br>Colambo. |
| 26. | इंगलैण्ड किस महाद्वीप में है ?<br>In which continent England is situated ? | योरोप में ।<br>In Europe. |
| 27. | गीता का मुख्य सार (उपदेश) क्या है ?<br>What is the (teachingist) of Gita ? | कर्म करो और फल की इच्छा न करो, कर्म की प्रधानता, कर्म ही सब कुछ है ।<br>Do your work and don't think about its result. Work is main. Work is all. |

Contd....

| Sr. No. | TEST QUESTIONS | ACCEPTABLE ANSWERS |
|---|---|---|
| 28. | मेघदूत नामक पुस्तक किसने लिखी ?<br>Who wrote the book named Meghdoot ? | कालिदास ।<br>Kalidas. |
| 29. | महाभारत किसने लिखी ?<br>Who wrote Mahabharat ? | वेदव्यास ।<br>Ved Vyas. |
| 30. | वेद क्या है ?<br>What is Veda ? | हिन्दुओं की प्राचीन धार्मिक ग्रंथ, ऋषियों के कहे हुए वाक्यों की पुस्तक ।<br>The ancient religious book of Hindus. The book of the sentences pronounced by saints. |
| 31. | दिल्ली से बम्बई कितनी दूर है ?<br>What is the distance between Bombay and Delhi ? | 700-1000 मील, 1000-1500 कि. मी. ।<br>700-1000 miles, 1000-1500 Km. |
| 32. | लोक सभा में कितने सदस्य होते हैं ?<br>How many members are there in Parliament ? | 540-560 तक का कोई भी उत्तर ।<br>Any answer between 540 to 560. |
| 33. | मनुष्य के शरीर में कितने प्रकार की खून की नसें होती हैं ?<br>How many types of bood veins are there in human body ? | साफ खून की, गंदे खून की/ arteries, veins and capillaries नीली और लाल खून की नसें ।<br>Of clean bood, of dirty blood, veins of blue and red blood. |

## B. DIGIT SPAN SUBTEST

Two parts of this test-Digit Forward and Digit Backward have already been administered under subtest IV-Attention and Concentration of The PGI Memory Scale, therefore, the items and procedure are not reproduced here again. The scoring procedure, however, is different then for the memory scale, hence it is given below.

इस परीक्षण के दो भाग हैं—अगली डिजिट (इकाई) और पिछली डिजिट (इकाई) उप-परीक्षण IV के अन्तर्गत दी जा चुकी है जो स्मृति माप के ध्यान और चित्र की एकाग्रता से संबद्ध है, इसलिये वे विषय एवं प्रक्रिया यहाँ पुन: नहीं दिये गये हैं । फिर भी अंक देने की प्रक्रिया स्मृति माप की अंकन प्रक्रिया से भिन्न है इसलिये यह आगे दी गई है ।

### Scoring

Each of the item is scored as 2, 1, or 0 as follows :

2 points if the reproduction on both the sets of an item is correct.

1 point if the reproduction is correct only on one of the two sets.

0 point if reproduction is not correct on any of the two sets.

There are six items on Digit Forward and seven items on Dight Backward therefore the maximum score would be 12 ($6 \times 2$) + 14 ($7 \times 2$) = 26.

## C. ARITHMETIC SUB-TEST

It consists of 15 items. Start with item 3 and give credit for items 1 and 2 if the subject passes either item 3 or item 4. If both items 3 and 4 are failed, administe item 1 and 2 before proceeding further. A problem may be repeated once if the subejct requests it, or if it is apparent that the subject did not understand. The subject may not be permitted to use pencil and paper for any problem. However subject should not be discouraged from using a finger to 'write' on the table.

For each item enter the subject's answer in the appropriate space on th Record Form.

Discontinue : After 5 consecutive failures.

### Scoring

1 point for each correct response. If the numerical quantity of a response i correct, give credit even if units, such as rupees and paise, are not stated. Als give credit if the subject spontaneously corrects a wrong answer.

*Maximum Score* : 15 points

| Sr. No. | TEST QUESTIONS | ANSWERS |
|---|---|---|
| 1. | यह कुल कितने ब्लाक हैं ? □ □ □ □ □ □ □<br>How many blocks are these in total ? | 7 |
| 2. | यदि आपके पास तीन किताबें हैं, उनमें से एक किताब किसी को दे दी, तो आपके पास कितनी बच रही ?<br>If you have three books, you gave one of these book to someone then how many books y ou have now ? | 2 |
| 3. | चार रुपए और पाँच रुपए कितने रुपए हुए ?<br>What is the total of Rs. 4 and Rs. 5 ? | 9 |
| 4. | आपने 56 रु. का कपड़ा खरीदा और 60 रु. दुकानदार को दिए तो आपको कितने रुपए वापिस मिलेंगे ?<br>You bought cloth for Rs. 56 gave Rs. 60 to the seller, how many Rs. will you get back ? | 4 |
| 5. | एक फल बेचने वाले ने 25-25 पैसे के हिसाब से 6 ग्राहकों से पैसे इकट्ठे किए तो कुल कितने पैसे इकट्ठे कर लिए ?<br>A fruit seller collected 25 paise each from six customers, how many paise did he collect ? | 1 रु. 50 पैसे |
| 6. | यदि एक रूमाल का मूल्य 6 रु. है तो 36 रु. में कितने रूमाल खरीदे जा सकते हैं ?<br>If the cost of a handkerchief is Rs. 6, how many handkerchieves can be bought for Rs. 36 ? | 6 |
| 7. | एक आदमी ने दो-दो रुपए वाले 7 पैन खरीदे और दुकानदार को 50 रु. का नोट दिया तो उसे कितने रुपए वापिस मिलेंगे ?<br>A man purchased seven pens for Rs. 2 each and gave the shopkeeper a note of Rs. 50, how many Rs. will he get back ? | 36 |
| 8. | $2\frac{1}{2}$ फुट में कितने इंच होते हैं ? या<br>How many inches are there in $2\frac{1}{2}$ feet ? or<br>($2\frac{1}{2}$ मीटर में कितने से. मी. होते हैं ?)<br>How many cm. are there in $2\frac{1}{2}$ meter. | 30 इंच<br><br>(250 से. मी.) |

*Contd.....*

| Sr. No. | TEST QUESTIONS | ANSWERS |
|---|---|---|
| 9. | तीन मील प्रति घण्टे की चाल से 24 मील जाने में कितना समय लगेगा ?<br>How much time it will take in going 24 miles with the speed of 3 miles per hour? | 8 घण्टे |
| 10. | एक आदमी ने 18 रु. में से $7\frac{1}{2}$ रु. खर्च कर दिए तो उसके पास कितने रुपए बच रहे ?<br>A man spent Rs. $7\frac{1}{2}$ from Rs. 18, how many Rs. left with him? | 10 रु. 50 पैसे |
| 11. | 31 पैसे की दो पैंसिलें आती हैं तो एक दर्जन कितने में मिलेंगी ?<br>The cost of two pencils is 31 paise. What would be the cost of one dozen pencils? | 1 रु. 86 पैसे |
| 12. | एक कार ने पाँच घण्टों में 275 मील की दूरी पार की तो उसकी औसत रफ्तार कितनी थी ?<br>A car crossed the distance of 275 miles in 5 hours, what was its average speed? | 55 मील प्रति घण्टा |
| 13. | एक आदमी ने पुराना फर्नीचर नये के दो तिहाई (2/3) मूल्य पर खरीदा। यदि उसने इसके लिए 400 रुपये का भुगतान किया तो नया फर्नीचर कितने में मिल जाता ?<br>A man purchased second hand furniture at the two third price of the new. If he paid Rs. 400 for it, what will be the cost of new furnitures? | 600 रुपये |
| 14. | एक आदमी की तनख्वाह 6000 रु. प्रति महीना है अगर उसकी तनख्वाह से 15% टैक्स काट लिया जाये तो उसे हर महीने कितनी तनख्वाह नगद मिलेगी ?<br>The salary of a man is Rs. 6000 per month. If 15% tax cut down from his salary, how much salary in cash will he get per month? | 5100 रुपये |
| 15. | आठ आदमी किसी काम को 6 दिनों में कर सकते हैं तो उसी काम को आधा दिन में खत्म करने के लिये कितने आदमियों की जरूरत पड़ेगी ?<br>Eight men can do a work in 6 days, how many men will be needed to complete the same work in a half day. | 96 आदमी |

## D. COMPREHENSION SUB-TEST

It consists of 18 items. Start with item 1. Read each question slowly to the subject. Some subjects find it difficult to remember an entire question. You may repeat the question, but do not alter the wording or abbreviate it. It is good practice to repeat the question if no response is obtained but no further urging should be given, except as indicated below :

इसमें 18 पद सम्मिलित हैं। पद 1 से प्रारम्भ कीजिये। प्रत्येक प्रश्न को रोगी के समक्ष धीरे-धीरे पढ़िये। कुछ रोगियों को सम्पूर्ण प्रश्न याद करना कठिन लगता है। आप प्रश्न दुहरा सकते हैं लेकिन उसे संक्षिप्त करना या शब्दों को बदलना नहीं है। यदि कोई उत्तर प्राप्त नहीं होता तो प्रश्न को दुहराया जा सकता है लेकिन पुनः निवेदन करना है सिवाय उस प्रक्रिया के जैसी कि नीचे दी गई है—

If the subject is hesitant, give encouragement by saying ;
यदि रोगी संकोच कर रहा हो तो उसे यह कहकर प्रोत्साहित कीजिये ;

Yes, tell more (हाँ और बताइये)।

If a response is unclear or ambiguous, you may say ;
यदि उत्तर अस्पष्ट या अनिश्चित हो तो आप कह सकते हैं ;

Tell something more about it (इसके बारे में और बताइये)।

If the subject does not give a 2-point response to item 1, say ;
यदि रोगी पद 1 से संबंधित 2 अंक वाला उत्तर नहीं देता तो कहिये ;

Engine is attached to draw the train (इंजन रेलगाड़ी खींचने के लिए लगाते हैं)।

This help may be given only for item 1. Items 3, 7, 10, 15 require 2 correct response for full credit. If the subject gives one correct response but does not give another spontaneously, ask for a second response. Say, 'Tell any other way to keep the healh it'.

यह सहायता केवल पद 1 के लिये दी जा सकती है। पद 3, 7, 10, 15 में पूर्ण अंक (श्रेय) प्राप्ति के लिये 2 सही उत्तरों की आवश्यकता है। यदि रोगी 1 उत्तर सही देता है लेकिन तुरंत ही दूसरा उत्तर नहीं देता तो उससे दूसरा उत्तर देने के लिये कहिये, यह कहते हुये कि 'सेहत ठीक रखने का कोई और उपाय बताओ'।

(or a similar restatement of the question), the examiner may request a second response *only once* during the administration of each relevant item. That is to say, if the second response elicited from the subject is either incorrect or an elaboration of the first response, the examiner may not ask for an additional response.

(या प्रश्न का एक बार पुन: उसी प्रकार वर्णन) परीक्षक प्रत्येक संबंधित विषय का परीक्षण करते समय केवल एक बार दूसरे उत्तर के लिये कह सकता है । अर्थात् यदि रोगी द्वारा दिया गया द्वितीय उत्तर गलत है या प्रथम उत्तर का ही विस्तृत रूप है तो परीक्षक आगे उत्तर देने के लिये नहीं कह सकता ।

Record *verbatim* the subject's response to each item in the appropriate space on the Record Form.

प्रत्येक विषय के लिये दिये गये छात्र में उत्तर को रिकार्ड फार्म (प्रपत्र) पर उपयुक्त स्थान पर शब्दश: रिकार्ड किया जाना है ।

**DISCONTINUE :** After 6 consecutive failures (response scored 0).

**बन्द करना :** लगातार छह बार असफल होने के बाद (उत्तरों पर प्राप्त अंक 0)

**SCORING** : Each item is scored 2, 1 or 0, depending on the degree of understanding expressed and the quality of the response. In scoring each item, the examiner should match the subject's response against the general criteria and the sample answers given below each question. Undoubtedly, some subjects will give unusual responses which are not typified by the sample answers. In such instances the score should be determined by the examiner's own judgement. Most of the 0-point examples given typify marginal resonses, those which contain evidence of some understanding may be queried in the natural manner prescribed above.

**अंकन :** प्रत्येक पद को 2, 10 अंक प्रदत्त किये गये हैं जो कि समझ के स्तर और उत्तरों की गुणवत्ता पर निर्भर है । प्रत्येक पद में अंक प्रदान करने के लिये परीक्षक को रोगी के उत्तर को दिये गये सामान्य सिद्धान्त से और प्रतिदर्श उत्तर जो प्रत्येक पद के नीचे दिये हैं, उनसे मिलाना चाहिये । निस्संदेह कुछ रोगी असाधारण उत्तर देंगे जो प्रतिदर्श उत्तरों से मेल नही खाते हैं । इस प्रकार के उदाहरणों में अंक निर्धारण परीक्षक के निर्णयानुसार होना चाहिए । अधिकांश 0 अंके के उदाहरण जो हाशिये पर दिये गये उत्तरों पर प्रदान किये जाते हैं, वें होते हैं जिनमें कि ऊपर दिये गये स्वाभाविक तरीके के अनुसार इस बात का प्रमाण होता है कि शायद प्रश्न को समझा नहीं गया है ।

Particular care should be exercised in scoring items which require two correct responses for full credit (these items are indicated by an asterisk preceding the item number). In order to receive full credit, the subject must express at least two of the *general ideas* which are indicated. Full credit cannot be given if the subject gives two answers, both of which express the same idea (e.g. only one point is earned if the subject responds to item 7 by saying that many foods need to be cooked because "cooking kills the germs" and "it is not healthy to eat uncooked food").

उन पदों के मूल्यांकन में विशेष सावधानी रखनी चाहिये जिनमें पूर्ण अंकों की प्राप्ति के लिये दो सही उत्तरों की आवश्यकता होती है (इन पदों को दिखाने के लिये Asterisk का चिह्न प्रत्येक विषय की क्रम संख्या से पूर्व लगाया गया है) पूर्ण अंक प्राप्ति के लिये छात्र को दर्शाये गये सामानय विचारों में से कम से कम दो को बताना चाहिये। यदि छात्र दो उत्तर देता है जिनमें प्रत्येक वही एक समान विचार प्रकट करता हो तो पूर्ण अंक नहीं दिये जा सकते। (अर्थात् यदि छात्र विषय 7 के लिये यह उत्तर देता है कि बहुत से भोजनों को पकाये जाने की आवश्यकता होती है क्योंकि "पकाने से कीटाणु नष्ट होते हैं" और "बिना पके भोजन को करना स्वास्थ्यप्रद नहीं है।"

*Maximum score* : 36 points (*अधिकतम अंक :* 36)

1. रेलगाड़ी में इंजन क्यों लगाते हैं ? **(Why the engine is attached with train ? )**

**2 points** – Response showing knowledge that engine provides power to move the train.

**2 अंक** — उत्तर इस ज्ञान को दिखाता है कि इंजन गाड़ी को चलाने की शक्ति प्रदान करता है।

To draw, to move, to draw the compartments, to impart the power to move.

खींचने के लिए, चलाने के लिए, डिब्बे खींचने के लिए, चलाने के लिए शक्ति प्रदान करता है।

**0 point** – Response not including idea of moving the train. To keep the engine warm......To lead the passengers or freight ; you could not go anywhere without a leader, like a captain in the army......To pay attention to signals.......To whistle when they come to a cross road or a curve.

**0 अंक**—उत्तर में गाड़ी चलाने से संबंधित कोई विचार नहीं है। इंजन को गर्म रखने के लिये —— यात्रियों को ले जाने के लिये ; यह सेना में कप्तान की भाँति है, बिना नेता के आप कहीं नहीं जा सकते ; सिग्नल की तरफ ध्यान देने के लिये —— चौराहे या मोड़ पर सीटी देने के लिये।

2. हम कपड़े क्यों धोते हैं ? **(Why do we wash the clothes ? )**

**2 points** – Any response which includes the idea of cleaning.

**2 अंक**—ऐसा उत्तर जिसमें स्वच्छता का विचार शामिल हो।

To clean, because they become dirty, to throw out their dirt.

साफ करने के लिए, मैले हो जाते हैं इसलिए, गंद निकालने के लिए।

**1 point** – Idea of attractiveness.

**1 अंक** — आकर्षक दिखने का विचार ।

So that we should appear to be smart, to draw the attention of others towards ourselves.

अच्छे लगें इसलिए, दूसरों को अपनी तरफ आकर्षित करने के लिए ।

**0 point** – Response not including inference to cleaning. Because it is Sunday.......It is good thing to do.......To keep out of mischief.

**0 अंक**—प्रतिक्रिया में स्वच्छता का विचार शामिल नहीं है । क्योंकि रविवार है — ऐसा करना अच्छा होता है — शरारत से बचने के लिये ।

***3. सेहत ठीक रहे इसके लिए क्या-क्या उपाय करने चाहिए ? (What we should do to keep our health fit ? )**

**General** : Have physical examinations at regular interval, have regular medical checkups......Go to the doctor once a year.

**सामान्य** : मियमित रूप से शारीरिक परीक्षण होना चाहिये ; नियमित चिकित्सकीय परीक्षण करवाना — वर्ष में एक बार चिकित्सक की राय लेना

**General** : Nutrition. (**सामान्य** : पोषण)

Take care of diet, eat clean things, take regular food, take balanced diet.

खाने-पीने का ध्यान रखना चाहिए, ठीक साफ-सुथरी चीजें खानी चाहिए, नियमित भोजन करना चाहिए, संतुलित खाना खाना चाहिए ।

**General** : Exercise. (**सामान्य** : व्यायाम)

Yoga should be practiced. Exercise should be done everybody. There should be walking everyday.

योगाभ्यास करना चाहिए, प्रतिदिन कसरत/वरजिश करनी चाहिए, घूमना/सैर करना चाहिए ।

Try to keep your body in shape ....... Take up jogging.

अपने शरीर को संतुलित रखने का प्रयास करिये — जॉगिंग करिये ।

**General** : Avoid prolonged stress and anxiety.

**सामान्य** : अत्यधिक तनाव तथा चिंता से दूर रहिये ।

Don't worry, be happy

ज्यादा चिंता न करना, खुश रहना ।

Don't let your tensions build up.......Release anxiety.......Avoid situations that aggrevate you.

अपने अन्दर तनावों को जन्म मत लेने दीजिये — चिंताओं को मुक्त कीजिये — ऐसी स्थितियों को टाल दीजिये जो आपको उत्तेजित करती हों ।

**General** : Eliminate bad habits. (**सामान्य** : बुरी आदतों को दूर कीजिये)

Don't take drugs that are bad for you......Don't drink alcohol to excess.

नशीली दवायें मत लीजिये जो आपके लिये बुरी हैं — अधिक शराब मत पीजिये ।

Don't take wine, cigeratte, Biree.

शराब, बीड़ी, सिगरेट नहीं पीनी चाहिए ।

**General** : Maintain an active mind. (**सामान्य** : अपने दिमाग को चुस्त (सक्रिय) रखिये)

Develop interests.......Have some hobbies.......Don't let yourself become dull......Get involved in activities.......Stay alert.

रुचियाँ विकसित कीजिये — कुछ हॉबीज बनाइये — अपने आपको सुस्त मत होने दीजिये — सक्रिया रहिये — चुस्त रहिये ।

Keep yourself busy in work.

अपने आप को काम काज में लगाए रखो ।

**2 points** – A response recognising at least two of six general ideas given above.

**2 अंक**—एक ऐसा उत्तर जिसमें ऊपर दिये गये छह सामान्य विचारों में से कम से कम दो विचार आ जायें ।

**1 point** – A response recognising one of the above ideas.

**1 अंक**—एक ऐसा उत्तर जिसमें उपर्युक्त विचारों में से 1 विचार सम्मिलित हो ।

**0 point** – A vague response, one that heavily suggests the need for maintaining health, or one that is not applicable to people in general.

**0 अंक**—एक अनिश्चित उत्तर, जो स्वास्थ्य सही रखने की आवश्यकता को बताता हो, या एक ऐसा उत्तर जो सामान्य रूप से लोगों के लिये व्यावहारिक न हो ।

Stay healthy.......Don't burn your candle at both ends......Stay active....... Practice preventive medicine.......Live in moderation........Practise clean living........ Take your medication........Don't let your emotions get the best of you.

स्वस्थ रहिये — अत्यधिक परिश्रम मत कीजिये — सक्रिया रहिये — रोगों से बचाव करने वाल दवायें लीजिये — संयम से रहिये — स्वच्छता से रहने का अभ्यास कीजिये — दवाओं का प्रयोग कीजिये — भावनाओं को नियंत्रण में रखिये ताकि वे आपको हानि न पहुँचा सके।

4. **हमको बुरी संगत से क्यों दूर रहना चाहिए ? (Why should we keep ourselves away from bad company ? )**

**2 points** – Any response containing the idea that a person is changed for the worse, corrupted or improperly influenced by bad company.

**2 अंक**—ऐसा उत्तर जिसमें यह विचार सम्मिलित हो कि बुरी संगति एक व्यक्ति में बुरा परिवर्तन ला सकती है, भ्रष्ट कर सकती है या गलत रूप में प्रभावित कर सकती है।

So that we may not be bad ....... bad habits may not be adopted ....... bad company makes bad affect ....... bad company makes us worse. To avoid bad habits and ideas.

हम भी बुरे न बन जाएँ — बुरी आदतें न पड़ जाएँ — बुरों की संगत से बुरा असर होता है — बुरी संगत हमको भी खराब कर देती है — गलत आदतों, विचारों से बचने के लिए।

**1 point** – Reference to specific outcome rather than a generalization of the effects of bad company.

**1 अंक**—विशेष प्रकार के उत्तरों को निकालना न कि बुरी संगति के प्रभावों को सामान्यीकरण ही हो।

To be good ...... to keep ourselves well ....... for our benefit ....... not to get bad result ....... may not be called bad.

अच्छे बने रहने के लिए — अपने आप को ठीक रखने के लिए — अपने फायदे के लिए — बुरा फल न मिले — बुरे न कहलावें।

**0 point** – Reference to badness alone without further explanation or idea of keeping out explanation of why person would be in trouble.

**0 अंक**—केवल बुराई को दर्शाना बजाय उस विचार या व्याख्या को पुनः प्रकट करना कि एक व्यक्ति कठिनाई में क्यों न पड़ सकता है।

Others do not consider it good. (दूसरे लोग अच्छा नहीं समझते ........ )

Bad company is no good.......They give you trouble......Get involved in a situation we could not control......So we would not be picked up......Keep better company.

बुरी संगति अच्छी नहीं है —— यह आपको कठिनाई में डालती है —— हम ऐसी स्थिति में फँस जाते हैं जो कि हम नियंत्रित नहीं कर सकते —— इसलिये हम ऊपर नहीं उठ सकते —— अच्छी संगति में रहो ।

5. **यदि सिनेमाघर में आप पहले व्यक्ति हैं जोकि धुआँ और आग निकलता देखते हैं, तो आपको क्या करना चाहिए ? (If in theater, you are the first person who see smoke and fire comingout, what you should do ?)**

**2 points** – Recognition that a person in authority on the scene, such as a manager or usher, should be notified.

**2 अंक**—यह स्वीकृति कि अधिकारी जो भी वहाँ उपस्थित हो जैसे कि प्रबंधक या गार्ड को सूचना दी जायेगी ।

Inform the manager ....... tell the gatekeeper ....... any employee of the picture hall will be imformed.

मैनेजर को बताओ —— गेटकीपर को कहेंगे —— सिनेमा के किसी कर्मचारी को सूचना देंगे ।

**1 point** – Recognition that responsible action, though not so immediately effective should be taken.

**1 अंक**—स्वीकृति कि कोई उत्तरदायित्वपूर्ण कार्य, यद्यपि वह उतना तुरंत प्रभावकारी न हो, किया जाना चाहिए ।

Extinguish it by throwing water or dust ....... should be entinguished ....... to do a phone call to fire brigade.

पानी-रेत डालकर बुझाएँगे —— बुझानी चाहिए —— आग बुझाने वाली मोटर को टेलीफोन करना चाहिए ।

**0 point** – Description of actions which would create a panic or would not avert disaster.

**0 अंक**—उन कार्यों को बताना जो भय उत्पन्न करें या जो संकट को न टालें ।

Run outside ....... run by making a noise ...... tell others too ....... will alert others by shouting fire-fire.

बाहर की तरफ भागो —— शोर मचाकर भागो —— दूसरों को भी बता दो —— आग आग चिल्ला कर सबको सावधान करेंगे ।

6. **शहर की भूमि का मूल्य गाँव की भूमि के मूल्य से अधिक क्यों होता है ? (Why does cost of land in cities higher than the cost of land in villages ? )**

**2 points** – Mention of demand with the implication of limited supply.

**2 अंक**—माँग के साथ-साथ सीमित पूर्ति का उल्लेख होना चाहिये ।

Law of supply, and demand ...... Land is available in short measure while the buyers are more in numbers ..... More people want city land ....Harder to get in the city (explained as demand)....Being used more, more people want the location.

पूर्ति का नियम और माँग — जमीन कम होती है और खरीदने वाले ज्यादा — अधिकांश लोग शहर में भूमि लेना चाहते हैं — शहर में भूमि कठिनाई से मिल पाती है (माँग की व्याख्या की जाये) — अधिक कार्यों के लिये प्रयुक्त होने के कारण अधिकांश लोग जमीन उपयुक्त स्थान पर लेना चाहते हैं ।

**1 point** – The idea of supply without the implication of demand, or mention of more than one convenience located in the city such as; theatres, stores, transportation, safety control, utilities, sanitation, streets.

**1 अंक**—पूर्ति का विचार बिना माँगे के झंझट (उल्लेख) के, या शहर में उपलब्ध एक से अधिक सुविधा का उल्लेख करना जैसे—सिनेमाघर, दुकानें, यातायात, सुरक्षा उपाय, उपयोगितायें, सफाई, सड़कें ।

In city lang is available in short measure, most people want to live in cities, the population is less in villages, in cities schools / hoapitals / cinema / business facilties are available.

शहर में जमीन कम होती है, सभी शहर में रहना चाहते हैं, गाँव में आबादी कम होती है, शहरों में स्कूल / अस्पताल/ सिनेमा आदि की सुविधाएँ होती हैं, व्यापार की सुविधा होती है ।

**0 point** – Response which shows little or no understanding of economic law involved.

**0 अंक**—ऐसा उत्तर जो आर्थिक नियम से संबंधित समझ को नहीं दिखाता या बहुत कम दिखाता है ।

Cities are better / Progressive, in cities all things cost more.

शहर ज्यादा अच्छे होते हैं/ उन्नत होते हैं, शहर में हर चीज की कीमत ज्यादा होती है ।

More work available in the city....Factories could be built on city land....Buildings are larger in the city, they cost more.

शहर में अधिक कार्य उपलब्ध होता है — शहर की भूमि पर कारखाने निर्मित हो सकते हैं — शहर की इमारतें अधिक बड़ी होती हैं, उनकी कीमत अधिक होती है ।

***7. अधिकतर खाने की चीजें पका कर तैयार की जाती हैं इसके क्या कारण हैं ? (Mostly eating marterials are prepared by cooking them. What are the reasons of it ?)**

**General** : Improves flavours. (**सामान्य** : सुगंध बढ़ जाती है ।)

flavour/tastes better....
(सुगन्धित होता है / अधिक स्वादिष्ट होता है ---)

Food becomes tasty, it flavours good.
(खाना स्वादिष्ट हो जाता है, अच्छी महक आने लगती है ।)

**General** : Permits or eases digestion. (**सामान्य** : पचाना आसान होता है ।)

Digestios becomes easy, digestes soon, increases strength.
पचाना आसान हो जाता है, जल्दी पच जाता है, ताकत बढ़ती है ।

Helps breakdown food.
भोजन के breakdown में सहायक होता है ।

**General :** Prevents disease. (**सामान्य** : बीमारियों से बचाता है ।)

Germs are killed by cooking.
पकाने में कीटाणु मर जाते हैं ।

......It is not healthy to eat some foods if they are not cooked......Uncooked foods make you sick......

---- बिना पकाये भोजन को करना स्वास्थ्यप्रद नहीं है ---- बिना पका भोजन आपको बीमार कर सकता है ----

You may be saved from a lot of diseases.
कितनी ही बीमारियों से बच सकते हैं ।

**General** : Improves texture of the food. (**सामान्य** : भोजन की रचना में सुधार होता है ।)

It is difficult to eat raw food stuff.
कच्चा खाना खाया नहीं जाता ।

...Cooking makes meat (Vegetables) tender...Makes it easier to chew.

---- पकाने से भोजन मुलायम हो जाता है ---- इससे यह चबाने में आसान हो जाता है ।

**2 points** – A response recognising at least two of the four general ideas given above.

**2 अंक**—एक ऐसा उत्तर जिसमें ऊपर दिये गये चार सामान्य विचारों में से दो विचार सम्मिलित हों ।

**1 point**—A response recognizing one of the above ideas.

**1 अंक**—एक ऐसा उत्तर जिसमें उपर्युक्त विचारों में से एक विचार सम्मिलित हो ।

**0 point**—A response suggesting that food is cooked for some trivial reason (e.g., to conform to custom)....

**0 अंक**—एक ऐसा उत्तर जो यह दर्शाता हो कि भोजन किसी अनावश्यक कारण से पकाया जाता है (जैसे—परंपरा के कारण) ——

It is prepared for eating, vitamins are inceased

पकाने के खाने के लिए तैयार हो जाता है, विटामिन बढ़ जाते हैं ।

Recipes tell you to cook food So you can mix things together...... For nutrition.......So you can eat the food hot.......Most people like their food hot.

खाना पकाने की विधियों के अनुसार यह पकाया जाता है इसलिये चीजें आपस में मिलायी जा सकता हैं —— पोषण के लिये —— अत: आप गर्म खाना खा सकते हैं —— अधिकतर लोग गर्म खाना पसंद करते हैं ।

**8. कुछ लोग जरूरत पड़ने पर रुपए दोस्तों से उधार न लेकर बैंक से उधार लेना अधिक पसन्द करते हैं—ऐसा क्यों ? (There are some people who like it more to borrow from bank than friends. Why ?**

**2 points**—Recognition of a specific way in which a friendship can be strained when money is borrowed, or a specific advantage of borrowing from the bank.

**2 अंक**—एक विशेष कारण की स्वीकृति जिसमें उल्लेख हो कि रुपया उधार लेने से दोस्ती में तनाव उत्पन्न हो जाता है, या बैंक से उधार लेने के पीछे कोई विशेष लाभ ।

Other friends can not know, there can not be dispute with friends if it is not returned at correct time, honour is increased by borrowing from bank, do not take obligation of friends.

दूसरे दोस्तों को पता नहीं लगता, ठीक समय पर वापिस न किया जाए तो दोस्तों में झगड़ा नहीं हो सकता है, बैंक से लेने में इज्जत बढ़ती है, दोस्तों का अहसान नहीं लेते ।

A friend may need the money for an emergency and you may not have it ..... A bank is impersonal and objective......Borrowing from a bank lets you keep your personal independence......Borrowing from a bank establishes credit.

एक मित्र को आवश्यक कार्य के लिये धन की आवश्यकता है और हो सकता है आपके पास न हो —— बैंक अव्यक्तिगत और निरपेक्ष होता है —— बैंक से उधार लेने से आपकी व्यक्तिगत आत्म-निर्भरता बनी रहती है —— बैंक से उधार लेने से साख स्थापित होती है ।

**1 point** – Recognition that borrowing from a friend may strain friendship but wothout explanation.

**1 अंक**—यह स्वीकृति कि दोस्त से पैसा उधार लेने से दोस्ती में तनाव आ सकता है लेकिन इसकी व्याख्या नहीं की गई ।

Friendship becomes not so deep ....... there may be dispute between to friends ....... Friendship may be brokenout ....... friend may demand hsi money whenever he wishes.

दोस्ती फीकी पड़ जाती है —— दोस्तों में झगड़ा हो सकता है —— दोस्ती टूट सकती है —— दोस्त कभी भी माँग सकता है ।

It gets sticky when you borrow from a friend......good to avoid personal difficulties with friends......Money brings problems into a friendship.

मित्र से पैसा उधार लेना परेशानी का कारण बन जाता है —— मित्रों के साथ व्यक्तिगत परेशानी टालना अच्छा होता है —— पैसा मित्रता में परेशानी का कारण बन जाता है ।

**0 point** – Vague response or one that suggests that borrowing the money is sure to end a friendship.

**0 अंक**—अनिश्चित उत्तर या ऐसा उत्तर जो यह दर्शाता है कि पैसा उधार लेने से मित्रता की समाप्ति निश्चित है ।

Friendship comes to an end ...... friend should not feel any problem ...... it is trade of time that money can not be borrowed from friends...... it is not fair to borrow from friends ...... freinds take more interests.

दोस्ती खत्म हो जाती है —— दोस्त को तकलीफ न हो...जमाना ऐसा ही है कि दोस्तों से नहीं ले सकते —— दोस्तों से लेना अच्छा नहीं —— दोस्त ब्याज ज्यादा लेते हैं ।

**9. यदि आपको एक लिफाफा मिलता है जो बन्द है, उस पर पता लिखा है और नए टिकट लगे हैं तो आपको क्या करना चाहिए ? (If you get an envelope, which is sealed and address is written on it and new stamps are sticked, what you should do ? )**

**2 points** – Any response which shows recognition that the letter should be put into the post / mail box / letter box.

**2 अंक**—ऐसा उत्तर जो यह दिखाता हो कि पत्र पोस्ट किया जाय / मेल बॉक्स / लैटर बॉक्स में डाला जाय ।

Post it ..... put it in letter box .... give it in post office ..... give it to postman, he will send it to the related man.

पोस्ट कर दो ---लेटर बक्स में डाल दो ---पोस्ट आफिस में दे दो ---डाकिए को दे दो जिसका होगा उसे पहुँचा देगा ।

**1 point** – Recognition that letter is the property of someone else but poor idea as to the disposition of it.

**1 अंक**—उत्तर में यह स्वीकृति कि पत्र किसी और की संपत्ति है लेकिन इसकी उपयुक्त व्यवस्था के बारे में विचार कुछ दुर्बल है ।

Give it to police .... find out the man whose address is written on the envelope.

पुलिस को दे दो ---जिसका पता लिखा है उसे तलाशो ।

**0 point** – No idea of what to do with letter or that the letter is the property of someone else.

**0 अंक**—इस बात का कोई विचार न हो कि पत्र का क्या किया जाना चाहिये या यह कि पत्र किसी अन्य की संपत्ति है ।

We'll not pick it, disclose it and see .... let it lie down, give it to anyone ..... throw the stamp of it ..... take it if anything is useful otherwise throw it.

उठाएँगे नहीं खोलकर देखो ---पड़ा रहने दो किसी को दे दो ---टिकिट उतार कर फैंक दो ---कोई मतलब की चीज हो तो ले लो नहीं तो फैंक दो ।

***10. लेन-देन में नकद की जगह चेक का इस्तेमाल करना क्यों अच्छा है ? (Why is it better to use cheque in place of cash ?)**

**General** : Record of payment. (**सामान्य** : देने का लेखा होता है ।)

The receiver may not deny, there remained a kind of receipt.

मुकर न जाए, एक प्रकार की रसीद रह जाती है ।

Use of the cheques to back up tax deductions.....helps you keep track of your expenses.

कर-कटौती के लिये चेक का प्रयोग करना — अपने व्यय का हिसाब रखने में सहायक होता है।

**General** : Safety. (**सामान्य** : सुरक्षा)

It may not be robbed by anyone, no worry if it is lost.

कोई लूट न ले, अगर खो जाए तो खास चिंता नहीं होती।

You can keep your money in the bank where it is safe......You don't have to carry a lot of cash around......You can stop payment.

आप अपना पैसा बैंक में रख सकते हैं जहाँ यह सुरक्षित है — आपको कहीं नकद पैसा अधिक मात्रा में नहीं ले जाना पड़ता — आप भुगतान रोक सकते हैं।

**General** : Convenience. (**सामान्य** : सुविधा)

It is vonvenient in exchange of money, we can pay money immediately, if necessary.

देन-लेन में सुविधाजनक है, एक दम रुपए पैसे देने हों तो दे सकते हैं।

You can buy something on the spur of the moment.

आवश्यक होने पर तुरंत किसी वस्तु को खरीद सकते हैं।

**2 points** – A response recognising at least two of the three general ideas given above.

**2 अंक**—एक ऐसा उत्तर जिसमें कि दिये गये उपर्युक्त तीन सामान्य विचारों में से कम से कम दो विचार हों।

**1 point** – A response recognising one of the above ideas.

**1 अंक**—ऐसा उत्तर जिसमें उपर्युक्त विचारों में से एक विचार दिया हो।

**0 point** – A response recognising only a trivial advantage of cheques, or one unrelated to their value in paying bills. Because you can fill out the exact amount quickly......You can get them clashed for money. Because cheques are cheaper than money orders.

**0 अंक**—एक ऐसा उत्तर जिसमें चेक का कोई महत्वहीन उपयोगिता बताई गई हो या बिलों के भुगतान में चेक का महत्व उसमें शामिल नहीं हो। क्योंकि आप निश्चित रकम शीघ्रता से भर सकते हैं — आप उन्हें रुपया प्रापत करने के लिये भुना सकते हैं क्योंकि चेक मनीऑर्डर की तुलना में सस्ते होते हैं।

**11. सरकार टैक्स क्यों लगाती है ?(Why does the Government impose tax ? )**

**2 points**—Support of the government institutions (city or government community or state may be given in place of government).

**2 अंक**—सरकारी संस्थाओं को चलाने के लिये (शहर या सरकारी समुदाय या राज्य को सरकार के स्थान पर दिया जा सकता है ।)

So that government expenses may be fulfilled, government may work properly ...... government does various works for the welfare of public , to fulfil this expenditure.

सरकार का खर्च निकलता रहे सरकार ठीक तरह से चलती रहे ---- सरकार जनता की भलाई के लिए भिन्न-भिन्न काम पर खर्च करती है उसे पूरा करने के लिए ।

**1 point**—Specific mention of two or more institutions supported by the Government.

**1 अंक**—सरकारी सहायता प्राप्त दो या अधिक संस्थाओं के नाम विशेष रूप से उल्लिखित हों ।

Money is needed for the development of the country, for fulfilling the expenses done on the construction of schools, hospitals, roads, bridges etc., for the welfare of public

देश के विकास के लिए पैसा चाहिए, लोगों की भलाई के लिए स्कूल, अस्पताल, सड़कें, पुल आदि बनाने में होने वाले खर्च को पूरा करने के लिए ----

Pay for police departments, roads etc......For the upkeep of institutions of all kinds......Schools and prisons.

पुलिस विभाग, सड़कों आदि के खर्च के लिये ---- सभी संस्थाओं के रखरखाव के लिये ----स्कूल और जेल ।

**0 point**—Reference to one specific institution, organization, or job, showing no grasp of the idea that taxes are used for the entire government structure.

**0 अंक**—किसी विशेष संस्था, संगठन या सेवा का संदर्भ जो इस विचार की ग्रहण क्षमता प्रकट न करें कि करों का प्रयोग संपूर्ण सरकारी ढाँचे के लिये होता है ।

Government opens hospitals, schools ..... they should pay who have much ..... it is government rule.

सरकार अस्पताल खोलती है ---- स्कूल खोलती है ---- जिनके पास ज्यादा है उन्हें देना चाहिए ---- सरकार का नियम है ।

**12. इस कहावत का क्या अर्थ है ?—"अकेला चना भाड़ नहीं फोड़ सकता।" (What is the meaning of this proverb ? – "A single gram can not burst parcher's oven.)**

If the response is another proverb, ask for an explanation and score it according to the criteria.

यदि उत्तर दूसरी कहावत से सम्बन्धित हो तो स्पष्टीकरण के लिये कहा जाय और दिये गये उत्तर सिद्धान्त (नियम) के अनुसार अंक प्रदान किये जायें।

**2 points** – An abstract generalization.

**2 अंक**—एक आदर्श सामान्यीकरण।

A person can not do anything alone, more persons are needed to do a big work ..... a large work can be done unitedly ..... a single person can do nothing ..... to do a large work unity is required.

अकेला आदमी कुछ नहीं कर सकता बड़ा काम करने के लिए ज्यादा लोगों की जरूरत होती है ..... एक साथ मिल कर बड़े से बड़ा काम हो सकता है ..... अकेले से कुछ नहीं हो सकता ..... बड़े काम के लिए एकता चाहिए।

**1 point** – A specific instance, or a related but not quite equivalent generalization.

**1 अंक**—एक विशिष्ट उदाहरण या एक संबंधित परन्तु एकदम समान सामान्यीकरण न हो।

There is no strength in a single thing.

एक चीज में ताकत नहीं होती।

**0 point** – No recognition that the statement is a proverb.

**0 अंक**—इस बात की स्वीकृति नहीं मिलती कि दिया गया कथन एक कहावत है।

A single person can not do many works ..... a gram can not break a parcher's oven.

अकेला आदमी बहुत से काम नहीं कर सकता ..... एक चने से भाड़ नहीं टूट सकता।

**13. इस कहावत का क्या अर्थ है ? —"थोथा चना बाजे घना।" (What is the meaning of this proverb ? – "An empty vessel sounds more much.")**

If the response is another proverb, ask for an explanation and score it according to the criteria.

यदि उत्तर दूसरी कहावत से सम्बन्धित हो तो स्पष्टीकरण के लिये कहा जाय और दिये गये उत्तर पर नियमानुसार अंक प्रदान किये जायें।

**2 points** – An abstract generalization.

**2 अंक**—एक आदर्श सामान्यीकरण ।

The person who has little knowledge, boasts much ..... they who know little speak much.

जिसमें ज्ञान कम होता है वह ज्यादा सेखी मारता है — जिनको कम आता है वे ज्यादा बोलते हैं ।

**1 point** – A specific instance, or a related but not quite equivalent generalization (if the answer is not properly worded or there is a doubt about the suitability of the answer for 2 points, give a point – generally a response gets a score of 2 points or 0 point).

**1 अंक**—एक विशिष्ट उदाहरण या फिर ऐसा सामान्यीकरण जो संबंधित तो हो परन्तु बिल्कुल समान न हो (यदि उत्तर में सही शब्द प्रयुक्त न किये गये हों या उत्तर के सही होने में संदेह हो तो 2 अंक के स्थान पर १ अंक दिया जाय—आमतौर पर किसी भी उत्तर को या तो 2 अंक दिये जाते हैं या 0 अंक) ।

**0 point** – No recognition that the statement is a proverb.

**0 अंक**—इस बात की स्वीकृति नहीं मिलती कि दिया गया कथन एक कहावत है ।

It is more noisy to make a noise in any empty room ..... empty vessel sound more.

खाली कमरे में शोर मचाने से ज्यादा आवाज आती है — खोखला चना ज्यादा बजता है — हिलाने से आवाज होती है ।

**14. यदि दिन के समय आप जंगल में रास्ता भूल जाएँ तो रास्ता कैसे तलाशेंगे ? (If at the time of day you lose your way in the forest, how will you find out the way ?)**

**2 points** – Any explained use of natural phenomena in order to find a way out or a systemic approach to the problem.

**2 अंक**—प्राकृतिक दृष्टि का प्रयोग जिसका वर्णन हो चुका हो जिससे कि रास्ता ढूँढा जा सके या समस्या का क्रमबद्ध हल ।

After knowing the direction with the help of Sun.

सूर्य की सहायता से दिशा ज्ञात करके ।

Walk along a river......use the watch as a compass (explained fully).

एक नदी के साथ-साथ चलिये — घड़ी का प्रयोग कुतुबनुमा की तरह कीजिये (पूर्ण वर्णन किया गया)

**1 point** – Mention of a haphazard means of getting out or a partial 2 points response unexplained. Search for a footpath and follow it...Look for landmarks for bearing.

**1 अंक**—बाहर निकलने के लिये संयोग का उल्लेख या आंशिक रूप से २ अंक का उत्तर जिसका वर्णन न हुआ हो। फुटपाथ को ढूँढिये और इसका अनुसरण कीजिये —— सीमा चिह्नों को ढूँढकर प्रयत्न करना।

We'll see climbing a high tree ..... will go along with footpath.

ऊँचे पेड़ पर चढ़कर देखेंगे...पगडण्डी के सहारे चलेंगे By the Sun.

**0 point** – Use of unreliable or senseless phenomena or reliance on people.

**0 अंक**—अविश्वसनीय या विवेकहीन वस्तु का उल्लेख या लोगों पर विश्वास।

Will ask anyone ..... will go forward, will use our brain ..... will go back to the way from where we have come ..... cry loudly any one will come for help.

किसी से पूछेंगे —— चलते जाएँगे दिमाग से काम लेंगे —— जिस रास्ते से आए थे उसी से वापिस चले जाएँगे —— जोर-जोर से चिल्लाओ कोई न कोई तो मदद के लिए आएगा।

***15. बच्चों के बारे में एक कानून है कि 18 वर्ष से कम उम्र वालों को नौकरी पर न रखा जाये, इस कानून की क्या जरूरत है ? (There is a law about childen that children below 18 years should not be given jobs. What is the need of this law ? )**

**General** : Protect the health of children. (**सामान्य** : बच्चों के स्वास्थ्य की सुरक्षा के लिये।)

Some jobs are too dangerous (strenuous) for children......Children need fresh air and sunshine......Growing children can't be made to work like adults.

कुछ नौकरियाँ बच्चों के लिये अत्यधिक खतरनाक होती हैं —— बच्चों को ताजा हवा और धूप की आवश्यकता होती है —— बढ़ते हुये बच्चों को बड़ों की तरह काम में नही लगाना चाहिये।

**General** : Education. (**सामान्य** : शिक्षा।)

Schooling should come first...work should not interfere with a child's education.

बच्चों का विद्यालय जाना सर्वप्रथम होना चाहिये —— एक बच्चे की शिक्षा में कार्य का हस्तक्षेप नहीं होना चाहिये।

This is the time of the education of children.

यह समय बच्चों की पढ़ाई का है।

**General** : Avoid exploiting children. (**सामान्य** : बच्चों का शोषण रोकना ।)

So that children are not abused......some employers would take advantage of children......children might be over-worked and under-paid.

ताकि बच्चों के साथ दुर्व्यवहार न हो पाये — कुछ नियोक्ता बच्चों का लाभ उठाते हैं — बच्चों से अधिक काम लिया जाता है और उन्हें कम तनख्वाह दी जाती है ।

**General** : Stability of adult job market. (**सामान्य** : वयस्कों के लिये श्रम-बाजार की स्थिरता ।)

To make jobs available for adults. Child workers undermined the pay scale.... So children don't flood the job market.

वयस्कों के लिये रोजगार उपलब्ध कराना, बाल श्रमिकों ने वेतनमानों को घटा दिा — जिससे बच्चे श्रम-बाजारों को अपनी भीड़से न भर सकें ।

**2 points** – A response recognizing at least two of the four general ideas given above.

**2 अंक**—ऐसा उत्तर जिसमें दिये गये चार सामान्य विचारों में से कम से कम दो विचार हों ।

**1 point** – A response recognizing one of the above ideas.

**1 अंक**—ऐसा उत्तर जिसमें उपर्युक्त विचारों में से एक विचार हो ।

**0 point** – A vague idea of protection alone, or no explanation of why child labour is bad.

**0 अंक**—केवल सुरक्षा से संबंध रखने वाला अनिश्चित उत्तर, या ऐसा उत्तर जिसमें इस बात का उल्लेख न हो कि बाल-श्रम बुरा क्यों है ।

Can not understand the work, are small, are minor, are weak.

काम नहीं समझ सकते, छोटे होते हैं, नाबालिग होते हैं, कमजोर होते हैं ।

16. **किसी व्यक्ति को गाड़ी चलाने के लिए राज्य की ओर से लाइसेंस लेना क्यों जरूरी है ? (Why is it necessary for a person to take licence of state government to drive a vehicle ? )**

**2 points** – a response recognizing that a licence-holder is a person who is a trained driver and acquainted with traffic rules.

**2 अंक**—ऐसा उत्तर जो दर्शाता हो कि लायसेंसधारक वह व्यक्ति है जो एक प्रशिक्षित चालक है ओर जो यातायात के नियमों से पूर्ण रूप से अवगत हो ।

गाड़ी चलाना आता है — Driving Test pass किया हुआ है ।

Knows driving ...,.. Passed driving test.

Safe and responsible driver...Training in drive – Knows traffic rules, that he knows driving is known.

सुरक्षित और जिम्मेदार चालक — वाहन चलाने का प्रशिक्षण — यातायात नियम के बारे में पता है, पता लता है कि चलाना आता है ।

**1 point** – A response not properly worded but implies that a licence- holder is a trained driver.

**1 अंक**—ऐसा उत्तर जिसमें उपर्युक्त शब्दों का प्रयोग नहीं है पर यह बताता है कि लायसेंसधारक एक प्रशिक्षित चालक है ।

If there is no licence, accident may be happened, to avoid accident.

Licence नहीं हो तो Accident कर सकता है, Accident से बचने के लिए ।

**0 point** – A response showing no recognition that a licence-holder is a trained driver.

**0 अंक**—ऐसा उत्तर जिसमें लायसेंसधारक के प्रशिक्षित चालक होने के बारे में कोई उल्लेख न हो ।

It is the rule of government, will be caught, may be caught if accident happens, government takes advantages of licence fee.

सरकार का नियम है, पकड़ा जाएगा, दुर्घटना हो जाय तो पकड़ा जा सकता है, सरकार को लाइसेंस की फीस का लाभ होता है ।

17. **ऐसा क्यों होता है कि जनम से बहरा व्यक्ति बोल भी नहीं सकता ?) Why is it so that a born deaf can not speak ?**

**2 points** – Recognition that hearing other people talk facilitates the production of speech...

**2 अंक**—यह स्वीकृति कि दूसरे लोगों को बातें करते हुये सुनना बोलने की क्षमता उत्पन्न करने में सुविधा प्रदान करता है ।

Does not hear so did not learn to speak, speaking is learnt by hearing, it is necessary to have the ciapcity to hear for speaking. How will he speak if does not hear.

सुनता नहीं है, इसीलिए बोलना नहीं सीखा, सुनकर ही बोलना सीखते हैं, बोलने के लिए सुनाई देना जरूरी है। सुनता नहीं है तो बोलेगा कैसे।

Child developes speech by hearing......have to be able to hear to initiate sounds.

बालक सुनकर बोलना सीखता है — प्रारंभिक ध्वनियों को सुनने योग्य होना।

**1 point** – The idea that hearing and talking are connected without the knowledge of how.

**1 अंक**—यह विचार कि सुनना और बात करना आपस में सम्बन्धित है लेकिन इसके कारण का ज्ञान उत्तर से स्पष्ट नहीं होता।

How could he reaply when did not hear others' conversation.

दूसरों की बात सुनी नहीं तो जबाब कैसे देगा।

Can't hear how sounds are produced......They are never able to tell correct form of words.

यह नहीं सुन सकता कि किस प्रकार ध्वनि उत्पन्न होती है — वे कभी शब्दों के सही रूप को नहीं बता पाते।

**0 point** – No idea of the interdependence of hearing others and learning to talk.

**0 अंक**—इस बात का कोई विचार प्रकट नहीं होता कि दूसरों की बातें सुनना और बात करना सीखने में परस्पर निर्भरता है।

God knows, result of doing, any vein may be wrong, might be born in the condition as it us now, there may be any disease, may be mental weakness.

भगवान जाने, कर्मों का फल है, कोई नस गलत होगी, पैदा ही ऐसा हुआ हो, बीमारी होगी कोई, दिमागी कमजोरी होगी।

**18.** **इस कहावत का क्या अर्थ है ?—" जब लोहा गर्म हो तभी चोट करो" । (What is the meaning of this proverb ? )**

If the response is another proverb, ask for an explanation and score it according to the criteria.

यदि उत्तर का सम्बन्ध दूसरी कहावत से हो तो स्पष्टीकरण के लिये कहा जाय और नियमानुसार अंक प्रदान किये जायें ।

**2 points** – An abstract generalization.

**2 अंक**—एक आदर्श स्पष्टीकरण ।

Do you work according to right opportunity, converse at right time, take advantage of opportunity, don't miss the chance.

ठीक मौके पर काम करो, ठीक समय पर बात करो, अवसर का लाभ उठाओ, मौके को मत चूको ।

**1 point** – A specific instance, or a related but not quite equivalent generalization. You can get the maximum benefit when circumstances are favourable to you.....Don't delay doing your work.....If you have a good idea, go ahead with it before some one changes his mind.

**1 अंक**—एक विशिष्ट या संबंधित उदाहरण किन्तु बिल्कुल समान सामान्यीकरण नहीं । जब परिस्थितियाँ आपके अनुकूल हों तो आप अधिकतम लाभ उठा सकते हैं । कार्य करने में विलंब न करें — अगर आपके पस कोई अच्छा विचार है तो आग बढ़ें — इससे पूर्व कि कोई अपना इरादा बदल दे ।

**0 point** – No recognition that the statement is a proverb ; a vague explanation of taking advantage of a situation on a personal level, a literal translation or elaboration.

**0 अंक**—इस बात की कोई स्वीकृति नहीं मिलती कि दिया गया कथन एक कहावत है, एक अनिश्चित उत्तर जैसे—किसी की स्थिति का व्यक्तिगत स्तर पर लाभ उठाना या एक शाब्दिक अनुवाद या विस्तृत रूप ।

Good man can do good deeds...... when the iron is hot, it curves soon ...... always strike at hot iron ...... iron melts when it is hot.

अच्छा आदमी अच्छे काम कर सकता है — जब लोहा गर्म होता है तो जल्दी मुड़ जाता है — गर्म पर ही चोट करनी चाहिए — गर्म होने पर लोहा पिघलता है।

When iron is hot, just a little strike will break it ...Melts so can cut it...Iron your clothes when the iron is hot...Go by your impulse...If somebody is mad at you, get even with them when you are both mad.

जब लोहा गर्म होता है तो थोड़ी-सी चोट करने पर ही टूट जाता है — यह पिघल जाता है अत: आसानी से कट सकता है। अपने कपड़ों पर इस्तिरी करो जब लोहा गर्म हो — अपनी भावनाओं के अनुसार कार्य करो — यदि कोई दूसरा तुम पर पागल हो तुम भी ऐसा ही करो जब तुम दोनों पागल हो।

## How to Determine T.Q. and V.Q. (T.Q. और V.Q. निर्धारित करना)

Test Quotient (T. Q.) may be considered similar to Verbal Quotient (V.Q.) or Intelligence Quotient (I.Q.) except for the difference that T.Q. is determined on the basis of responses on a single test and V. Q./I. Q. is an arithmetical mean of the 4 T. Qs. One can argue whether mean of 4 T.Qs. be labelled as I.Q. or V.Q. ? It is a matter of convention, convenience and precision. If one calls V.Q. it is implied that one is talking about verbal I.Q., but if one calls it I.Q., one needs to specify whether it is verbal I.Q. or Performance I.Q. However, if intelligence functioning is based both on performance and verbal tests of intelligence then it seems to be legitimate to call it I.Q. Therefore, for the sake of precision, the term V. Q. is used here to denote verbal I.Q.

In the subsequent tables norms are provided for male and female separately for 5 age groups and 3 educational levels. The following example will make it clear as to how to determine T.Qs. and V.Q. Suppose there is a female, 42 years old with a schooling of 4th class, she obtains raw scores on Verbal Adult Intelligence Scale (VAIS) as follows :

परीक्षण लब्धि (T.Q.) को शाब्दिक लब्धि (V.Q.) या बुद्धि लब्धि (I.Q.) की भाँति निर्धारित किया जा सकता है। अंतर केवल इतना है कि परीक्षण लब्धि का निर्धारण केवल एक परीक्षा में दिये गये उत्तरों के आधार पर होता है और V.Q. / I.Q. 4 T.Q. का समान्तर माध्य होता है। यह तर्क किया जा सकता है कि क्या 4 T.Q. का माध्य I.Q. या V.Q. के समान स्तर का होता है ? सयह एक सहमति का विषय है, सुविधा और यर्थाथता का मामला है। यदि कोई इसे V.Q. कहता है तो यह माना जाता है शाब्दिक I.Q. की बात हो रही है लेकिन यदि I.Q. कहा जाता है तो व्यक्ति को यह स्पष्ट करने की आवश्यकता होती है कि यह शाब्दिक I.Q. है या उपलब्धि I.Q. । हालाँकि, यदि बुद्धि की क्रिया प्रणाली उपलब्धि और बुद्धि के शाब्दिक परीक्षणों पर आधारित है तो इसे I.Q. कहना न्यायोचित प्रतीत होता है। यही कारण है कि शुद्धता के लिये यहाँ V.Q. को शाब्दिक I.Q. के लिये प्रयुक्त किया है।

आगे दी गयी तालिका में महिला एवं पुरुषों के अलग-अलग 5 आयु समूहों और 3 शैक्षिक स्तरों के नमूने दिये गये हैं। दिया गया उदाहरण यह स्पष्ट कर देगा कि T.Q. और V.Q. को किस प्रकार निर्धारित किया जाये। मतान लीजिये एक 42 वर्षीय महिला जिसने कक्षा 4 तक स्कूली शिक्षा प्राप्त की है, वह निम्नलिखित अनुभवरहित अंक शाब्दिक प्रौढ़ बुद्धि स्तरीय परीक्षा (VAIS) में प्राप्त करती है:

| | **Raw Score**<br>**अनुभवरहित अंक** | **T.Q.**<br>**परीक्षण लब्धि** |
|---|---|---|
| On Information<br>सूचना में | 9 | 103 |
| On Digit Span<br>डिजिट स्पॉन (अंगुलियों से गणना) | 10 | 114 |
| On Arithmetic<br>अंकगणित में | 6 | 97 |
| On Comprehension<br>ग्राह्यता में | 17 | 114 |

For this lady, table (V.Q. 7) of norms meant for Age 40-49, Education 0-5 is seen. In this table T.Qs for males are given towards the left hand side and for females towards right hand side of the page. Consult right hand side table which gives T.Qs. of 103, 114, 97 and 114 for the raw scores of 9, 10, 6, 17 on information, digit span, arithmetic and comprehension subtests respectively. The total of 4 T.Qs would be 428 and its mean would be $\frac{428}{4} = 107$. Thus her V.Q. is 107.

After obtaining T.Qs on 4 subtests of VAIS dysfunction rating may be assigned separately for each of the subtests as follows :

| T.Q. | Dysfunction of Rating |
|---|---|
| 81 and above | 0 |
| 61 to 80 | 2 |
| Less than 61 | 3 |

Since there are 4 subtests, so there could be a maximum dysfunction rating of $3 \times 4 = 12$ on this test.

**N. B. :** Conversion of raw score into T.Q. is based on the assumed mean of 100 and standard deviation of 15 using $100 + \left(\frac{X - M}{SD}\right) 15$.

100 and 15 are assumed mean and SD. X is a raw score for which T.Q. is to be determined, M and SD are the mean and SD of the raw scores of the group.

# TABLES OF NORMS FOR VERBAL ADULT INTELLIGENCE SCALE

## TABLE : VQ 1 ; AGE : 20–29 ; EDU : 0–5

| Raw Score | Male | | | | Female | | | |
|---|---|---|---|---|---|---|---|---|
| | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. |
| 1 | 62 | 56 | 47 | 33 | 69 | 63 | 74 | 51 |
| 2 | 66 | 62 | 54 | 38 | 75 | 69 | 80 | 56 |
| 3 | 71 | 67 | 62 | 43 | 80 | 74 | 85 | 61 |
| 4 | 75 | 74 | 69 | 48 | 86 | 80 | 90 | 66 |
| 5 | 80 | 79 | 77 | 53 | 92 | 85 | 96 | 71 |
| 6 | 84 | 85 | 84 | 58 | 97 | 91 | 101 | 76 |
| 7 | 88 | 91 | 92 | 63 | 103 | 96 | 106 | 81 |
| 8 | 93 | 97 | 99 | 68 | 108 | 102 | 112 | 86 |
| 9 | 97 | 103 | 107 | 73 | 114 | 108 | 117 | 91 |
| 10 | 102 | 108 | 114 | 78 | 120 | 113 | 122 | 96 |
| 11 | 106 | 114 | 122 | 84 | 125 | 119 | 128 | 101 |
| 12 | 111 | 120 | 129 | 89 | 131 | 124 | 133 | 106 |
| 13 | 115 | 126 | 137 | 94 | 136 | 130 | 139 | 111 |
| 14 | 120 | 132 | 144 | 99 | 142 | 135 | 144 | 117 |
| 15 | 124 | 137 | 152 | 104 | 148 | 141 | 150 | 122 |
| 16 | 129 | 143 | | 109 | 153 | 146 | | 127 |
| 17 | 133 | 149 | | 114 | 159 | 152 | | 132 |
| 18 | 138 | 155 | | 119 | 164 | 158 | | 137 |
| 19 | 142 | 161 | | 124 | 170 | 163 | | 142 |
| 20 | 147 | 167 | | 129 | 176 | 169 | | 147 |
| 21 | 151 | 172 | | 134 | 181 | 174 | | 152 |
| 22 | 155 | 178 | | 139 | 187 | 180 | | 157 |
| 23 | 160 | 184 | | 144 | 192 | 185 | | 162 |
| 24 | 164 | 190 | | 149 | 198 | 191 | | 167 |
| 25 | 169 | 196 | | 154 | 204 | 196 | | 172 |
| 26 | 173 | 201 | | 160 | 209 | 202 | | 177 |
| 27 | 178 | | | 165 | 215 | | | 182 |
| 28 | 182 | | | 170 | 220 | | | 187 |
| 29 | 187 | | | 175 | 226 | | | 192 |
| 30 | 191 | | | 180 | 232 | | | 197 |
| 31 | 196 | | | 185 | 237 | | | 202 |
| 32 | 200 | | | 190 | 243 | | | 207 |
| 33 | 205 | | | 195 | 248 | | | 212 |
| 34 | | | | 200 | | | | 217 |
| 35 | | | | 205 | | | | 222 |
| 36 | | | | 210 | | | | 227 |
| Mean | 9·56 | 8·54 | 8·08 | 14·23 | 6·47 | 7·62 | 5·78 | 10·69 |
| SD | 3·36 | 2·58 | 2·00 | 2·96 | 2·68 | 2·70 | 2·78 | 2·99 |

## TABLE : VQ 2 ; AGE : 20–29 ; EDU : 6–9

| Raw Score | Male | | | | Female | | | |
|---|---|---|---|---|---|---|---|---|
| | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. |
| 1 | 56 | 39 | 36 | 43 | 60 | 47 | 42 | 52 |
| 2 | 60 | 46 | 43 | 47 | 64 | 53 | 50 | 56 |
| 3 | 63 | 53 | 51 | 50 | 68 | 59 | 58 | 59 |
| 4 | 67 | 60 | 58 | 54 | 72 | 66 | 66 | 63 |
| 5 | 71 | 67 | 66 | 57 | 76 | 72 | 75 | 66 |
| 6 | 75 | 74 | 73 | 61 | 80 | 78 | 83 | 69 |
| 7 | 79 | 81 | 80 | 64 | 84 | 84 | 91 | 73 |
| 8 | 82 | 88 | 88 | 67 | 89 | 91 | 99 | 76 |
| 9 | 86 | 94 | 95 | 71 | 93 | 97 | 107 | 80 |
| 10 | 90 | 101 | 102 | 74 | 97 | 103 | 115 | 83 |
| 11 | 94 | 108 | 110 | 78 | 101 | 110 | 123 | 87 |
| 12 | 98 | 115 | 117 | 81 | 105 | 116 | 131 | 90 |
| 13 | 101 | 122 | 125 | 85 | 109 | 122 | 139 | 94 |
| 14 | 105 | 129 | 132 | 88 | 113 | 128 | 148 | 97 |
| 15 | 109 | 136 | 139 | 92 | 117 | 135 | 156 | 101 |
| 16 | 113 | 143 | | 95 | 121 | 141 | | 104 |
| 17 | 116 | 150 | | 98 | 126 | 147 | | 108 |
| 18 | 120 | 157 | | 102 | 130 | 153 | | 111 |
| 19 | 124 | 164 | | 105 | 134 | 160 | | 115 |
| 20 | 128 | 171 | | 109 | 138 | 166 | | 118 |
| 21 | 132 | 178 | | 112 | 142 | 172 | | 122 |
| 22 | 135 | 185 | | 116 | 146 | 179 | | 125 |
| 23 | 139 | 192 | | 119 | 150 | 185 | | 128 |
| 24 | 143 | 199 | | 123 | 154 | 191 | | 132 |
| 25 | 147 | 206 | | 126 | 158 | 197 | | 135 |
| 26 | 151 | 213 | | 129 | 163 | 204 | | 139 |
| 27 | 154 | | | 134 | 167 | | | 142 |
| 28 | 158 | | | 136 | 171 | | | 146 |
| 29 | 162 | | | 140 | 175 | | | 149 |
| 30 | 166 | | | 143 | 179 | | | 153 |
| 31 | 170 | | | 147 | 183 | | | 156 |
| 32 | 173 | | | 150 | 187 | | | 160 |
| 33 | 177 | | | 154 | 191 | | | 163 |
| 34 | | | | 157 | | | | 167 |
| 35 | | | | 160 | | | | 170 |
| 36 | | | | 164 | | | | 174 |
| Mean | 12·63 | 9·78 | 9·65 | 17·42 | 10·76 | 9·47 | 8·12 | 14·77 |
| SD | 3·95 | 2·15 | 2·03 | 4·36 | 3·65 | 2·39 | 1·65 | 4·32 |

**TABLE : VQ 3 ; AGE : 20–29 ; EDU : 10 and above**

| Raw Score | Male | | | | Female | | | |
|---|---|---|---|---|---|---|---|---|
| | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. |
| 1 | 45 | 49 | 25 | 34 | 52 | 43 | 39 | 43 |
| 2 | 48 | 53 | 32 | 37 | 54 | 49 | 45 | 46 |
| 3 | 50 | 58 | 39 | 40 | 57 | 55 | 51 | 49 |
| 4 | 53 | 63 | 45 | 43 | 60 | 60 | 58 | 51 |
| 5 | 56 | 67 | 52 | 46 | 63 | 66 | 64 | 54 |
| 6 | 59 | 72 | 59 | 49 | 65 | 71 | 70 | 57 |
| 7 | 61 | 77 | 66 | 51 | 68 | 77 | 76 | 59 |
| 8 | 64 | 81 | 73 | 54 | 71 | 83 | 83 | 62 |
| 9 | 67 | 86 | 80 | 57 | 74 | 88 | 89 | 65 |
| 10 | 70 | 91 | 87 | 60 | 77 | 94 | 95 | 67 |
| 11 | 72 | 95 | 94 | 63 | 79 | 100 | 102 | 70 |
| 12 | 75 | 100 | 101 | 66 | 82 | 105 | 108 | 73 |
| 13 | 78 | 104 | 107 | 69 | 85 | 111 | 114 | 75 |
| 14 | 81 | 109 | 114 | 72 | 88 | 116 | 120 | 78 |
| 15 | 83 | 114 | 121 | 75 | 90 | 122 | 127 | 81 |
| 16 | 86 | 118 | | 77 | 93 | 128 | | 83 |
| 17 | 89 | 123 | | 80 | 96 | 133 | | 86 |
| 18 | 92 | 128 | | 83 | 99 | 139 | | 89 |
| 19 | 94 | 132 | | 86 | 101 | 145 | | 91 |
| 20 | 97 | 137 | | 89 | 104 | 150 | | 94 |
| 21 | 100 | 141 | | 92 | 107 | 156 | | 97 |
| 22 | 103 | 146 | | 95 | 110 | 161 | | 99 |
| 23 | 106 | 151 | | 98 | 112 | 167 | | 102 |
| 24 | 108 | 155 | | 101 | 115 | 173 | | 105 |
| 25 | 111 | 160 | | 103 | 118 | 178 | | 107 |
| 26 | 114 | 165 | | 106 | 121 | 184 | | 110 |
| 27 | 116 | | | 109 | 123 | | | 113 |
| 28 | 119 | | | 112 | 126 | | | 115 |
| 29 | 122 | | | 115 | 129 | | | 118 |
| 30 | 124 | | | 118 | 132 | | | 121 |
| 31 | 127 | | | 121 | 134 | | | 124 |
| 32 | 130 | | | 124 | 137 | | | 126 |
| 33 | 133 | | | 127 | 140 | | | 129 |
| 34 | | | | 129 | | | | 132 |
| 35 | | | | 132 | | | | 134 |
| 36 | | | | 135 | | | | 137 |
| Mean | 21·00 | 12·03 | 11·91 | 23·77 | 18·50 | 11·06 | 10·73 | 22·17 |
| SD | 5·46 | 3·24 | 2·18 | 5·19 | 5·45 | 2·67 | 2·39 | 5·61 |

## TABLE : VQ 4 ; AGE : 30–39 ; EDU : 0–5

| Raw Score | Male | | | | Female | | | |
|---|---|---|---|---|---|---|---|---|
| | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. |
| 1 | 67 | 54 | 53 | 52 | 65 | 57 | 65 | 53 |
| 2 | 71 | 61 | 60 | 55 | 71 | 64 | 71 | 57 |
| 3 | 74 | 67 | 66 | 58 | 76 | 71 | 78 | 62 |
| 4 | 78 | 73 | 73 | 62 | 81 | 78 | 85 | 66 |
| 5 | 81 | 80 | 79 | 65 | 86 | 84 | 91 | 71 |
| 6 | 85 | 86 | 86 | 69 | 92 | 91 | 98 | 75 |
| 7 | 88 | 92 | 92 | 72 | 97 | 98 | 104 | 79 |
| 8 | 92 | 99 | 99 | 76 | 102 | 105 | 111 | 84 |
| 9 | 95 | 105 | 105 | 79 | 108 | 111 | 118 | 88 |
| 10 | 99 | 111 | 112 | 82 | 112 | 118 | 124 | 93 |
| 11 | 102 | 117 | 118 | 86 | 117 | 125 | 131 | 97 |
| 12 | 106 | 124 | 123 | 89 | 122 | 132 | 137 | 102 |
| 13 | 109 | 130 | 131 | 93 | 128 | 138 | 144 | 106 |
| 14 | 112 | 136 | 138 | 96 | 133 | 145 | 150 | 111 |
| 15 | 116 | 143 | 144 | 99 | 138 | 152 | 157 | 115 |
| 16 | 119 | 149 | | 103 | 143 | 159 | | 119 |
| 17 | 123 | 155 | | 106 | 148 | 167 | | 124 |
| 18 | 126 | 162 | | 110 | 153 | 172 | | 128 |
| 19 | 130 | 168 | | 113 | 159 | 179 | | 133 |
| 20 | 133 | 174 | | 117 | 164 | 186 | | 137 |
| 21 | 137 | 180 | | 120 | 169 | 193 | | 142 |
| 22 | 140 | 187 | | 123 | 174 | 199 | | 146 |
| 23 | 144 | 193 | | 127 | 179 | 206 | | 151 |
| 24 | 147 | 199 | | 130 | 184 | 213 | | 155 |
| 25 | 151 | 206 | | 134 | 189 | 220 | | 160 |
| 26 | 154 | 212 | | 137 | 195 | 226 | | 164 |
| 27 | 158 | | | 140 | 200 | | | 168 |
| 28 | 161 | | | 144 | 205 | | | 173 |
| 29 | 164 | | | 147 | 210 | | | 177 |
| 30 | 168 | | | 151 | 215 | | | 182 |
| 31 | 171 | | | 154 | 222 | | | 186 |
| 32 | 175 | | | 158 | 226 | | | 191 |
| 33 | 178 | | | 161 | 231 | | | 195 |
| 34 | | | | 164 | | | | 198 |
| 35 | | | | 168 | | | | 204 |
| 36 | | | | 171 | | | | 209 |
| Mean | 10·37 | 8·22 | 8·16 | 15·13 | 7·62 | 7·29 | 6·31 | 11·60 |
| SD | 4·33 | 2·38 | 2·31 | 4·39 | 2·91 | 2·22 | 2·27 | 3·37 |

## TABLE : VQ 5 ; AGE : 30–39 ; EDU : 6–9

| Raw Score | Male | | | | Female | | | |
|---|---|---|---|---|---|---|---|---|
| | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. |
| 1 | 55 | 45 | 39 | 49 | 50 | 40 | 52 | 49 |
| 2 | 59 | 51 | 45 | 52 | 54 | 48 | 58 | 52 |
| 3 | 62 | 58 | 52 | 55 | 59 | 55 | 65 | 55 |
| 4 | 66 | 64 | 59 | 58 | 63 | 62 | 71 | 59 |
| 5 | 70 | 71 | 66 | 61 | 67 | 70 | 77 | 62 |
| 6 | 73 | 78 | 73 | 64 | 72 | 77 | 84 | 65 |
| 7 | 77 | 84 | 80 | 67 | 76 | 85 | 90 | 69 |
| 8 | 81 | 91 | 87 | 70 | 81 | 92 | 97 | 72 |
| 9 | 84 | 97 | 94 | 73 | 85 | 99 | 103 | 75 |
| 10 | 88 | 104 | 101 | 76 | 90 | 107 | 110 | 79 |
| 11 | 92 | 110 | 108 | 79 | 94 | 114 | 116 | 82 |
| 12 | 95 | 117 | 115 | 82 | 99 | 121 | 123 | 85 |
| 13 | 99 | 123 | 122 | 85 | 103 | 129 | 129 | 89 |
| 14 | 103 | 130 | 129 | 88 | 108 | 136 | 136 | 92 |
| 15 | 106 | 137 | 136 | 91 | 112 | 143 | 142 | 95 |
| 16 | 109 | 143 | | 94 | 117 | 151 | | 99 |
| 17 | 113 | 150 | | 97 | 121 | 158 | | 102 |
| 18 | 117 | 156 | | 100 | 125 | 165 | | 105 |
| 19 | 120 | 163 | | 103 | 130 | 173 | | 109 |
| 20 | 124 | 169 | | 106 | 134 | 180 | | 112 |
| 21 | 127 | 176 | | 109 | 139 | 187 | | 115 |
| 22 | 131 | 182 | | 112 | 143 | 195 | | 119 |
| 23 | 135 | 189 | | 115 | 148 | 202 | | 122 |
| 24 | 138 | 195 | | 118 | 152 | 210 | | 125 |
| 25 | 142 | 202 | | 121 | 157 | 217 | | 129 |
| 26 | 145 | 209 | | 124 | 161 | 224 | | 132 |
| 27 | 149 | | | 127 | 166 | | | 135 |
| 28 | 153 | | | 130 | 170 | | | 139 |
| 29 | 156 | | | 133 | 175 | | | 142 |
| 30 | 160 | | | 136 | 179 | | | 145 |
| 31 | 163 | | | 139 | 184 | | | 149 |
| 32 | 167 | | | 142 | 188 | | | 152 |
| 33 | 171 | | | 145 | 192 | | | 155 |
| 34 | | | | 148 | | | | 159 |
| 35 | | | | 151 | | | | 163 |
| 36 | | | | 154 | | | | 166 |
| Mean | 13·39 | 9·41 | 9·80 | 18·04 | 12·27 | 9·09 | 8·45 | 16·31 |
| SD | 4·16 | 2·29 | 2·15 | 5·03 | 3·36 | 2·04 | 2·31 | 4·47 |

**TABLE : VQ 6 ; AGE : 30–39 ; EDU : 10+**

| Raw Score | Male | | | | Female | | | |
|---|---|---|---|---|---|---|---|---|
| | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. |
| 1 | 35 | 48 | 10 | 13 | 45 | 45 | 39 | 33 |
| 2 | 38 | 53 | 18 | 17 | 48 | 50 | 45 | 36 |
| 3 | 41 | 58 | 26 | 20 | 51 | 56 | 52 | 39 |
| 4 | 44 | 63 | 34 | 24 | 54 | 62 | 58 | 42 |
| 5 | 47 | 68 | 42 | 28 | 57 | 67 | 64 | 45 |
| 6 | 50 | 73 | 50 | 31 | 60 | 73 | 70 | 48 |
| 7 | 54 | 78 | 58 | 35 | 63 | 78 | 76 | 51 |
| 8 | 57 | 83 | 66 | 38 | 66 | 84 | 83 | 53 |
| 9 | 60 | 88 | 73 | 42 | 70 | 89 | 89 | 56 |
| 10 | 63 | 93 | 81 | 46 | 73 | 95 | 95 | 59 |
| 11 | 66 | 98 | 89 | 49 | 76 | 101 | 102 | 62 |
| 12 | 69 | 103 | 97 | 53 | 79 | 107 | 108 | 65 |
| 13 | 72 | 108 | 105 | 56 | 82 | 112 | 114 | 68 |
| 14 | 75 | 113 | 113 | 60 | 85 | 118 | 120 | 72 |
| 15 | 79 | 118 | 121 | 64 | 88 | 123 | 126 | 75 |
| 16 | 82 | 123 | | 67 | 91 | 129 | | 78 |
| 17 | 85 | 128 | | 71 | 94 | 135 | | 81 |
| 18 | 88 | 133 | | 75 | 97 | 140 | | 84 |
| 19 | 91 | 138 | | 78 | 101 | 146 | | 87 |
| 20 | 94 | 143 | | 82 | 104 | 152 | | 90 |
| 21 | 97 | 148 | | 85 | 107 | 157 | | 94 |
| 22 | 101 | 153 | | 89 | 110 | 163 | | 97 |
| 23 | 104 | 158 | | 93 | 113 | 168 | | 100 |
| 24 | 107 | 163 | | 96 | 116 | 174 | | 103 |
| 25 | 110 | 169 | | 100 | 119 | 179 | | 106 |
| 26 | 113 | 174 | | 103 | 122 | 185 | | 109 |
| 27 | 116 | | | 107 | 125 | | | 112 |
| 28 | 119 | | | 111 | 128 | | | 115 |
| 29 | 122 | | | 114 | 132 | | | 118 |
| 30 | 126 | | | 118 | 135 | | | 121 |
| 31 | 129 | | | 121 | 138 | | | 124 |
| 32 | 132 | | | 125 | 141 | | | 127 |
| 33 | 135 | | | 129 | 144 | | | 130 |
| 34 | | | | 132 | | | | 133 |
| 35 | | | | 136 | | | | 136 |
| 36 | | | | 139 | | | | 139 |
| Mean | 21·80 | 11·37 | 12·33 | 25·04 | 18·80 | 10·81 | 10·77 | 23·06 |
| SD | 4·79 | 2·98 | 1·89 | 4·16 | 4·84 | 2·67 | 2·41 | 4·92 |

## TABLE : VQ 7 ; AGE : 40–49 ; EDU : 0–5

| Raw Score | Male | | | | Female | | | |
|---|---|---|---|---|---|---|---|---|
| | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. |
| 1 | 66 | 55 | 52 | 50 | 69 | 62 | 56 | 60 |
| 2 | 69 | 61 | 59 | 54 | 73 | 68 | 64 | 64 |
| 3 | 73 | 67 | 65 | 57 | 77 | 74 | 72 | 67 |
| 4 | 76 | 74 | 71 | 60 | 82 | 79 | 81 | 70 |
| 5 | 80 | 80 | 79 | 64 | 86 | 85 | 89 | 74 |
| 6 | 84 | 86 | 84 | 67 | 90 | 91 | 97 | 77 |
| 7 | 87 | 92 | 91 | 70 | 94 | 97 | 105 | 80 |
| 8 | 91 | 98 | 97 | 74 | 99 | 102 | 113 | 84 |
| 9 | 94 | 104 | 103 | 77 | 103 | 108 | 121 | 87 |
| 10 | 98 | 110 | 110 | 80 | 107 | 114 | 129 | 90 |
| 11 | 102 | 116 | 116 | 84 | 111 | 120 | 137 | 94 |
| 12 | 105 | 122 | 123 | 87 | 116 | 126 | 145 | 97 |
| 13 | 109 | 128 | 129 | 90 | 121 | 132 | 153 | 101 |
| 14 | 112 | 135 | 135 | 94 | 125 | 136 | 162 | 104 |
| 15 | 116 | 141 | 142 | 97 | 129 | 142 | 170 | 107 |
| 16 | 120 | 147 | | 100 | 134 | 149 | | 110 |
| 17 | 123 | 153 | | 104 | 138 | 155 | | 114 |
| 18 | 127 | 159 | | 107 | 142 | 161 | | 117 |
| 19 | 130 | 165 | | 111 | 147 | 167 | | 121 |
| 20 | 134 | 171 | | 114 | 151 | 172 | | 124 |
| 21 | 138 | 177 | | 117 | 155 | 178 | | 127 |
| 22 | 141 | 183 | | 121 | 159 | 184 | | 131 |
| 23 | 145 | 189 | | 124 | 163 | 189 | | 134 |
| 24 | 148 | 196 | | 127 | 168 | 195 | | 138 |
| 25 | 152 | 202 | | 131 | 173 | 200 | | 141 |
| 26 | 156 | 208 | | 134 | 177 | 206 | | 144 |
| 27 | 159 | | | 137 | 181 | | | 148 |
| 28 | 163 | | | 141 | 186 | | | 151 |
| 29 | 166 | | | 144 | 190 | | | 154 |
| 30 | 170 | | | 147 | 194 | | | 158 |
| 31 | 174 | | | 151 | 199 | | | 161 |
| 32 | 177 | | | 154 | 203 | | | 164 |
| 33 | 181 | | | 157 | 207 | | | 168 |
| 34 | | | | 161 | | | | 171 |
| 35 | | | | 164 | | | | 174 |
| 36 | | | | 168 | | | | 178 |
| Mean | 10·52 | 8·32 | 8·45 | 15·82 | 8·22 | 7·55 | 6·39 | 12·77 |
| SD | 4·17 | 2·46 | 2·34 | 4·49 | 3·46 | 2·60 | 1·85 | 4·47 |

## TABLE : VQ 8 ; AGE : 40–49 ; EDU : 6–9

| Raw Score | Male | | | | Female | | | |
|---|---|---|---|---|---|---|---|---|
| | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. |
| 1 | 54 | 47 | 38 | 37 | 22 | 48 | 46 | 53 |
| 2 | 57 | 53 | 45 | 41 | 29 | 54 | 53 | 56 |
| 3 | 60 | 59 | 51 | 44 | 35 | 60 | 61 | 59 |
| 4 | 63 | 64 | 57 | 47 | 42 | 66 | 69 | 62 |
| 5 | 66 | 70 | 64 | 51 | 49 | 72 | 76 | 65 |
| 6 | 69 | 76 | 70 | 54 | 55 | 78 | 84 | 68 |
| 7 | 72 | 82 | 77 | 57 | 62 | 84 | 91 | 71 |
| 8 | 75 | 88 | 83 | 61 | 69 | 91 | 99 | 74 |
| 9 | 78 | 94 | 89 | 64 | 75 | 97 | 106 | 77 |
| 10 | 82 | 100 | 96 | 67 | 82 | 103 | 114 | 80 |
| 11 | 85 | 105 | 102 | 71 | 89 | 109 | 121 | 83 |
| 12 | 88 | 111 | 109 | 74 | 95 | 115 | 129 | 87 |
| 13 | 92 | 117 | 115 | 77 | 102 | 121 | 137 | 90 |
| 14 | 95 | 123 | 121 | 81 | 109 | 127 | 145 | 93 |
| 15 | 97 | 129 | 128 | 84 | 115 | 133 | 152 | 96 |
| 16 | 100 | 135 | | 87 | 122 | 140 | | 99 |
| 17 | 103 | 141 | | 91 | 129 | 146 | | 102 |
| 18 | 106 | 146 | | 94 | 135 | 152 | | 105 |
| 19 | 110 | 152 | | 97 | 142 | 158 | | 108 |
| 20 | 113 | 158 | | 101 | 149 | 164 | | 111 |
| 21 | 116 | 164 | | 104 | 155 | 170 | | 114 |
| 22 | 119 | 170 | | 108 | 162 | 176 | | 117 |
| 23 | 122 | 176 | | 111 | 169 | 182 | | 121 |
| 24 | 125 | 182 | | 114 | 175 | 188 | | 124 |
| 25 | 128 | 187 | | 118 | 182 | 195 | | 127 |
| 26 | 131 | 193 | | 121 | 189 | 201 | | 130 |
| 27 | 134 | | | 124 | 195 | | | 133 |
| 28 | 138 | | | 128 | 202 | | | 136 |
| 29 | 141 | | | 131 | 209 | | | 139 |
| 30 | 144 | | | 134 | 215 | | | 142 |
| 31 | 147 | | | 138 | 222 | | | 145 |
| 32 | 150 | | | 141 | 229 | | | 148 |
| 33 | 153 | | | 144 | 235 | | | 151 |
| 34 | | | | 148 | | | | 154 |
| 35 | | | | 151 | | | | 157 |
| 36 | | | | 154 | | | | 161 |
| Mean | 15·89 | 10·06 | 10·65 | 19·71 | 12·70 | 9·52 | 8·11 | 16·31 |
| SD | 4·82 | 2·56 | 2·35 | 4·94 | 4·53 | 2·45 | 1·97 | 4·87 |

**TABLE : VQ 9 ; AGE : 40–49 ; EDU : 10+**

| Raw Score | Male | | | | Female | | | |
|---|---|---|---|---|---|---|---|---|
| | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. |
| 1 | 22 | 40 | | 18 | 41 | 42 | 30 | 33 |
| 2 | 26 | 46 | | 22 | 44 | 48 | 37 | 36 |
| 3 | 29 | 51 | | 25 | 47 | 53 | 44 | 39 |
| 4 | 33 | 57 | 11 | 28 | 50 | 59 | 50 | 42 |
| 5 | 36 | 63 | 21 | 32 | 53 | 65 | 57 | 45 |
| 6 | 40 | 68 | 31 | 35 | 56 | 71 | 63 | 48 |
| 7 | 43 | 74 | 41 | 38 | 59 | 76 | 70 | 51 |
| 8 | 46 | 79 | 51 | 42 | 62 | 82 | 77 | 54 |
| 9 | 50 | 85 | 61 | 45 | 65 | 88 | 83 | 57 |
| 10 | 53 | 91 | 71 | 48 | 68 | 93 | 90 | 60 |
| 11 | 56 | 96 | 81 | 52 | 71 | 99 | 97 | 62 |
| 12 | 60 | 102 | 91 | 55 | 74 | 105 | 103 | 65 |
| 13 | 64 | 108 | 101 | 58 | 77 | 111 | 110 | 68 |
| 14 | 67 | 115 | 111 | 62 | 80 | 116 | 117 | 71 |
| 15 | 70 | 119 | 121 | 65 | 83 | 122 | 123 | 74 |
| 16 | 74 | 125 | | 68 | 86 | 128 | | 77 |
| 17 | 77 | 130 | | 72 | 89 | 133 | | 80 |
| 18 | 81 | 136 | | 75 | 92 | 139 | | 82 |
| 19 | 84 | 142 | | 78 | 95 | 145 | | 85 |
| 20 | 88 | 147 | | 82 | 98 | 151 | | 88 |
| 21 | 91 | 153 | | 85 | 101 | 156 | | 91 |
| 22 | 95 | 158 | | 88 | 104 | 162 | | 94 |
| 23 | 98 | 164 | | 92 | 107 | 168 | | 97 |
| 24 | 101 | 170 | | 95 | 110 | 173 | | 100 |
| 25 | 105 | 175 | | 98 | 113 | 179 | | 102 |
| 26 | 108 | 181 | | 102 | 116 | 185 | | 105 |
| 27 | 112 | | | 105 | 119 | | | 108 |
| 28 | 115 | | | 109 | 122 | | | 111 |
| 29 | 119 | | | 112 | 125 | | | 114 |
| 30 | 122 | | | 115 | 128 | | | 117 |
| 31 | 126 | | | 119 | 131 | | | 121 |
| 32 | 129 | | | 122 | 134 | | | 123 |
| 33 | 132 | | | 125 | 137 | | | 126 |
| 34 | | | | 129 | | | | 129 |
| 35 | | | | 132 | | | | 132 |
| 36 | | | | 135 | | | | 135 |
| Mean | 23·55 | 11·62 | 12·86 | 25·38 | 20·72 | 11·12 | 11·51 | 23·88 |
| SD | 4·36 | 2·66 | 1·49 | 4·48 | 4·98 | 2·62 | 2·27 | 5·16 |

## TABLE : VQ 10 ; AGE : 50–59 ; EDU : 0–5

| Raw Score | Male | | | | Female | | | |
|---|---|---|---|---|---|---|---|---|
| | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. |
| 1 | 64 | 48 | 58 | 49 | 70 | 66 | 58 | 56 |
| 2 | 67 | 55 | 64 | 52 | 75 | 71 | 66 | 60 |
| 3 | 71 | 62 | 70 | 56 | 79 | 77 | 74 | 64 |
| 4 | 74 | 69 | 76 | 60 | 83 | 83 | 82 | 68 |
| 5 | 78 | 76 | 82 | 63 | 88 | 88 | 90 | 72 |
| 6 | 81 | 83 | 87 | 67 | 92 | 94 | 98 | 76 |
| 7 | 85 | 90 | 93 | 71 | 96 | 100 | 106 | 80 |
| 8 | 88 | 97 | 99 | 74 | 100 | 105 | 114 | 84 |
| 9 | 92 | 104 | 105 | 79 | 105 | 111 | 122 | 88 |
| 10 | 95 | 111 | 111 | 81 | 109 | 117 | 130 | 92 |
| 11 | 99 | 118 | 117 | 85 | 113 | 122 | 138 | 95 |
| 12 | 102 | 124 | 123 | 89 | 118 | 128 | 146 | 99 |
| 13 | 106 | 132 | 129 | 92 | 122 | 134 | 154 | 103 |
| 14 | 109 | 139 | 135 | 96 | 126 | 140 | 162 | 107 |
| 15 | 113 | 146 | 141 | 100 | 130 | 145 | 170 | 111 |
| 16 | 116 | 153 | | 103 | 135 | 151 | | 115 |
| 17 | 120 | 160 | | 107 | 139 | 157 | | 119 |
| 18 | 123 | 167 | | 111 | 143 | 163 | | 123 |
| 19 | 127 | 174 | | 114 | 148 | 168 | | 127 |
| 20 | 130 | 181 | | 118 | 152 | 174 | | 131 |
| 21 | 134 | 188 | | 121 | 156 | 180 | | 135 |
| 22 | 137 | 194 | | 125 | 160 | 185 | | 139 |
| 23 | 141 | 201 | | 129 | 165 | 191 | | 143 |
| 24 | 144 | 208 | | 132 | 169 | 197 | | 147 |
| 25 | 148 | 215 | | 136 | 173 | 202 | | 151 |
| 26 | 151 | 222 | | 140 | 178 | 208 | | 156 |
| 27 | 155 | | | 143 | 182 | | | 159 |
| 28 | 158 | | | 147 | 186 | | | 163 |
| 29 | 162 | | | 151 | 190 | | | 167 |
| 30 | 165 | | | 154 | 195 | | | 170 |
| 31 | 169 | | | 158 | 199 | | | 174 |
| 32 | 172 | | | 161 | 203 | | | 178 |
| 33 | 176 | | | 165 | 208 | | | 182 |
| 34 | | | | 169 | | | | 186 |
| 35 | | | | 172 | | | | 190 |
| 36 | | | | 176 | | | | 194 |
| Mean | 11·28 | 8·42 | 8·10 | 15·09 | 7·87 | 7·03 | 6·24 | 12·12 |
| SD | 4·28 | 2·16 | 2·53 | 4·13 | 3·49 | 2·64 | 1·87 | 3·80 |

## TABLE : VQ 11 ; AGE : 50–59 ; EDU : 6–9

| Raw Score | Male | | | | Female | | | |
|---|---|---|---|---|---|---|---|---|
| | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. |
| 1 | 54 | 44 | 36 | 42 | 58 | 37 | 51 | 52 |
| 2 | 57 | 50 | 42 | 45 | 62 | 45 | 58 | 55 |
| 3 | 60 | 56 | 48 | 48 | 65 | 52 | 64 | 58 |
| 4 | 63 | 62 | 54 | 51 | 69 | 59 | 70 | 61 |
| 5 | 66 | 68 | 61 | 55 | 72 | 67 | 77 | 64 |
| 6 | 69 | 75 | 67 | 58 | 75 | 74 | 83 | 67 |
| 7 | 72 | 81 | 74 | 61 | 79 | 82 | 89 | 70 |
| 8 | 75 | 87 | 80 | 64 | 82 | 89 | 96 | 73 |
| 9 | 79 | 93 | 86 | 67 | 86 | 96 | 102 | 76 |
| 10 | 82 | 99 | 93 | 70 | 89 | 104 | 109 | 79 |
| 11 | 85 | 105 | 99 | 73 | 93 | 111 | 115 | 82 |
| 12 | 88 | 112 | 105 | 76 | 96 | 119 | 121 | 85 |
| 13 | 91 | 136 | 112 | 79 | 100 | 126 | 128 | 88 |
| 14 | 94 | 124 | 118 | 82 | 103 | 133 | 134 | 91 |
| 15 | 97 | 130 | 124 | 85 | 106 | 141 | 141 | 94 |
| 16 | 100 | 136 | | 88 | 110 | 148 | | 97 |
| 17 | 103 | 142 | | 92 | 113 | 156 | | 100 |
| 18 | 106 | 149 | | 95 | 116 | 163 | | 103 |
| 19 | 109 | 155 | | 98 | 120 | 170 | | 106 |
| 20 | 112 | 161 | | 101 | 124 | 178 | | 109 |
| 21 | 115 | 167 | | 104 | 127 | 185 | | 112 |
| 22 | 118 | 173 | | 107 | 131 | 192 | | 115 |
| 23 | 121 | 180 | | 110 | 134 | 199 | | 118 |
| 24 | 125 | 186 | | 113 | 137 | 207 | | 121 |
| 25 | 127 | 192 | | 116 | 141 | 214 | | 124 |
| 26 | 130 | 198 | | 119 | 144 | 222 | | 127 |
| 27 | 133 | | | 122 | 148 | | | 130 |
| 28 | 136 | | | 125 | 151 | | | 133 |
| 29 | 139 | | | 128 | 155 | | | 136 |
| 30 | 142 | | | 132 | 158 | | | 139 |
| 31 | 145 | | | 135 | 162 | | | 142 |
| 32 | 147 | | | 138 | 165 | | | 145 |
| 33 | 152 | | | 141 | 169 | | | 148 |
| 34 | | | | 144 | | | | 151 |
| 35 | | | | 147 | | | | 154 |
| 36 | | | | 150 | | | | 157 |
| Mean | 16·02 | 10·10 | 11·14 | 19·72 | 13·10 | 9·47 | 8·63 | 16·97 |
| SD | 4·93 | 2·43 | 2·37 | 4·87 | 4·35 | 2·03 | 2·35 | 5·03 |

**TABLE : VQ 12 ; AGE : 50–59 ; EDU : 10+**

| Raw Score | Male Information T.Q. | Male Digit Span T.Q. | Male Arithmetic T.Q. | Male Comprehension T.Q. | Female Information T.Q. | Female Digit Span T.Q. | Female Arithmetic T.Q. | Female Comprehension T.Q. |
|---|---|---|---|---|---|---|---|---|
| 1 | 21 | 34 | | 34 | 43 | 36 | 28 | 45 |
| 2 | 24 | 40 | | 36 | 46 | 42 | 35 | 48 |
| 3 | 28 | 46 | 09 | 39 | 49 | 48 | 42 | 50 |
| 4 | 31 | 53 | 18 | 42 | 52 | 55 | 49 | 53 |
| 5 | 35 | 59 | 27 | 45 | 55 | 61 | 56 | 55 |
| 6 | 38 | 65 | 37 | 47 | 58 | 67 | 63 | 58 |
| 7 | 42 | 71 | 46 | 50 | 60 | 74 | 70 | 60 |
| 8 | 46 | 77 | 55 | 53 | 63 | 80 | 77 | 62 |
| 9 | 49 | 83 | 64 | 56 | 66 | 86 | 84 | 65 |
| 10 | 53 | 90 | 74 | 59 | 69 | 93 | 91 | 67 |
| 11 | 56 | 98 | 83 | 61 | 72 | 99 | 98 | 70 |
| 12 | 60 | 103 | 92 | 64 | 75 | 105 | 105 | 72 |
| 13 | 63 | 109 | 101 | 66 | 78 | 112 | 111 | 75 |
| 14 | 66 | 115 | 111 | 69 | 81 | 118 | 118 | 77 |
| 15 | 70 | 121 | 120 | 72 | 84 | 124 | 125 | 80 |
| 16 | 74 | 127 | | 75 | 87 | 131 | | 82 |
| 17 | 77 | 133 | | 78 | 90 | 137 | | 84 |
| 18 | 81 | 139 | | 81 | 93 | 143 | | 87 |
| 19 | 85 | 145 | | 83 | 96 | 150 | | 89 |
| 20 | 88 | 152 | | 86 | 98 | 156 | | 92 |
| 21 | 92 | 158 | | 89 | 101 | 162 | | 94 |
| 22 | 96 | 164 | | 92 | 104 | 169 | | 97 |
| 23 | 98 | 170 | | 94 | 107 | 175 | | 99 |
| 24 | 102 | 176 | | 97 | 110 | 181 | | 102 |
| 25 | 106 | 182 | | 100 | 113 | 187 | | 104 |
| 26 | 109 | 188 | | 103 | 116 | 194 | | 107 |
| 27 | 113 | | | 105 | 119 | | | 109 |
| 28 | 117 | | | 108 | 122 | | | 111 |
| 29 | 120 | | | 111 | 125 | | | 114 |
| 30 | 124 | | | 114 | 128 | | | 116 |
| 31 | 128 | | | 116 | 131 | | | 119 |
| 32 | 131 | | | 119 | 134 | | | 121 |
| 33 | 134 | | | 122 | 136 | | | 124 |
| 34 | | | | 125 | | | | 126 |
| 35 | | | | 127 | | | | 129 |
| 36 | | | | 130 | | | | 131 |
| Mean | 23·32 | 11·69 | 12·83 | 25·04 | 20·50 | 11·16 | 11·33 | 23·29 |
| SD | 4·23 | 2·44 | 1·62 | 5·45 | 5·13 | 2·37 | 2·17 | 6·11 |

**TABLE : VQ 13 ; AGE : 60–69 ; EDU : 0–5**

| Raw Score | Male | | | | Female | | | |
|---|---|---|---|---|---|---|---|---|
| | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. |
| 1 | 71 | 62 | 62 | 64 | 68 | 67 | 68 | 62 |
| 2 | 74 | 68 | 67 | 67 | 73 | 74 | 75 | 66 |
| 3 | 77 | 73 | 73 | 69 | 78 | 80 | 83 | 70 |
| 4 | 81 | 79 | 78 | 72 | 83 | 86 | 90 | 74 |
| 5 | 84 | 85 | 83 | 75 | 89 | 92 | 98 | 77 |
| 6 | 87 | 91 | 89 | 77 | 94 | 99 | 106 | 81 |
| 7 | 90 | 96 | 94 | 80 | 99 | 105 | 113 | 85 |
| 8 | 94 | 102 | 99 | 83 | 104 | 111 | 121 | 89 |
| 9 | 97 | 108 | 105 | 86 | 109 | 117 | 129 | 93 |
| 10 | 100 | 114 | 110 | 88 | 114 | 123 | 136 | 97 |
| 11 | 104 | 119 | 115 | 91 | 119 | 130 | 144 | 101 |
| 12 | 107 | 125 | 121 | 94 | 124 | 136 | 151 | 104 |
| 13 | 110 | 131 | 126 | 96 | 129 | 142 | 159 | 108 |
| 14 | 113 | 137 | 131 | 99 | 134 | 148 | 167 | 112 |
| 15 | 117 | 142 | 136 | 102 | 139 | 155 | 174 | 116 |
| 16 | 120 | 148 | | 104 | 144 | 161 | | 120 |
| 17 | 123 | 154 | | 107 | 150 | 167 | | 124 |
| 18 | 127 | 160 | | 110 | 155 | 173 | | 127 |
| 19 | 130 | 165 | | 113 | 160 | 180 | | 131 |
| 20 | 133 | 171 | | 115 | 165 | 186 | | 135 |
| 21 | 136 | 177 | | 118 | 170 | 192 | | 139 |
| 22 | 140 | 183 | | 121 | 175 | 198 | | 143 |
| 23 | 143 | 188 | | 123 | 180 | 204 | | 147 |
| 24 | 146 | 194 | | 126 | 185 | 211 | | 151 |
| 25 | 150 | 200 | | 129 | 190 | 217 | | 154 |
| 26 | 153 | 206 | | 131 | 195 | 223 | | 158 |
| 27 | 156 | | | 134 | 200 | | | 162 |
| 28 | 160 | | | 137 | 205 | | | 166 |
| 29 | 163 | | | 140 | 211 | | | 170 |
| 30 | 166 | | | 142 | 216 | | | 174 |
| 31 | 169 | | | 145 | 221 | | | 177 |
| 32 | 173 | | | 148 | 226 | | | 181 |
| 33 | 176 | | | 150 | 231 | | | 185 |
| 34 | | | | 153 | | | | 189 |
| 35 | | | | 156 | | | | 193 |
| 36 | | | | 159 | | | | 197 |
| Mean | 9·88 | 7·61 | 8·12 | 14·31 | 7·24 | 6·21 | 5·24 | 10·83 |
| SD | 4·56 | 2·61 | 2·82 | 5·55 | 2·95 | 2·41 | 1·97 | 3·90 |

## TABLE : VQ 14 ; AGE : 60–69 ; EDU : 6–9

| Raw Score | Male | | | | Female | | | |
|---|---|---|---|---|---|---|---|---|
| | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. |
| 1 | 62 | 49 | 49 | 54 | 62 | 50 | 45 | 44 |
| 2 | 65 | 56 | 55 | 56 | 65 | 57 | 53 | 48 |
| 3 | 68 | 62 | 60 | 59 | 69 | 64 | 61 | 52 |
| 4 | 70 | 68 | 65 | 61 | 72 | 70 | 68 | 55 |
| 5 | 73 | 74 | 71 | 64 | 75 | 77 | 76 | 59 |
| 6 | 76 | 80 | 76 | 67 | 78 | 84 | 83 | 63 |
| 7 | 78 | 87 | 81 | 69 | 81 | 91 | 91 | 66 |
| 8 | 81 | 93 | 87 | 72 | 85 | 98 | 99 | 70 |
| 9 | 84 | 99 | 92 | 74 | 88 | 105 | 106 | 74 |
| 10 | 86 | 105 | 97 | 77 | 91 | 112 | 114 | 78 |
| 11 | 89 | 111 | 103 | 80 | 94 | 119 | 122 | 81 |
| 12 | 92 | 118 | 108 | 82 | 97 | 125 | 128 | 85 |
| 13 | 94 | 124 | 113 | 85 | 101 | 132 | 137 | 89 |
| 14 | 97 | 130 | 119 | 87 | 104 | 139 | 144 | 92 |
| 15 | 100 | 136 | 124 | 90 | 107 | 146 | 152 | 96 |
| 16 | 102 | 142 | | 93 | 110 | 153 | | 100 |
| 17 | 105 | 149 | | 95 | 113 | 160 | | 104 |
| 18 | 108 | 155 | | 98 | 117 | 167 | | 107 |
| 19 | 110 | 161 | | 100 | 120 | 174 | | 111 |
| 20 | 113 | 167 | | 103 | 123 | 181 | | 115 |
| 21 | 116 | 173 | | 105 | 126 | 187 | | 119 |
| 22 | 118 | 180 | | 108 | 130 | 194 | | 122 |
| 23 | 121 | 186 | | 111 | 133 | 201 | | 126 |
| 24 | 124 | 192 | | 113 | 136 | 208 | | 130 |
| 25 | 126 | 198 | | 116 | 139 | 215 | | 133 |
| 26 | 129 | 204 | | 118 | 142 | 222 | | 137 |
| 27 | 132 | | | 121 | 146 | | | 141 |
| 28 | 134 | | | 124 | 149 | | | 145 |
| 29 | 137 | | | 126 | 152 | | | 148 |
| 30 | 140 | | | 129 | 155 | | | 152 |
| 31 | 142 | | | 131 | 158 | | | 156 |
| 32 | 145 | | | 134 | 162 | | | 159 |
| 33 | 148 | | | 137 | 165 | | | 163 |
| 34 | | | | 139 | | | | 167 |
| 35 | | | | 142 | | | | 171 |
| 36 | | | | 144 | | | | 174 |
| Mean | 15·09 | 9·14 | 10·49 | 18·85 | 12·76 | 8·28 | 8·16 | 16·00 |
| SD | 5·61 | 2·42 | 2·81 | 5·80 | 4·68 | 2·18 | 1·97 | 4·03 |

**TABLE : VQ 15 ; AGE : 60–69 ; EDU : 10+**

| Raw Score | Male | | | | Female | | | |
|---|---|---|---|---|---|---|---|---|
| | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. | Information T.Q. | Digit Span T.Q. | Arithmetic T.Q. | Comprehension T.Q. |
| 1 | 33 | 50 | 23 | 31 | 42 | 42 | 40 | 35 |
| 2 | 36 | 55 | 30 | 34 | 45 | 48 | 46 | 38 |
| 3 | 39 | 60 | 36 | 37 | 48 | 55 | 52 | 41 |
| 4 | 42 | 65 | 42 | 40 | 51 | 61 | 58 | 44 |
| 5 | 45 | 71 | 49 | 43 | 54 | 68 | 64 | 47 |
| 6 | 48 | 76 | 55 | 46 | 57 | 75 | 71 | 50 |
| 7 | 51 | 81 | 62 | 48 | 60 | 81 | 77 | 53 |
| 8 | 54 | 86 | 68 | 51 | 63 | 88 | 83 | 56 |
| 9 | 57 | 91 | 75 | 54 | 66 | 94 | 89 | 59 |
| 10 | 60 | 96 | 81 | 57 | 69 | 101 | 95 | 62 |
| 11 | 63 | 102 | 88 | 60 | 72 | 107 | 101 | 65 |
| 12 | 66 | 107 | 94 | 63 | 75 | 114 | 107 | 68 |
| 13 | 68 | 112 | 101 | 65 | 78 | 120 | 113 | 71 |
| 14 | 71 | 117 | 107 | 68 | 81 | 127 | 120 | 74 |
| 15 | 74 | 122 | 114 | 71 | 84 | 134 | 126 | 77 |
| 16 | 77 | 128 | | 74 | 87 | 140 | | 80 |
| 17 | 80 | 133 | | 77 | 90 | 147 | | 83 |
| 18 | 83 | 138 | | 80 | 93 | 153 | | 86 |
| 19 | 86 | 143 | | 82 | 96 | 160 | | 89 |
| 20 | 89 | 148 | | 85 | 99 | 166 | | 92 |
| 21 | 92 | 154 | | 88 | 102 | 173 | | 95 |
| 22 | 95 | 159 | | 91 | 105 | 179 | | 98 |
| 23 | 98 | 164 | | 94 | 108 | 186 | | 101 |
| 24 | 101 | 169 | | 97 | 111 | 192 | | 104 |
| 25 | 104 | 174 | | 99 | 114 | 199 | | 107 |
| 26 | 107 | 180 | | 102 | 117 | 206 | | 110 |
| 27 | 110 | | | 105 | 120 | | | 113 |
| 28 | 113 | | | 108 | 124 | | | 115 |
| 29 | 116 | | | 110 | 127 | | | 118 |
| 30 | 119 | | | 113 | 130 | | | 121 |
| 31 | 122 | | | 116 | 133 | | | 124 |
| 32 | 125 | | | 119 | 136 | | | 127 |
| 33 | 128 | | | 122 | 139 | | | 130 |
| 34 | | | | 125 | | | | 133 |
| 35 | | | | 128 | | | | 136 |
| 36 | | | | 131 | | | | 139 |
| Mean | 23·51 | 10·66 | 12·86 | 25·17 | 20·19 | 9·87 | 10·79 | 22·76 |
| SD | 5·02 | 2·89 | 2·32 | 5·30 | 4·96 | 2·29 | 2·46 | 5·04 |

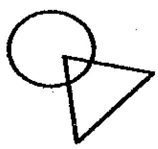
## 4. NAHOR-BENSON TEST (N. B. Test)

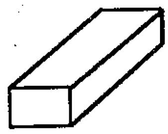
It consists of 8 cards. Out of these five cards contain a design each and three cards contain the instructions to be followed, i. e., subjects are required to draw shape of objects (Nahor and Benson. 1970 ; Pershad and Verma, 1978). Five designs are besed on developmental pettern. One learns to draw simple drawings early in life and then drawing with depth perception. Cards IV and V represent copying designs with depth. It is assumed that whenever organic brain dysfunction occurs, it will show regression in ability and behaviour and that it follows *Ribots* law, i. e., the things learned earlier in life will diminish in the end and the things learned later in life will start diminishing earlier. Cards IV and V provide for this assumption.

Generally, drawing tests measure functioning of the right hemisphere and that too related with parieto-occipital lobe. Parietal lobe function includes spatial relationship and occipital includes visual perceptual acuity. Therefore, Nahor and Benson test performance, specially on designs I-V, will be disturbed depending upon the extent of lesion in the right parieto occipital region. The last three cards where instructions are to be read, consolidated, understood and executed, will involve left hemisphere and disturbances of transferance from left to right hemisphere.

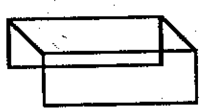
• Thus, this test provides for simultaneous assessment of both the hemispheres. Its 8 designs (5 drawings + 3 instructions) are given below :

I


II


V


III


IV


VI DRAW A CIRCLE

VII DRAW A CUBE

VIII DRAW A CLOCK SHOWING THE TIME OF 5.20

### Material

1. Eight cards with drawings and instructions as given above. Size of the cards may vary slightly from the standard size of 15 × 11 cm. but all cards should be of identical size. Drawings and Instructions for the drawings should be printed boldly, horizontally and separately on each card.
2. A full scale sheet of white paper of good quality.
3. Properly sharpened HB pencil.
4. A rubber.

### Administration

Having established the workable rapport, place the blank-sheet of paper on the table infront of the subject. Paper is to be placed horizontally. Put a pencil and a rubber on the paper. Place the first card at the top of the paper. Ask the subject to "copy it as precisely as possible" (बिल्कुल ऐसा ही बनाओ). After getting an indication that he has copied it, remove the card, place Card II. If his drawing of previous card deviated much from the original, then instructions may be repeated, otherwise there is no need of repeating the instruction. Continue like this till card VI is presented. Present Card VI and ask the patient to "*read the instruction as written on the Card and do as required*" (कार्ड में जो लिखा है पढ़ो, फिर जो लिखा है वैसा करो) (Instructions on the card may be in two or three languages depending upon the regional preference, Card VII may be translated in Hindi as ईंट की शक्ल बनाओ, because it is difficult to translate cube in Hindi for low literates). Card VII and Card VIII may then be presented one by one. On card VIII if a digital clock is made then ask the subject to make a complete clock with arms. If it is not clear from his drawing, ask him which one is the little arm, at what number it is set and likewise ask him to show the long arm and at what number it is set (इसमें छोटी सुई कौन-सी है ? और कहाँ है ? और बड़ी सुई ?). Finally, ask him what time is shown therein.

### Precautions

1. Subject is not to be told whether he can use erasure (rubber) or not. If he asks 'can I use rubber', reply him as you please (like) (जैसा चाहो करो ।).

2. Instructions on the card is not to be read by the tester-except in case of low literate subjects
3. If he enquires about the accuracy of his copied design, ask him what you think. Do as precisely as you can ? (तुम्हें क्या लगता है । जितना ठीक हो सके बनाओ ।).
4. Do not pass any judgement about the accuracy of the drawings during the process of testing or at the end of the administration.
5. If the subject and his attendants wish to know about the outcome of the testing, say politely it all will be evaluated, may take little more time before any conclusion is drawn.

## Scoring

Scoring is based on the principle of 'all or none', i. e., either a design is correct or it is incorrect. Each incorrect drawing is given a score of one. There are eight designs so the maximum error score could be 8 and minimum *zero*. While scoring for incorrect drawing, keep in mind the educational and cultural background of the patient. A little departure in the copying design, if over all configuration is maintained, in an individual belonging to deprived culture (rural) able to read instruction with considerable difficulty, may be passed as correct. Over all size of the copying design, tremulous lines and space at joints are not considered meaningful for scoring except for overall configuration and proportions of parts.

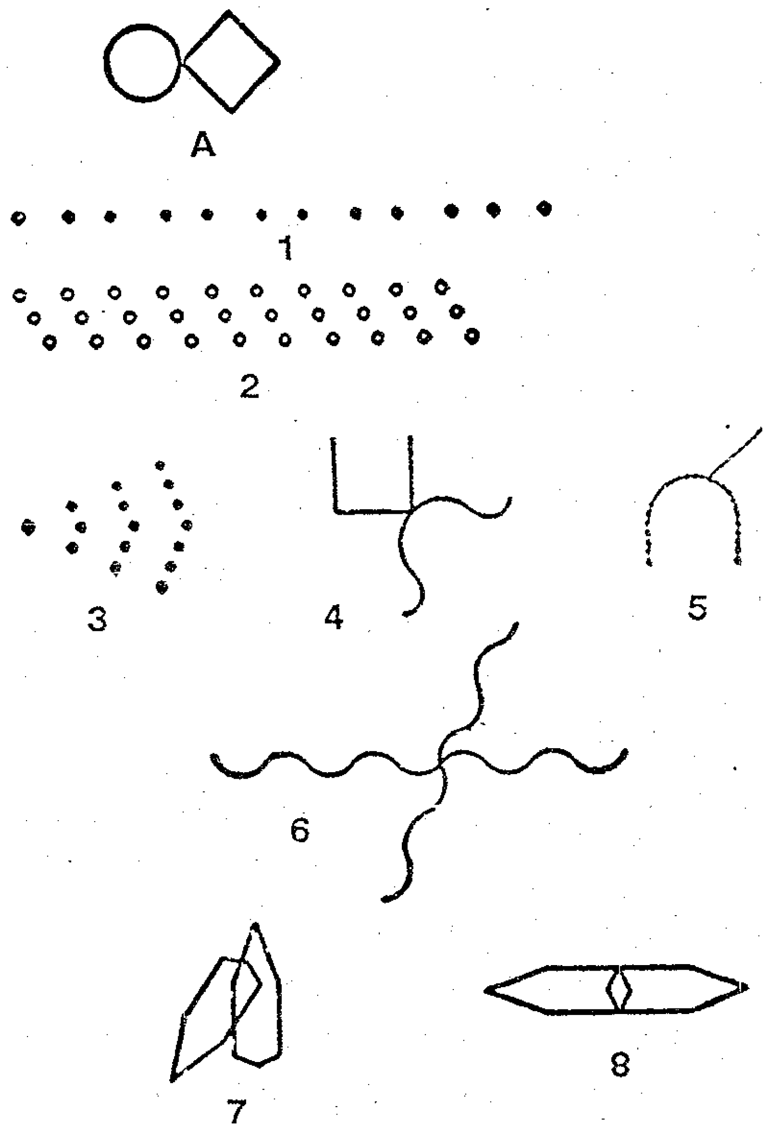
## Interpretation

| Educational | Error Score | Dysfunction Rating |
|---|---|---|
| Those who can read with considerable difficulty | 0 – 3 | 0 |
| | 4 – 5 | 2 |
| | 6 – 8 | 3 |
| Those who can read fluently | 0 – 2 | 0 |
| | 3 – 4 | 2 |
| | 5 – 8 | 3 |

## 5. BENDER VISUAL-MOTOR GESTALT TEST (BVMGT)

BVMGT test consists of nine figures characterized by their gestalt. It is not a test of visual memory / imagery, rather it is a test of perceptual and Visual motor functioning. Perceptual behaviour is regarded, here, as involving three broad stages – (1) sensory reception of the figures, (2) interpretation or organization of the lines (figures) at the central level of the nervous system and (3) motor performance of the figure (drawing). This whole process of reception, interpretation (consolidation or gestalt formation or organization) and execution, i.e., performance is distorted by brain injury, chemical imbalnce, toxicity, and degenerative processes of the brain cells and nervous systems. Thus, in case of brain dysfunction, emerging from any of the sources, the reproduction of the Bender's designs will be affected.

Based on Wertheimer's classical work in the field of perception, the Bender Visual Motor Gestalt Test was introduced by Bender in 1938, as a test of visuo-motor coordination. Soon, however, she discovered its utility as a psycho-diagnostic tool in the field of clinical psychiatry/psychology. This test since then (inception) is enjoying a high degree of acceptance amongst clinical psychologists both in India and abroad. It stands among first 10 psychological tests as far as its rssearch output is concerned, and its rank ranged between 1 and 5th in different surveys conducted separately as far as its use in Practice is concerned (Dubey, Pershad and Verma, 1981). The test has been used in various psychiatric conditions like neurosis, psychosis. delinquent, aggressive behaviour, maturational/developmental level, emotional disturbances, suicides etc., but most often it is preferred and used for the diagnosis of brain dysfunction.




**The Bender Visual-Motor Gestalt Test.**
**A Visual-motor Gestalt test and its clinical use.**
*Research Monograph,* **American Orthopsychiatric Association, 1938, No. 3**

**Material**

1. Nine cards of the Bender Visual Motor Gestalt Test.
2. A sheet of full-scape bond paper or any other good quality paper of similar size.

3. An erasure (rubber).
4. One ordinary HB pencial.

**Procedure**

Place a paper (in horizontal position), a rubber and a pencil on the table After making the patient comfortable give the following instructions :

"मेरे पास कुछ कार्ड हैं। मैं इन्हें आपको दिखाऊँगा। जो कुछ भी इनमें बना है, बिल्कुल वैसा ही कागज पर नकल करके बनाना होगा।"

"Here are a few cards (figures). I will show them one-by-one. You are required to copy them. Make your copy as much as alike the drawing on the card".

If subject asks any question, he should be replied properly in a noncommital language, for example, if he asks 'should I measure the length of the lines', 'should I count the dots'; Experimenter should answer, 'just make your drawing as much like this one as possible'. Do not instruct the subject, whether he can use the erasure (rubber) or not. If he asks, 'can I use the rubber', say as you please. Place the paper before the subject in a horizontal position, and the card just on the top of the paper and ask the subject to copy it. If he picks up the card, let him do, but if he turns the card put it again in a correct direction and when he places the card on the table, keep it again at the top of the paper in centre. Do not allow the subject to rotate the paper, however, if he changes his posture, let him do so. Remove the card when he gives an indication that he has copied it. If his copied design deviates markedly, ask him, "what is this" (indicating towards the copied design) ? If he names some object note it down on a separate paper. In any case, if copied design is not similar either in size or shape, than before presenting the next card repeat the instruction pointing towards the presented card say :

"जैसा इसमें बना है, बिल्कुल वैसा ही बनाना है।"

## Scoring

Here Hain's method (1964) of scoring is given which was also followed by Bhargava and Sandhu (1987). this is a simple and more quantitative procedure. In this method protocol is searched for the following 15 signs. Each sign is scored only once. Each sign is given a numerical weight. Each sign is scored on the basis of 'all or none', i. e., either a sign is observed in the protocol or is absent. There is no scope for a sign to be partially present. Numerical weight or score for a sign remains the same whether that particular sign is observed in only one design or in all the nine designs.

1. **Perseveration**–There are two types of perseveration 'intra' and 'inter'. Intra perseveration when a design is enlarged, i. e., when more than required number of dots or loops are made. Inter perseveration when image of the earlier card still persists inspite of the fact that another fresh card is presented. In such a case the earlier design is repeated by the subject. Give a score of 4 whenever there is perseveration of any type or of both types either on one or more cards.
2. **Rotation**–Whenever one or more complete designs are rotated more than 45 degrees from its axis, score 4. This sign is not to be scored if only one sub part of a design is rotated.
3. **Concretism**–Whenever there is marked deviation in the reproduction of the subject, he may be enquired as to what he has drawn, if he names the objects like: bow, bird, tree, etc., then this sign is to be scored and a numerical value of 4 may be given.
4. **Added angles**–This sign is scored when an extra angle is added on cards A, 4, 7 and 8. Give a numerical score of 3.
5. **Separation of lines**–When a line appears to be broken and forms two separate lines which do not touch each other, give a score of 3.

6. **Overlap**—This is to be scored when a part of the design runs into the 'space' or 'area' of another design. If overlap is present give a score of 3.
7. **Distortion**—Score it when there is a destruction of the gestalt of any one or more designs leading to an extreme departure from the stimulus card. Give a weight of 3 points.
8. **Embellishments**—When extra meaningless small line is added in the design, this sign is scored as 2.
9. **Partial Rotation**—When the whole design is rotated 20 to 45 degrees from its axis or any of the sub part is tilted more than 20 degrees ; this sign is considered to be present, and a score of 2 is given.
10. **Omission**—Score one point when either of the two sub-parts of a design is omitted.
11. **Abbreviation**—Score one, when number of dots or circles are reduced in reproduction.
12. **Separation**—Score one, when there is a separation between subparts of a design.
13. **Absence of Erasure**—Score one, when there is a failure to attempt to erase the errors. If there was no error, there was no need to erase, in such condition this sign is not to be scored. If the subject has used erasure on any of the designs and has not used on others, even then this sign is not to be scored.
14. **Closure**—Score one when in any of the designs more than one angle is not closed.
15. **Point of Contact**—Score one point, when on design 'A' the square and circle do not touch each other adequately.

**Range of the Scores**

Maximum = 34

Minimum = 0

**Interpretation**

| Raw Score | Dysfunction Rating |
|---|---|
| 0–4 | 0 |
| 5–8 | 2 |
| 9–34 | 3 |

**Scoring Sheet**

| Sign | Score |
|---|---|
| Perseveration (intra/inter) | 4 |
| Rotation or Reversal | 4 |
| Concretism | 4 |
| Added angles | 3 |
| Separation of lines | 3 |
| Overlap | 3 |
| Distortion | 3 |
| Embellishments | 2 |
| Partial Rotation | 2 |
| Omission | 1 |
| Abbreviation | 1 |
| Separation | 1 |
| Absence of Erasure | 1 |
| Closure | 1 |
| Point of Contact | 1 |

**Total Score**
**(out of 34)**

# PART III

# A WORKED-UP CASE

## MATERIAL FOR P. G. I. BATTERY OF BRAIN DYSFUNCTION

PGI Battery of Brain Dysfunction consists of following materials :

1. 10 Page test blank for recording responses.
2. 5 cards for Visual Retention Sub-test of memory (Item IX).
3. 2 cards of picture of common objects—one with ten objects and 2nd with 20 objects for Recognition sub-test of memory (Item X).
4. Kohs' Block Design test as used in Bhatia's Battery (10 cards and 16 cubes)
5. Pass-A-Long Test as used in Bhatia's Battery (8 cards, Blocks and Boxes).
6. 8 cards of Nahor-Benson Test.
7. 9 cards of Bender Visual Motor Gestalt Test.
8. Handbook of the PGI Battery of Brain Dysfunction—for (1) the items of Verbal Adult Intelligence Scale and (2) for the procedure of Administration, scoring and interpretation of each test separately and jointly.

## RESPONSE SHEET OF PGI-BBD

**Dwarka Pershad** *(Chandigarh)*
**S. K. Verma** *(Chandigarh)*

नाम (Name) –

आयु (Age) – **32** लिंग (Age) – **F**

शिक्षा (Education) – **B. A.** दिनांक (Date) – **10.11.88**

| Sr. No. | Test Variables ⟶ Subtests ↓ | Dysfunction Rating Score | | |
|---|---|---|---|---|
| | | 3 | 2 | 0 |
| 1. | Remote Memory | 0–2 | 3–4 | 5+ |
| 2. | Recent Memory | 0–2 | 3–4 | 5+ |
| 3. | Mental Balance | 0–2 | 3–4 | 5+ |
| 4. | Attention and Concentration | 0–2 | 3–4 | 5+ |
| 5. | Delayed Recall | 0–2 | 3–4 | 5+ |
| 6. | Immediate Recall | 0–2 | 3–4 | 5+ |
| 7. | Retention for Similar Pairs | 0–2 | 3–4 | 5+ |
| 8. | Retention for Dissimilar Pairs | 0–2 | 3–4 | 5+ |
| 9. | Visual Retention | 0–2 | 3–4 | 5+ |
| 10. | Recognition | 0–2 | 3–4 | 5+ |
| 11. | $*\frac{P}{K} \times 100$ | 251+ | 200–250 | upto 199 |
| 12. | Performance Quotient | 60 | 61–80 | 81+ |
| 13. | T. Q. on Information | 60 | 61–80 | 81+ |
| 14. | T. Q. on Digit Span | 60 | 61–80 | 81+ |
| 15. | T. Q. on Arithmetic | 60 | 61–80 | 81+ |
| 16. | T. Q. on Comprehension | 60 | 61–80 | 81+ |
| 17. | Difference in P. Q. and V. Q. | 24+ | 15–23 | upto 14 |
| 18. | Nahor and Benson Test | 6+ | 4–5 | 0–3 |
| 19. | Bender-Gestalt Test | 9+ | 5–8 | 0–4 |
| | **TOTAL** | **6 × 3 = 18** | **3 × 2 = 6** | **10 × 0 = 0** |

Remarks

Dysfunction Rating Score = **24**

**National Psychological Corporation**
**4/230, Kacheri Ghat, AGRA – 282 004 (U.P.) INDIA**

[2]

## SUB-TEST – I : MEMORY SCALE

**I. पूर्वकालीन स्मृति (Remote Memory)**

√ 1. आपकी उम्र कितनी है ? ........32 वर्ष........

√ 2. आपका जन्म कहाँ हुआ ? ........लखनऊ (उ.प्र.)........

3. (a) आपकी शादी कब हुई ? ........N. A........

(b) आपने नौकरी या व्यवसाय करना कब से चालू किया ? ........N. A........

3 × (c) आपने पढ़ना कब छोड़ा या हाई स्कूल कब पास किया ? ........याद नहीं........

√ 4. आपके सबसे छोटे बच्चे या भाई, बहन की उम्र कितनी है ? ........12 साल........

× 5. आप इस विभाग (अस्पताल) में पहली बार अपने इस इलाज के लिये कब आये ? ........बहुत दिन हो गए........

× 6. पिछली बार आप इस विभाग (अस्पताल) में कब आये थे ? ........याद नहीं........

**II. वर्तमान स्मृति (Recent Memory)**

× 1. कल आपने रात के खाने में क्या खाया ? ........याद नहीं........

√ 2. आज सुबह आपने नाश्ते में क्या खाया ? ........परौंठा, चाय........

2 × 3. इस महीने का क्या नाम है ? ........अक्टूबर 87........

√ 4. आज कौन सा दिन है ? ........वीरवार........

× 5. कल आपसे कौन-कौन मिलने आया या कल आप किससे मिलने गये ? ........कोई नहीं........

**III. मानसिक सन्तुलन (Mental Balance)**

6 | 3 √ 1. A, B, C, से Z तक (Alphabet of any language) बोलें। **Correct 10 Seconds**

3 √ 2. 20 से 0 तक उल्टी गिनती बोलें। **Correct 12 Seconds**

0 × 3. 40 से तीन घटाते हुये उल्टी गिनती बोलें। **37, 36, 33, 30, 28, 25, 22, 17, 14, 11, 8, 5, 2**

**IV. ध्यान एवं एकाग्रता (Attention and Concentration)**

| (1) मैं कुछ अंक पढ़कर सुनाने वाला हूँ। ध्यान से सुनें। जब मैं अंक पढ़ चुकूँ तो आप उन अंकों को दुहरायेंगे। | | (2) अब मैं कुछ और अंक पढ़कर सुनाने वाला हूँ; परन्तु ये अंक आपको विरुद्ध क्रम में (अन्त से प्रारम्भ करके) सुनाने होंगे, जैसे - यदि मैं कहूँ '2-5' तो आप कहें '5-2'। | |
|---|---|---|---|
| √ 1. 5–7–3 | √ 4–1–7 | √ 1. 8–5 | √ 2–8 |
| √ 2. 5–3–8–7 | × 6–1–5–8 | √ 2. 4–3–7 | √ 8–5–1 |
| √ 3. 1–6–4–9–5 | √ 2–9–7–6–3 | × 3. 8–5–6–3 | √ 3–7–5–9 |
| × 4. 3–4–1–7–9–6 | √ 6–1–5–8–3–9 | × 4. 4–7–2–9–1 | × 9–2–5–8–4 |
| × 5. 7–2–5–9–4–8–3 | × 4–7–1–5–3–8–6 | 5. 2–5–9–4–8–9 | 7–1–5–3–9–6 |
| 6. 4–7–2–9–1–6–8–5 | 9–2–5–8–3–1–7–4 | 6. 3–5–8–6–1–9–2 | 6–3–7–1–4–8–5 |
| | | 7. 8–5–2–3–6–1–9–4 | 2–8–4–5–9–7–1–3 |

**DF = 6** **DB = 4** = 10

[3]

**V. समयान्तरित स्मरण (Delayed Recall)**

मैं कुछ चीजों के नाम पढ़ूँगा, आप ध्यान से सुनें और जब मैं आपसे उन चीजों के नाम दुहराने के लिये कहूँ तो आप उन चीजों के नाम बतलावें जिनको मैंने पढ़ा था। (एक नाम प्रति सैकण्ड की रफ्तार से पढ़ें और एक मिनट बाद सब नामों को दुहराने के लिये कहें)

| 1 | 2 |
|---|---|
| √ छाता | √ मछली |
| √ फूल | √ लैम्प |
× घड़ी | √ रुपया |
| √ तस्वीर | √ ताज |
| √ पैन्सिल | √ खिलौना |
| 4 | 5 |

9

**VI. तत्कालीन स्मरण (Immediate Recall)**

मैं एक-एक करके कुछ वाक्य पढ़ूँगा, आप ध्यान से सुनें, जब मैं बोल चुकूँ तो आपको वही वाक्य दोहराना होगा।

9

1. राम कुर्सी से उठा, दरवाजा खोला और बाजार चला गया। 3
2. रोगी को मेज पर लिटाया, उसको देखा, दवा लिखी और कल आने के लिये कहा। 3
3. मोहन के घर पानी नहीं था, उसने बाल्टी उठाई, बाजार के नल पर गया, पानी भरा और वापिस लौट आया। 3

**VII. समान युग्मों का शाब्दिक धारण (Verbal Retention for Similar Pairs)**

मैं कुछ चीजों के जोड़े बताऊँगा। आप ध्यान से सुनें। इन जोड़ों में से जब एक चीज का नाम लिया जाये तो आपको जोड़े की दूसरी चीज का नाम बतलाना होगा। (दो सैकण्ड प्रति युग्म की रफ्तार से पढ़ें। प्रत्येक जोड़े के बाद लगभग 5 सैकण्ड का अन्तर रखें, अन्तिम युग्म के बाद लगभग 10 सैकण्ड का अंतर रखकर जोड़े की प्रथम वस्तु का नाम पढ़ें और इसके साथ की दूसरी चीज का नाम बतलाने के लिये कहें।)

1. पेड़ ..... फूल √
2. मीठा ..... नमकीन √
3. आदमी ..... औरत √ 5
4. दिन ..... रात √
5. काला ..... सफेद √

**VIII. असमान युग्मों का शाब्दिक धारण (Verbal Retention for Dissimilar Pairs)**

निर्देशन एवं परीक्षण प्रशासन उपर्युक्त भाँति रहेगा, लेकिन युग्म उनके सामने लिखे क्रम में पूछें। यदि प्रयोज्य सन्तोषजनक उत्तर न दे पाये तो उत्तर बतलाकर पुनः दूसरे युग्म का उत्तेजक शब्द बोलें; इस प्रकार तीन प्रयास करवायें। अगर प्रयोज्य प्रथम प्रयास में ही सब सही उत्तर दे तब भी दो प्रयास और करवाने अनिवार्य हैं।

| | | | | |
|---|---|---|---|---|
| मेज | काला | 4 × | 2 × | 1 × |
| पेड़ | ऊँचा | 2 × | 1 √ | 5 √ |
| लैम्प | खुरदरा | 1 × | 5 √ | 3 √ |
| बच्चा | कड़वा | 3 √ | 4 √ | 2 √ |
| सपना | गहरा | 5 √ | 3 √ | 4 √ |
| | | 2 | 4 | 4 |

10

[4]

## IX. दृष्टिक धारण (Visual Retention)

मैं आपको एक-एक करके कुछ कार्ड दिखाऊँगा, आप अच्छी तरह से देखें । कुछ देर (15 सैकण्ड) बाद में कार्ड उठा लूँगा और फिर जब मैं आपसे कहूँ (30 सैकण्ड बाद) तब आप वैसा ही चित्र कागज पर बनायेंगे जैसा कि आपने देखा था । (एक कागज, एक पेन्सिल तथा एक रबड़ प्रयोज्य को दें लेकिन यह न कहें कि रबड़ का प्रयोग कर सकता है या नहीं )

| | | |
|---|---|---|
| 1) | + − + − + − + | 2 |
| 2) | ○ △ ○ △ ○ △ | 2 |
| 3) | △ ▭ △ ▭ △ ▭ | 1 |
| 4) | ○ = × ○ = × ○ = × | 0 |
| 5) | ▭ △ + △ ▭ | 3 |
| | | 8 |

## X. प्रत्यभिज्ञान (Recognition)

मैं आपको एक कार्ड दिखलाऊँगा, जिसमें कुछ चीजें बनी हुई हैं । इन्हें आप ध्यान से (30 सैकण्ड तक) देखें ; कुछ देर (2 मिनट) बाद मैं आपको दूसरा कार्ड दिखलाऊँगा जिसमें से आपको पहले देखी हुई चीजें पहचान कर उनके नाम बतलाने होंगे । (प्रयोज्य को यह न बतलावें कि पहले वाले कार्ड में कितनी चीजें थी तथा अभी कितनी चीजें उसे और बतलानी हैं, बतलाये हुये नामों को नीचे लिखें)

**Responses :**

**Bed, Lock, table, Pen, Comb, fan**

**Book, Chair**

7 − 1 = 6

### SUMMARY OF MEMORY SCALE

| Sub-tests | I | II | III | IV | V | VI | VII | VIII | IX | X | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Raw Scores | 3 | 2 | 6 | 10 | 9 | 9 | 5 | 10 | 8 | 6 | |
| Converted Scores | 1 | 1 | 1 | 5 | 5 | 5 | 5 | 3 | 3 | 1 | |

[5]

## SUB-TEST – II : BATTERY OF PERFORMANCE TESTS OF INTELLIGENCE

| KOHS' BLOCK | | | PASS-A-LONG | | | Remarks |
|---|---|---|---|---|---|---|
| Item No. | Time taken | Score | Item No. | Time taken | Score | |
| $B_1$ | 55 Seconds | 3 | $B_1$ | 40" | 3 | $\frac{P}{K} \times 100 = \frac{22}{6} \times 100 = 367$ |
| $B_2$ | 70 " | 2 | $B_2$ | 45" | 3 | |
| $B_3$ | Fail / Demonstrated | – | $B_3$ | 50" | 3 | |
| $B_4$ | 110" | 1 | $B_4$ | 60" | 3 | |
| $B_5$ | Failed / Demonstrated | – | $B_5$ | 25" | 6 | |
| $B_6$ | Failed | – | $B_6$ | 85" | 4 | |
| $B_7$ | | | $B_7$ | 185" | 0 | |
| $B_8$ | | | $B_8$ | Failed | – | |
| $B_9$ | | | | | | |
| $B_{10}$ | | | | | | |
| | Total Score | 6 | | Total Score | 22 | $\frac{73 + 119}{2}$ |
| | T.Q. = | 73 | | T.Q. = | 119 | **= 96** |

Performance Quotient (P.Q.) = 96

[ 6 ]

## SUB-TEST – III : VERBAL ADULT INTELLIGENCE SCALE (V A I S)

### 1. INFORMATIONS

Discontinue after 7 consecutive failures response.

√ 1. गेंद गोल
√ 2. साल 12
√ 3. थर्मामीटर बुखार देखने का
√ 4. सूरज पूर्व से
√ 5. प्रधानमंत्री राजीव, नेहरू, इन्दिरा जी
√ 6. गुरु नानक/गालिब शायर
√ 7. झण्डा लाल / हरा / सफेद
√ 8. लोड़ी 13 जनवरी को
√ 9. रामायण बाल्मीकि
√ 10. कुरान/बाइबिल मुसलमानों की धार्मिक पुस्तक
√ 11. औरत-कद 5 फुट
× 12. आजाद 26 जनवरी
√ 13. बीरवल अकबर का मंत्री
√ 14. गाँधी जयन्ती 2 अक्टूबर
√ 15. रबड़ प्रेड़ों से
√ 16. ताजमहल शाहजहाँ ने
× 17. जनसंख्या 60 करोड़
√ 18. त्रिवेणी इलाहाबाद
× 19. बम्बई गुजरात
× 20. दिशा—दिल्ली-मद्रास पता नहीं
× 21. दही खट्टा लगाने से दूध में
× 22. गहरे रंग के कपड़े पता नहीं
× 23. साल-सप्ताह 48
√ 24. ताप 100°c
× 25. सीलोन लंका
× 26. इंग्लैण्ड पता नहीं
× 27. गीता अच्छे काम करो
× 28. मेघदूत No Response
× 29. महाभारत "
× 30. वेद चार हैं
× 31. दिल्ली-बम्बई 500 कि.मी.
32. लोक-सभा
33. नसें

TOTAL SCORE = 17

### 2. DIGIT SPAN

Discontinue after failure on BOTH TRIALS of any item

DIGITS FORWARD SCORE = 6

DIGITS BACKWARD SCORE = 5

TOTAL SCORE = 11

### 3. ARITHMETIC

Discontinue after 5 consecutive failures.

| Problem | Socre 1 or 0 | |
|---|---|---|
| 1. | √ | |
| 2. | √ | |
| 3. | √ | 9 रुपये |
| 4. | √ | 4 " |
| 5. | √ | 1.50 |
| 6. | × | No Response |
| 7. | √ | 36 |
| 8. | √ | 30 इंच |
| 9. | × | 6 घण्टे |
| 10. | × | 11.50 |
| 11. | × | करीब 2 रु. की |
| 12. | × | No Response |
| 13. | × | " |
| 14. | × | 585 |
| 15. | × | 16 |

TOTAL SCORE = 7

[6]

## SUB-TEST – III : VERBAL ADULT INTELLIGENCE SCALE (V A I S)

### 1. INFORMATIONS

Discontinue after 7 consecutive failures response.

√ 1. गेंद **गोल**
√ 2. साल **12**
√ 3. थर्मामीटर **बुखार देखने का**
√ 4. सूरज **पूर्व से**
√ 5. प्रधानमंत्री **राजीव, नेहरू, इन्दिरा जी**
√ 6. गुरु नानक/गालिब **शायर**
√ 7. झण्डा **लाल / हरा / सफेद**
√ 8. लोड़ी **13 जनवरी को**
√ 9. रामायण **बाल्मीकि**
√ 10. कुरान/बाइबिल **मुसलमानों की धार्मिक पुस्तक**
√ 11. औरत-कद **5 फुट**
× 12. आजाद **26 जनवरी**
√ 13. बीरवल **अकबर का मंत्री**
√ 14. गाँधी जयन्ती **2 अक्टूबर**
√ 15. रबड़ **पेड़ों से**
√ 16. ताजमहल **शाहजहाँ ने**
× 17. जनसंख्या **60 करोड़**
√ 18. त्रिवेणी **इलाहाबाद**
× 19. बम्बई **गुजरात**
× 20. दिशा—दिल्ली-मद्रास **पता नहीं**
× 21. दही **खट्टा लगाने से दूध में**
× 22. गहरे रंग के कपड़े **पता नहीं**
× 23. साल-सप्ताह **48**
√ 24. ताप **100°c**
× 25. सीलोन **लंका**
× 26. इंग्लैण्ड **पता नहीं**
× 27. गीता **अच्छे काम करो**
× 28. मेघदूत **No Response**
× 29. महाभारत **"**
× 30. वेद **चार हैं**
× 31. दिल्ली-बम्बई **500 कि.मी.**
32. लोक-सभा
33. नसें

**TOTAL SCORE = 17**

### 2. DIGIT SPAN

Discontinue after failure on BOTH TRIALS of any item

DIGITS FORWARD SCORE = **6**

DIGITS BACKWARD SCORE = **5**

TOTAL SCORE = **11**

### 3. ARITHMETIC

Discontinue after 5 consecutive failures.

| **Problem** | **Socre 1 or 0** | |
|---|---|---|
| 1. | √ | |
| 2. | √ | |
| 3. | √ | **9 रुपये** |
| 4. | √ | **4 "** |
| 5. | √ | **1.50** |
| 6. | × | **No Response** |
| 7. | √ | **36** |
| 8. | √ | **30 इंच** |
| 9. | × | **6 घण्टे** |
| 10. | × | **11.50** |
| 11. | × | **करीब 2 रु. की** |
| 12. | × | **No Response** |
| 13. | × | **"** |
| 14. | × | **585** |
| 15. | × | **16** |

**TOTAL SCORE = 7**

[7]

| 4. **COMPREHENSION**<br>Discontinue after 6 failures | Response | **Score 2, 1 or 0** |
|---|---|---|
| √ 1. इंजन | खींचने को | |
| √ 2. कपड़े | साफ करने को | |
| ×*3. सेहत | खूब खाओ पियो | |
| √ 4. बुरी संगति | बुरी आदतों से बचाने के लिए | |
| √ 5. सिनेमा घर | बुझाना चाहिए | |
| √ 6. शहर की भूमि | जमीन कम होती है खरीदने वाले ज्यादा | |
| √ *7. खाने की चीज | स्वाद लगती है जल्दी पचती है | **Difference of V.Q. & P.Q.**<br>88 – 96 = 8 |
| × 8. उधार | दोस्त ब्याज ज्यादा लेते हैं | |
| √ 9. लिफाफा | जिसका हो उसे पहुँचा दो। | |
| √*10. चैक | सुविधा रहती है। | |
| √ 11. टैक्स | स्कूल / अस्पताल चलाने को | |
| × 12. अकेला चना | एक चने से भाड़ नहीं टूटता | |
| √ 13. थोथा चना | ज्यादा बोलना | |
| × 14. जंगल में रास्ता | पता नहीं, किसी से पूछ कर | |
| ×*15. बच्चा-कानून | छोटे होते हैं | |
| × 16. गाड़ी लाइसेंस | सरकार का नियम है | |
| √ 17. बहरा व्यक्ति | सुनकर ही बोलना सीखते हैं | |
| × 18. लोहा-गर्म | गर्म होकर जल्दी मुड़ता है | |
| **Total** 17 | **(Max. 36)** | |

* If the subject replies with only one idea, ask for a second response. Rephrase the test item appropriately, saying, "Tell me another reason why (another thing people can do)

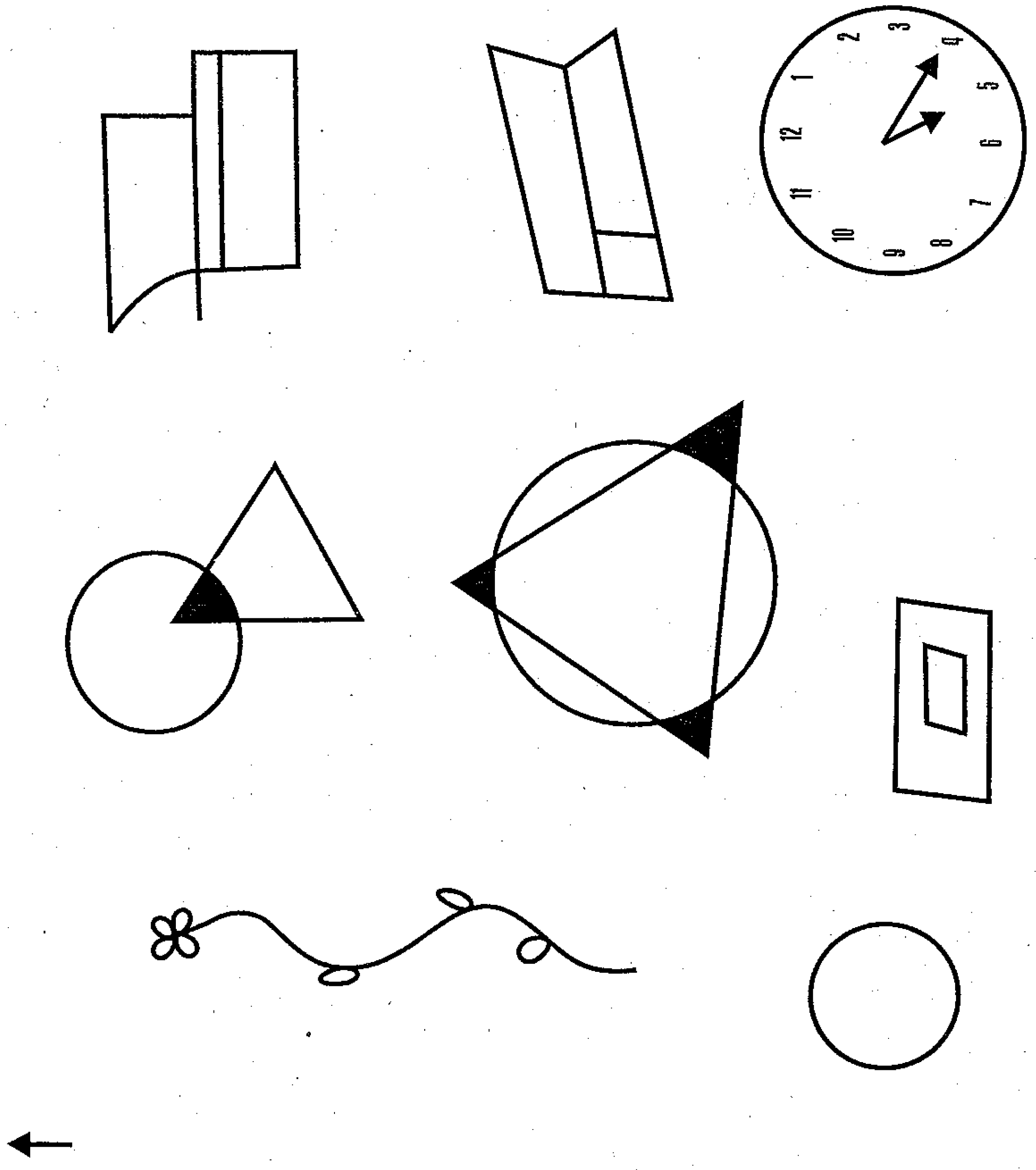
## SUMMARY OF V A I S

| Sub-tests | Information | Digit Span | Arithmetic | Comprehension |
|---|---|---|---|---|
| Raw Scores | 17 | 11 | 7 | 17 |
| Test Quotients | 94 | 101 | 76 | 81 |

$\frac{352}{4}$ Verbal I. Q. = **88**

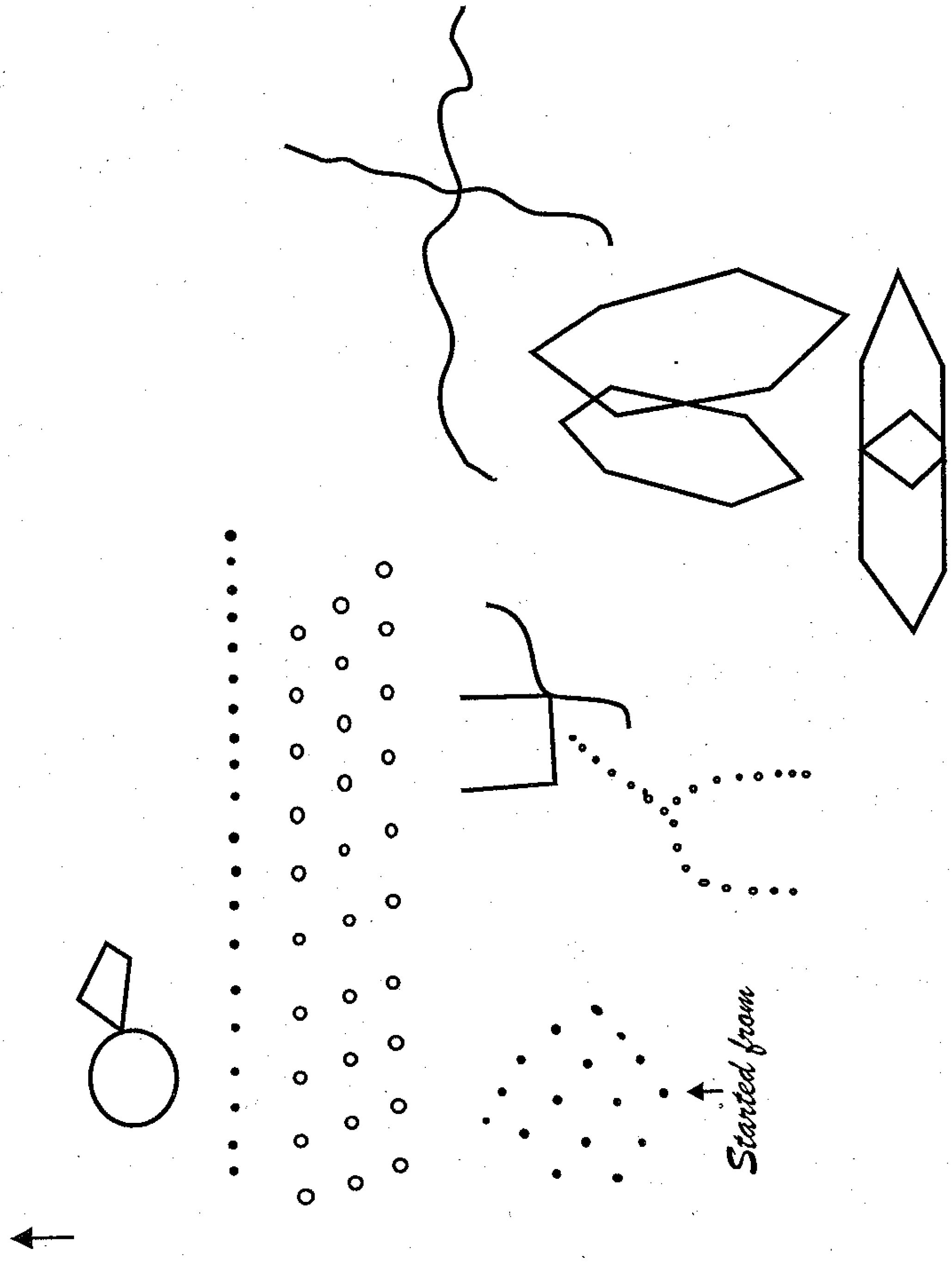
[ 8 ]

## RESPONSE FOR NAHOR BENSON TEST



[ 9 ]

## RESPONSE FOR BENDER VISUAL MOTOR GESTALT TEST



[ 10 ]

## SUB-TEST – IV : NAHOR BENSON TEST

| Item No. | I | II | III | IV | V | VI | VII | VIII | Total Score |
|---|---|---|---|---|---|---|---|---|---|
| Figures wrongly reproduced | √ | √ | √ | × | × | √ | √ | √ | 2 |

## SUB-TEST – V : BENDER GESTALT TEST

| | Score |
|---|---|
| – Perserveration | 4 |
| – Rotation or Reversal | 4 |
| – Concretism | 4 |
| – Added angles | 3 |
| – Separation of lines | 3 |
| – Overlap | 3 |
| – Distortion | 3 |
| – Embelishments | 2 |
| – Partial Rotation | 2 |
| – Omission | 1 |
| – Abbreviation | 1 |
| – Separation | 1 |
| – Absence of erasure | 1 |
| – Closure | 1 |
| – Point of contact | 1 |
| **TOTAL SCORE** | **14** |

## A WORKED-UP CASE

Miss 'X', 32 years old, B.A., unmarried was referred for psychometric evaluation for differential diagnosis (organic brain pathology). She had good academic record throughout. PGI Battary of Brain Dysfunction was administered on her. There was no marked oddity in her behaviour.

Detailed record of her testing is given in the beginning of this section with a view to appraise the user of the bettary as to how to write the responses, and scores on the record form (test blank) and how to interpret the various cognitive functioning and to rate them for the patient's cognitive 'Dysfunction Rating Score.

**Memory**

Since the patient is unmarried and had never undertaken any remunerative job, questions 3a and its alternative 3b were not applicable (NA) for her, therefore the next alternative 3c was asked. She could not recall, inspite of her best efforts, the year of leaving the college (sub test I). On this subtest only three of her responses, out of six were correct, thus her score became 3.

Likewise on subtest II her two responses were correct giving her a score of two (out of 5).

On subtest III she recited Alphabet of English correctly within 10 seconds (score = 3). She also recited 20 backward to 0—all correct within 12 seconds (score = 3). On the third item of this subtest, i. e., 40 minus 3 subtest she made three mistakes (mistakes are underlined in the test blank). Since she made mistakes to complete the task time was not required to be considered for scoring. She scored zero because mistakes exceed. Total score on this subtest for this patient thus became 3 + 3 + 0 = 6.

On subtest IV, patient could repeat 6 digits forward and 4 digits backward correctly thus her score became 6 + 4 = 10.

On subtest V, she could recall only 4 names on the 1st list after an interval of 60 seconds, and all the five from the 2nd list. Score on this test thus became 4 + 5 = 9.

On subtest VI, she recalled all the three clauses of sentence 1 correctly. On second sentence her last three clauses were correct whereas on first clause she substituted word बैठाया for लिटाया. Thus first clause is marked as wrong. On third sentence her first and last two clauses were correct. She omitted words उसने on second clause and केवल पर on 3rd clause. Thus she scored 3 + 3 + 3 = 9 on this test (out of 3 + 4 + 5 = 12).

On subtest VII, she recalled all the five response words correctly following stimulus words. Thus scored full 5 points.

On subtest VIII she could recall 2 response words correctly on first trial, 4 each on 2nd and 3rd trial, obtaining a score of 2 + 4 + 4 = 10 (out of 5 + 5 + 5 = 15).

On subtest IX, she reproduced designs (elements) of card I and II correctly setting maximum score of 2 on each. On card III, she replaced the dimond by triangle which can be considered perseveration of one of the elements of the earlier card. Number and position of rectangles were correct for which score of one is awarded on this card. On card 4 instead of making 0 = x 0 = x she made it 0 = x 0 = x0 = x, thus, there was perseveration of all the elements and the reproduction was considered wrong and zero score was given. On card 5 she forgot to make a design inside the rectangles, otherwise her reproduction was correct. Because of this omission score of 1 was deducted from the maximum score of 4 on this card. Her total score on this test thus became 2 + 2 + 1 + 0 + 3 = 8 (out of 2 + 2 + 2 + 3 + 4 = 13).

On subtest X, she identified 8 objects as seen earlier, out of which one was misidentified (table identified as having seen in the exposed card). Thus she recognized seven correct and one incorrect making a score of 6 (7 − 1 = correct − incorrect) out of 10.

All the scores are written on test blank in the table of summary of Memory Scale. These raw scores were then converted into weighted scores tables - 15 for appropriate educational level and written in the summary table in the column converted scores' whenever raw score falls in 0 - 2 quintile give a score of 1, for 2 – 4 a score of 3, for 4 to 6 a score of 5 or 5 +. These 1, 3, 5 are the midpoints or averages of 1st, 2nd and 3rd quintile. The converted scores for 10 subtests are then located on first page of the test blank and are encircled (or ticked). In present example a Dysfunction rating of 3 is given to four subtests, i. e., 1, 2, 3, and 10 rating of two to 2 subtests, i. e., 8 and 9 and rating of zero to remaining 4 subtests, i. e., 4, 5, 6 and 7.

**Performance Intelligence Tests**

Subject showed her inability to complete the $B_3$ item of Kohs' Block Design test after trying for one minute. She was, asked to try again with an assurance that she would be able to make a design similar to that of the design shown in the card. If she had produced the task (No encouragement was, however, needed if she had produced a wrong design saying that she had made one similar to that of the card). On the expiry of two minutes when she was unable to complete it correctly, the solution of the design was demonstrated slowly by the tester, giving her ample opportunity to watch. Thereafter, card $B_4$ was presented to her, blocks were reshuffled and she was asked to make a design with the help of 4 blocks similar to that of the card. On seeing her in motion the stop watch was started. On her failing on two consecutive designs this test was discontinued. Each of the items completed correctly were scored according to the time taken by her consulting scoring procedure given on page number 63.

Likewise Pass-A-Long test was administered. Six items B1 to B6 were passed by her within the respective time limits. On B7 she took more than maximum permissible time. Since she was trying to complete this item with interest she was not stopped even after 180 seconds which was maximum limit, however, a score of zero was assigned on this test. Since she completed this design herself, there was no necessity of demonstrating it again and design B8 was directly presented which she could not complete correctly. Scoring for all the items completed successfully was done on the basis of time required for each as given on page number 63.

Total scores on 'Kohs' Block Design Test' and 'Pass-A-Long Test' were 6 and 22 as mentioned in the Test Blank. These raw scores were then converted into T. Q. (Test Quotient) as given in table No. P. Q. VI (right hand side for female) which gave TOs of 73 and 119 for two tests respectively. The P. Q. (Performance Quotient = Performance I. Q.) was calculated by computing average of two test Quotients which came to be 96 $= \frac{(73+119)}{2}$ This figure was located on the first page of the test blank (test proforma) at S. No. 12. P. Q. of this patient (96) is located in the last column which had a dysfunction rating of zero. For the variable at S. No. 11 on this page ratio of raw score on Pass-A-Long and Kohs' Block Design tests was determined and the product was multiplied by 100 to get rid of decimal points. This was found to be 367 $= \left(\frac{22}{6} \times 100\right)$ in the present example which is located in the first column as 251 + under dysfunction rating score of 3 and encircled.

**Verbal Adult Intelligence Scale**

Items of *information subtest* were put to her one by one and her answers were noted down in the space provided on Page 5 of the test blank (this test could be started from item 5 also if tester assumed responsibility that the subject is capable of answering four unasked questions. In this case a tick mark is to be put on each of the four questions). Each answer is to be evaluated as pass or fail consulting the model answers provided on page 74 A tick is put against each item passed and a cross for fail. After seven consecutive failure test was discontinued. All the ticks were counted that constituted raw score for this test which is 17 in the present example.

*Digit span test* was not administered here again because it had already been administered under memory testing. The scoring procedure, however, is different for this subtest for the assessment of intelligence. While considering it for intelligence both the sets of the same length of digits were administered. A tick mark was put against a series when reproduced in sequence correctly, separately for both the alternates. When she failed on both alternates of the same length, test was discontinued. In the present example scores for digit forward was 6 and for digit backward was 5 (count the ticks both for digit forward and backward). The total score on this subtest thus became 11.

*Arithmetic subtest* was started from item 3 instead of item 1 assuming that the subject would pass items 1 and 2 if she could answer correctly on items 3 and 4. This assumption holds true here and a tick was put on item numbers 1 and 2 without writing the answers against these items. For items 6, 12, 13, she did not gave any response, even on repeating the items and waiting nearly for minute for each item. Neither she appeared to be working for the solution. Answers that did not tally with the model responses were crossed. Ticks were counted which constituted score for this subtest. Score on this test is 7 out of 15 in the present case.

*Comprehesion* subtest poses a little problem of eliciting responses and their scoring. Items 3, 7, 10 and 15 which are star marked, required two good reasons to get full credit of two points. On item 3, her response was 'eat and drink' which did not correspond to any of the concepts given in the model answers, hence a cross mark was put. On item 7, two concepts were given by the subject and both were right therefore, double tick was put on this item (indicating item is scored as two). On 10th item subject gave only one concept and therefore she was asked to tell another reason, but could not, hence one tick was put on this item meaning therefore that item is given only one point. Response of item 15 was wrong inspite of asking her to tell more about it. On remaining items, completely abstract and properly worded answers were given two score (two ticks were marked on items), partly abstract or poorly worded items were given one socre (one tick). Whenever spontaneous answer qualified for one score patient was put a question like, tell me more about it', 'can you tell me how' etc. All ticks were counted and that constituted the score. In the present example her score is 17 on this test.

Raw scores of all the four subtests were entered in the space provided on page 7 of the test blank under summary of VAIS. Test Quotients (T. Q.) were written below the raw score for each subtest consulting the appropriate table of norms which are numbered VQ1 to VQ15. In the present example TQs were arrived from table No. VQ6 (right hand side for female). These test quotients provided information for variable No. 13, 14, 15 and 16. On Page No. 1 of the Proforma. T. Qs on Information, Digit Span and Comprehension (Variables 13, 14 and 16) tests got dysfunction rating of zero and the numericals 81 + are encircled for each of these variable in the last column. T. Q. on arithmetic (variable No. 15) is

76 which comes in the middle column of the Page 1 under Dysfunction Rating Score of 2.

For the variable no. 17, i.e., diference between Verbal and Performance Quotients, just the difference is considered which in the present example is 8 (VQ = 88 and PQ 96). Difference of eight points was located in the last column under dysfunctions Rating Score of zero on Page 1 of the Test Blank as upto 14. This numerical value was encircled.

**Nahor Benson Test**

On this test, design number 1, 2 and 3 were copied correctly. Designs 4 and 5 (with depth perception) were wrong. Designs 6, 7 and 8 were sketched correctly as per instructions imprinted on the cards. Only 2 designs are wrong out of 8 designs. Therefore, her error score on this test becomes 2. For the error score of 2 on this subtest a dysfunction rating score of zero is given on test blank page No. 1 and the value given on the test blank was encircled.

**Bender Visual Motor Gestalt Test**

Drawings as copied by the subject on 'Bender Visual Motor Gestalt Test' are given on page 9 of the test blank and detailed scoring for different types of errors are given on page 10 of the test blank. An error was counted only once whether it was observed on one design or on all the designs. Her drawings suggested that there was perseveration of drawings of card 1 (19 dots instead of 12) and card 6 (6 crests instead of 3 on vertical line). *Rotation* was noticed on design 3. The drawing was rotated at 90 degree angle. *Overlap* was observed on card Nos 5 and 7 where area of design 5 penetrated in the overall area of design 4 and the drawing of card 7 penetrated in the area of design 6. *Partial rotation* was observed on Card A where rectangle was tilted upward. *Closure* sign was present on the drawings of card 7 and 8 where angles were not closed or points were spaced.

Thus on her drawings perseveration, rotation, overlap, partial rotation and closure, signs were observed. These signs were allotted scores as mentioned on page 10 of the test blank and patient obtained a total score of 14. On page 1 of the proforma at variable No. 19, 14 is located in the first column, i.e., 9 + under the heading of Dysfunction rating score of 3. At this point a circle was placed.

**Dysfunctioning** – There are six circles in first column of page 1, this mean a score of 18 (each circle is multiplied by 3). On 2nd column there are three circles each with a value of 2 points and in the last column there are 10 circles each with a value of zero. Thus total dysfunctioning rating score of the patient under consideration is 24 (18 + 6 = 24).

**Interpretation**

The dysfunction rating score is above cut off point (20 or more) suggesting that there is significant cognitive dysfunction, indicating organic brain pathology.

**Qualitative Analysis** – Her personal remote and recent memory is poor. Her score on mental balance and recognition subtests fall in the lowest quintile. Her Abstract Intelligence is significantly poor compared to practical ability (scores on Kohs' and Pass-A-Long tests). She had difficulty in copying the designs involving depth perception and her gestalt formation was disturbed suggesting poor perceptual acuity.

The patient in the present example, however, retains average verbal capacity, ability to learn new associates. Overall her verbal capacity is better than non-verbal capacity.

Significantly higher dysfunction score indicates cognitive disturbance, suggestive of organic brain pathology, poor non-verbal capacity points to a lesion in the right hemisphere.

## GENERAL CONSIDERATIONS FOR ADMINISTRATION OF PGI BATTERY OF BRAIN DYSFUNCTION

1. All the tests in this battery are to be administered to each subject, where organic brain dysfunction needs to be confirmed or ruled out on psychometry, except where only specific functions like intelligence or memory or perceptual acuity need to be evaluated for the therapeutic, rehabilitatory purposes or where there is need to assess the specific effect of clinical intervention.
2. The tests should be administered as far as possible in the same order as mentioned in Dysfunction Rating Score Sheet. The *only exception* is that if the subject does not cooperate on one or more tests the examiner should proceed with the remainder of the battery and then return to the uncompleted test or tests at the end.
3. Many of the tests are discontinued after the subject fails a specified number of items in succession. An item is considered 'failed' if the 'S' (subject) receives a score of Zero on it.
4. If the S answers an item and then changes the response spontaneously, the final response is the one that is to be scored.
5. Except where specifically forbidden, questions may be repeated when the S does not seem to understand. Extensive probing is to be avoided.
6. In all cases instructions and procedure of administration, scoring and interpretation, should be followed rigidly, to get the best result out of this battery.
7. In case of any deviation from the standard instructions a note must be made and interpretations made with due caution and reservations, mentioning the reasons for all such deviations in sufficient details.

## PRECAUTIONS DURING INTERPRETATION

Before interpreting poor performance as indicative of organic brain dysfunction, a number of other possible determinants of defective performance should be considered, viz.,

1. Lack of adequate effort on the part of hostilel a social or paranoid patients.
2. Inability of severely depressed patients to complete reproductions, particularly of the complex designs.
3. Inability of patients depleted by serious physical disease to complete reproduction.
4. Autistic preoccupation on the part of schizophrenic patients leading to irrelevant reproductions.
5. Defective graphomotor skill and poor task adjustment because of lack of education and relevant social experience.
6. Defective performance on the part of persons simulating mental incompetence.

In case of a non-cooperative patient or, where above cited behavioural observations are considered to be responsible for poor performance, such facts may be noted in sufficient details and the possibility of brain dysfunction may be offered with restraint. Repetition of test (s) after a reasonable time gap has been found to be of considerable help in such cases.

## USAGES OF THE BATTERY

1. This battery is standardised on the adult Neuro-psychiatric patients in the age range of 20-45 years but can be used upto 50 years of age irrespective of their educational level and sex. Verbal Adult Intelligence Scale and Bhatia's Short Battery, however, can be used on S in the age range of 20 – 69 and 20 – 59 respectively.
2. It can be used to rule out or confirm the diagnosis of Organic Brain Pathology.
3. It can be used to find out the degree of deficits if any in memory, intelligence and/or perceptuo-motor acuity, separately also. Cognitive retraining can be planned accordingly.
4. It can be used to plan rehabilitation strategy for the neuro-psychiatric patients.
5. It can be used in conjuncture with Dysfunction Analysis Questionnaire-DAQ (Pershad et al., 1985) to evaluate the influence of rehabilitatory measures, medicine and surgery in pre- and post-intervention paradigm in different categories of patients.
6. It can be used as a research tool to find out differential loss in varied categories of patients suffering from neurological and psychiatric illnesses.
7. It can be used to find out changes in these cognitive functions over a period of time, with and without treatment as well as with different types of therapies.
8. It can be used both qualitatively as well as quantitatively. Qualitative changes can be emphasized, even in the absence of effective quantitative improvements – so that subjective feelings of well being can be targetted and achieved. One can be resigned to fate and learn to live with the existing deficits. What cannot be altered has to be accepted, in order to maintain quality of life.

9. This Battery is routinely used at many hospitals, psychiatric nursing homes, research-cum-service clinics / institutes throughout Hindi speaking states in India – particularly at PGIMER, Chandigarh where it was originally developed, with varied clinic population – both psychiatric and non-psychiatric.

**Signs Suggestive of Organic Brain Dysfunction on PGI-BBD**

| | Quantitative | | Qualitative |
|---|---|---|---|
| **1.** | Higher Total Dysfunction Score, and | **1.** | Poorer Performance on Block Design Test |
| **2.** | Higher Dysfunction Scores on any or all five tests included : | **2.** | Misrecognitions on PGIMS |
| | (i) PGIMS | **3.** | Inability to repeat Alphabets on PGIMS |
| | (ii) BBPT | **4.** | Poor on Dissimilar Pair Test on PGIMS |
| | (iii) VAIS | **5.** | Perseveration on BVMGT |
| | (iv) BVMG | **6.** | Rotation / Reversal on BVMGT |
| | (v) Nahor and Benson | **7.** | Poor on Depth Items on N-B Test |
| **3.** | Marked Variability in Performance on Subtests | **8.** | Poor on Clock Test on N-B Test |
| | | **9.** | Poorer on Arithmatic Test (VAIS) |
| | | **10.** | Subjective report of declining memory |
| | | **11.** | Confusion – inability to follow simple instructions |
| | | **12.** | Disorientation to Time / Place / Person |

- **False Positives and False Negatives on PGI-BBD**–PGI-BBD is useful for assessing types of brain dysfunction – both in the same individual over a period of times, before and after intervention– as well as for group comparisons. However, it is not very useful for making a neurological diagnosis – since false positives and false negatives are present to some extent (in undesirable quantity).

- **Lateralization of Brain Lesion on PGI-BBD**–Behaviour on verbal items and subtests can be easily compared with non-verbal items and subtests – mainly perceptual subtests / items which can suggest which side of the brain is more involved.
- **Localization of Brain Lesion on PGI-BBD**–The data on PGI-BBD is suggestive, indicating the functions of different areas of brain. However, definite location and size of the lesion cannot be measured on this test.

Hence data on PGI-BBD has to be judiciosuly used while making judgements about individuals. A note of caution while interpreting results on PGI-BBD is hence required (versus quick, jumping to conclusions hastily without other supporting data).

## SUGGESTIONS TO IMPROVE MEMORY AND OTHER MENTAL FUNCTIONS

While it is not possible to improve general intelligence after a certain age, there is always some chance to improve the functioning levels of some of the mental functions, particularly the memory even in older age groups.

- **Intelligence**—Abilities do not improve after a certain age but optimum intelligent use of whatever resources are available to an individual. Acceptance of one's limitations helps in this process. There should not be any reluctance to take a minimum help from others, while not making a demand on others is likely to induce others to lend some help to an elderly person.

- **Memory**—There are many ways by which one can avoid forgetting little things while learning to retain things :—

  (1) Plan daily activities and write them in some order, e.g., according to time (Morning to night) and place (activities related to one place or direction and time).

  (2) Reading it to oneself, several time and stricking on activities completed.

  (3) Repeating whatever things are left to be done.

  (4) Preparing chits / handouts for guidance.

  (5) Over learning by repetitions and insightful digestion of whole things to the extent possible.

  (6) Never losing faith in oneself and one's abilities.

  (7) Not losing heart over failures.

  (8) Not brooding over possible failures. Remain optimistic.

  (9) Be ready to accept newer methods.

  (10) Have intention to retain and remember to smile whenever successful.

- **Perceptuo-motor Functions**

  (1) Writing, drawing things wherever difficulties are experienced. Over and over repititions help.

  (2) Learning to relax and follow rest-work-rest-work procedure in place of long learning trials.

  (3) Accept and welcome words of advice from others.

  Relaxation Techniques always help in these things.

**Suggestions for further Work on PGI-BBD**

1. Cross-validating test results on different populations in India.
2. Translation of the Battery (Verbal Parts) in different languages of India.
3. Preparation of local norms from different parts of India.
4. Modifications in sub-tests as and when required.
5. Combinations of the Battery with other tests of positive mental health like wellbeing, Quality of Life, Life Satisfaction, Adjustments, etc.

# REFERENCES

## A

Abraham, A (1981). Construction and standardization of smell matching test and its usefulness in the diagnosis of right temporal lobe lesion. Unpublished Ph. D. thesis, Madras University.

Abraham, A. and Mathai, K. V. (1983). The effect of right temporal lobe lesions on matching of smell. *Neuropsychologia*, **21**, 277-81.

Ahmad, S. et al. (1989). Cognitive impairement chronic heroin addicts. *J. of Personality and Clinical Studies,* **5 (2),** 237-40.

Alexander, D. A. (1970). The application of Grahm kendall Memory for design test to elderly normal and psychiatirc groups. *British Journal of Clinical Psychology*, **9**, 85-86.

Arora, D. (1988). Influence of general anaesthesia on cognitive functions of hypertensive on propranolol. Unpublished M. D. Dissertation, Postgraduate Institute of Medical Education and Research, Chandigarh.

## B

Bachrach, H. and Mintz, J. (1974). The Wechsler Memory Scale as a tool for the detection of mild cerebral dysfunction. *Journal of Clinical Psychology*, **30**, 58-60.

Baddeley, A. D. and Warrington, E. K. (1970). Amnesia and the distinction between long and short term memory. *Journal of Verbal Learning and Verbal Behaviour*, **9**, 176-89.

Bender, L. (1938). A Visual Motor Gestalt Test and its clinical use. *American Orthopsychiatric Association*, Research Monograph No. 3.

Benton, A. L. (1963). *A Revised Visual Retention Test*. Psychological Corporation, NY.

Bhargava, M. and Sandhu, R. (1987). *Guidelines for Bender Visual Motor Gestalt Test*. Ankur Psychological Agency, Lucknow.

Bhatia, C. M. (1955). *Manual for Bhatia's Battery of Performance Test of Intelligence*, Oxford Univ. Press, Bombay.

Bhatia, C. M. (1955). *Performance Tests of Intelligence*. Oxford University Press, Bombay.

**C**

Campbell, D. (1957). A study of some sensory motor functions in psychiatric patients. Unpublished Ph. D. thesis, University of London.

Cattell, R. B. (1970). *The Scientific Analysis of Personality*. Penguin Books, England.

Cattell, R. B. (1976). Theory of fluid crystallised intelligence : A critical experiment. *Journal of Educational Psychology*, **54**, 1-22.

Cattell, R. B. and Cattell, A. K. S. (1960). *IPAT Culture Free Intelligence Test*. Institute for Personality and Ability Testing. Champaign. Ill.

Chawla, H. M. (1973). A study of Lateralization and localization of cerebral lesion with the aid of psychological tests. Unpublished M. D. Thesis. All India Institute of Medical Sciences, New Delhi.

Cronholm, B. and Molander, L. (1957). Memory disturbances after electroconvulsive therapy. I. Conditions 6 hours after electroshock treatment. *Acta Psychiatrica Scandinavica*, **32**, 280-306.

## D

Delis, D. C. and Kaplan, E. (1983). Hazards of a standardized Neuropsychological Battery. *Journal of Consulting and Clinical Psychology*, **51**, 396-98.

Dhand, U.K. ; Chandra Shekhar, M. ; Chopra, J.S. and Pershad, D. (1990). Transient global amnesia with persistent nondominant hemispheric cognitive deficit. *Neurology India*, **38,** 195-98.

Dubey, B. L. Pershad, D. and Verma, S. K. (1981). An evaluation of Rorschach as clinical tool. *Indian Journal of Clinical Psychology*, **8**, 157-63.

## F

Filskow, W. B. and Goldstein, S. G. (1974). Diagnostic validity of the Halstead Reitan Neuropsychological Battery. *Journal of Consulting and Clinical Psychology*, **42**, 382-87.

Fitzhugh, K. B. ; Fitzhugh, L. G. and Reitan, R. M. (1961). Psychological deficits in relation to acuteness of brain dysfunction. *Journal of Consulting Psychology*, **25**, 61.

Fuller, F. and Lair, J. (1963). The Minnesota Percepto-diagnostic test. *Journal of Clinical Psychology, Monograph Supplement*, **16**, 1-33.

## G

Garg, D. (1989). Personal Communication-Based on M.D. research work. Department of Psychiatry, Dayanand Medical College, Ludhiana.

Golden, C. J. (1975). A group form of the stroop color and word test. *Journal of Personality Assessment*, **39**, 384-86.

Golden, C. J. (1977). Validity of the Halstead-Reitan Neuropsychological Battery in a mixed psychiatric and brain injured population. *Journal of Consulting and Clinical Psychology*, **45**, 1043-51.

Golden, C. J., Hammeke, T. A. and Purisch, A. D. (1978). Diagnostic validity of a standardized neurological battery derived from Luria's neuropsychological tests. *Journal of Consulting and Clinical Psychology*, **46**, 1258-65.

Golden, C. J., Hammeke, T. A. and Purisch, A. D. (1980). *A manual for the Luria-Nebraska Neuropsychological Battery*. Western Psychological Services, Los Angeles, Calif.

Goldstein, K. and Scheerer, M. (1941). Abstract and concrete behaviour : An experimental study with special tests. *Psychological Monogram*, **53(2)**.

Graham, F. K. and Kendall, B. S. (1960). Memory for Designs Test. A Revised General Manual. *Perceptual and Motor Skills*, **11**, 147-88.

Guilford, J. P. (1942). A simle scorin weight for test items and its reliability. *Psychometrika*, **6**, 367-74.

**H**

Hain, J. D. (1964). The Bender Gestalt Test : a scoring method for identifying brain damage. *Journal of Consulting Psychology*, **28**, 34-40.

Halstead, W. C. (1947). *Brain and Intelligence*, University of Chicago Press, Chicago.

Hammake, T. A., Golden, C. J. and Purisch, A. D. (1978). A standardized short and comprehensive neuropsychological test battery based on the Luria Neuropsychological Investigation. *International Journal of Neuro-Sciences*, **8**, 135-41.

Heaton, R. K. Baade, L. E. and Johnson, K. L. (1978). Neuropsychological test results associated with psychiatric disorders in adults. *Psychological Bulletin*, **85**, 141-62.

Heaton, R. K. Chelune, G. J. and Lehman, R. A. W. (1981). Relation of neuropsychological and personality test results to patients' complaints of disability. Unpublished manuscript, University of Colorado Health Centre

(Quoted from *Journal of consulting and Clinical Psychology*, **49(6)**, 807-21).

Heaton, R. K. and Pendleton, M. G. (1981). Use of neuropsychological tests to predict adult patients' every day functioning. *Journal of Consulting and Clinical Psychology*, **49**, 807-21.

Heaton, R. K. ; Vogt, A. T.; Hoehn, M. M. ; Lewis, J. A. ; Crowley, I. J. and Stallings, M. A. (1979). Neuropsychological impairment with schizophrenia Vs. acute and chronic cerebral lesions. *Journal of Clinical Psychology*, **35**, 49-53.

Hooper, H. E. (1966). *Manual, Hooper Visual Organization*, Western Psychological Services, Los Angeles.

**I**

International Classification of Diseases (1967). World Health Organization, Geneva.

Isaacs, B. and Akhtar, A. J. (1972). The set test : A rapid test of mental functioning in old people. *Age and Ageing*, **1**, 222-26.

**J**

Jensen, A. R. (1973). Level I and Level II abilities in three ethnic groups. *American Educational Research Journal*, **10**, 263-76.

Jensen, A. R. (1974). Interaction of Level I and II abilities with race and socio-economic status. *Journal of Educational Psychology*, **66**, 99-111.

**K**

Kak, V. K. ; Pershad, D. and Kumar, H. (1995). Status of cognitive and psycho-social functioning six months after head injury. *Indian J. of Clinical Psychology*, **22 (2),** 11-16.

Kapur, M. (1978). Measurement of organic brain dysfunction. *Indian Journal of Clinical Psychology*, **5**, 1-9.

Kapur, M. (1978). Validity of organic brain dysfunction in psychotic population. *Indian Journal of clinical Psychology*, **5**, 10-16.

Kapur, M. (1978). A short screening battery of tests to detect organic brain dysfunction. *Journal of Clinical Psychology*, **34**, 104-11.

Kar, B. C. (2006). Cognitive dysfunction in manic state in bipolar affective disorder, **33 (1),** 65-71.

Kar, B.R. ; Rao, S. ; Chandramouli, B.A. ; Thennarasu, K. (2004). NIMHANS Neuropsychological Battery for Children Manual. NIMHANS, Bangalore.

Kulshreshtha, S. K. (1971). *Hindi adaptation of Stanford- Binet Intelligence Scale*. Manas Sewa Sansthan Prakashan, Allahabad.

Kulshrestha, S. P. (1979). *Educational Psychology*, Loyal Book Depot, Meerut.

Kumar, R. and Sharma, R. G. (1991). Pattern of Memory deficit in paranoid and non-paranoid schizophrenia. *Indian J. of Clinical Psychology,* **18(2),** 55-57.

L

Lacks, P. B. ; Clobert, J. ; Harrow, M. and Levine, J. (1970). Further evidence concerning the diagnostic accuracy of the Halsteadorganic test battery. *Journal of Clinical Psychology*, **26**, 480-81.

Lenzer, I. I. (1980). Halstead-Reitan Test Battery : a problem of differential diagnosis. *Perceptual and Motor Skills*, **50**, 611-30.

Lewis, R. F. (1973). The memory deficit of Korsakoff patients as a function of material and modality. *Dissertation Abstracts International*.

Lewis, R. F. ; Nelson, R. W. and Eggertsen (1979). Neuropsychological test performances of paranoid schizophrenic and brain damaged patients. *Journal of Clinical Psychology*, **35**, 54-59.

Lishman, W. A. (1978). Research into dementia. *Psychological Medicine*, **8**, 353-56.

## M

Mahajan, A. ; Pershad, D. and Verma, S. K. (1988). Influence of age, education and sex on performance tests of intelligence. *Indian Journal of Psychometry and Education*, **19**, 65-71.

Matarazzo, J. D. (1972). *Wechsler's Measurement and Appraisal of Adult Intelligence*, The Williams & Willkins Co., Baltimore.

McQuitty, L. L. (1957). Elementary linkage analysis for isolating orthogonal and oblique tyes and typal relevancies. *Educational and Psychological Measurement*, **17**, 207-29.

Mendhiratta, S. S. ; Wig, N. N. and Verma, S K. (1978). Some psychological correlates of long term heavy cannabis users. *British Journal of Psychiatry*, **132**, 482-86.

Mishra, B. P. ; Nizami, S.H. ; Nizami, A. H. ; Shukla, T. R. ; Gautam, B. D. and Narang, R. L. (2000). Luria Nebraska Neuropsychological Battery – A construct validation study in India, **27 (2),** 273-77.

Morgan, C. T. (1966). Some structural factors in perception. *In David, C. Beardsless and M. Wertheimer (Eds) Readings in Perception*. Affiliated East-West Press Ltd., New Delhi.

Moses, J. A. ; Thompson, L. L. and Cardelline, J. P. (1983). Discrimination of brain damage from chronic psychosis by the Luria-Nebroska Neuropsychological Battery : a closer look. *Journal of Consulting and Clinical Psychology*, **51**, 441-49.

Mukundan, C. R. ; Reddy, G. N. N. ; Hegde, A. S. ; Shankar, J. and Kaliaperumal, V. G. (1987). Neuro-psychological and clinical recovery in patients with head trauma. *NIMHANS Journal*, **5(1)**, 23-31.

Murthy, H. N. (1966). Short scale of the Bhatia's Performance Tests. *Indian Psychological Review* **2**, 133-34.

## N

Nahor, A. and Benson, D. F. (1970). A screening test for orga nic brain disease in emergency psychiatric evaluation. *Behavioral Neuropsychiatry*, **2**, 23-26.

Narang, R. L. ; Pershad, D. ; Gupta, R. and Garg, D. (1991). Cognitive dysfunctions in alcoholics, *Indian J. of Psychiatry,* **33 (4)**, 297-301.

Natarajan, M. (1987). Adjustmental Problems of Head Injured Patients. Unpublished Report of I. C. M. R. Project. Department of Neurology and Neurosurgery, Madurai Medical College, Madurai.

## P

Panda, J. K. (1988). The neuropsychological performance of the schizophrenic patients on Luria's Test. *Indian Journal of Psychological Medicine*, **11(1)**, 11-12.

Pareek, U. (1966). *Bahaviour science Research in india : A directory* 1925-1966. Behaviour Science Centre, Delhi.

Parsons, O. A. and Prigatano, G. P. (1978). Methodological Considerations in clinical neuropsychological research. *Journal of Consulting and Clinical Psychology*, **46**, 608-19.

Payne, R. W. (1968). Cognitive abnormalities, *In H. J. Eysank (ed.)* Handbook of Abnormal Psychology. Pitman Medical Publishing Co., London.

Payne, R. W. and Hewlett, J. H. G. (1960). Thought disorder in psychotic patients. *In H. J. Eysenk (Ed.) Experiments in Personality*. Routledge & Kegan Paul, London.

Pershad, D. (1977). *Handbook for PGI-Memory Scale.* National Psychological Corporation, Agra.

Pershad, D. ; Mahajan, A. and Verma, S. K. (1988) Validation of BSB-R in Adult Psychiatric Patients. *Journal of Personality and Clinical Studies* **4(2)**, 223-26.

Pershad, D. ; Mahajan, A. and Verma, S. K. (1988). Bhatia's Short Scale of Intelligence—a suggested modification for adults. *Personality Study and Group Behaviour*, **8**, 43-48.

Pershad, D. and Verma, S. K. (1975). Evaluation of Boston Memory Scale. *Indian Journal of Clinical Psychology*, **2**, 51-56.

Pershad, D. and Verma, S. K. (1978). Cross Validation of a screening test for organic brain disease. *Neurology India*, **26**, 107-10.

Pershad, D. and Verma, S. K. (1978). A Preliminary report on the adaptation of Wechsler's Adult Intelligence Scale (Verbal section) In Hindi. *I. S. P. T., Journal of Research*, **2**, 81-91.

Pershad, D. and Verma, S. K. (1980). Clinical utility of Hooper's Visual Organization Test (VOT) : A preliminary investigation. *Indian Journal of Clinical Psychology*, **7**, 67-70.

Pershad, D. and Verma, S. K. (1987). Cross-validation of psychomatric evaluation of organic brain pathology. *Journal of Personality and Clinical studies*, **3(1)**, 23-26.

Pershad, D. and Verma, S. K. (1988). *The Concept and Assessment of Intelligence (Indian Perspective)*. National Psychological Corporation, Agra.

Pershad, D. and Verma, S. K. (1988). *Translation and adaptation of WAIS-R Verbal Scale in Hindi*. In the Concept and Assessment of Intelligence. National Psychological Corporation, Agra.

Pershad, D. ; Verma, S. K. ; Malhotra, A. K. and Malhotra, S. (1985). *Measurement of Dysfunction and Dysfunction Analysis Questionnaire* (DAQ). National Psychological Corporation, Agra.

Pershad, D. ; Verma, S. K. ; Malhotra, S. and Prabhakar, S. (1984a). Screening of organic brain dysfunction. *Indian Journal of Psychiatry*, **26**, 349-55.

Pershad, D. ; Verma, S. K. ; Malhotra, S. and Prabhakar, S. (1984b). Cognitive rating in the identification of organic brain dysfunction : A psychometric approach. *Bulletin PGI*. **18**, 129-36.

Pershad, D. ; Verma, S. K. ; Malhotra, S. and Prabhakar, S. (1984c). Cross validity of the battery of cognitive tests for the assessment of brain damage cases. *Indian Journal of Psychology*, **59(3 & 4)**, 36-44.

Pershad, D. ; Verma, S. K. ; Malhotra, S. ; Prabhakar, S. and Khan, H. A. (1984). Utility of Psychological tests in the diagnosis of brain dysfunction cases. *Indian Journal of Applied Psychology*, **21**, 25-32.

Pershad, D. ; Verma, S. K. and Wig, N. N. (1980). Education as a variable in the clinico-behavioral researches, *Bulletin, PGI*, **14**, 49-54.

Pershad, D. and Wig, N. N. (1974). A visuomotor retention test for clinical use in India. *Indian Journal of Psychometry and Education*, **5**, 60-66.

Pershad, D. and Wig, N. N. (1976). A battery of simple tests of memory for use in India. *Neurology India*, **24**, 86-93.

Pershad, D. and Wig, N. N. (1980). Normative study of visual retention test. *Indian Journal of Psychology*, **55**, 12-19.

Pershad, D. and Wig, N. N. (1988). *Revised Manual for PGI Memory Scale (PGIMS)*. National Psychological Corporation, Agra.

## R

Ramalingaswami, P. (1975). *Measurement of Intelligence Among Adults Indians*. NCERT, New Delhi.

Rao, S.L. ; Subbakrishna, D.K. and Gopukumar, K. (2004). NIMHANS Neuropsychology Battery – Manual. NIMHANS, Bangalore.

Raven, J. C. (1958). *Manual Standard Progressive Matrices*. H. K. Lewis & Co. Ltd., London.

Reitan, R. M. (1958). Validity of the Trail Making Test as an indication of Organic brain damage. *Perceptuo-Motor skills*, **8**, 271-76.

Reitan, R. M. (1966). A research programme on the psychological effects of brain lesions in human beings. *In N. R. Ellis (Ed.) International Review of Research in Mental Retardation*. Academic Press, New York.

Reitan, R. M. and Davison, L. A. (1974). *Clinical Neuropsychology*, D. C. Winston, Washington.

## S

Sabhesan, S. ; Arumugham, R. and Natarajan, M. (1991). Psychopathology of confabulations in the head injury. *Indian J. of Psychiatry,* **33 (4),** 309-13.

Seidenberg, M. ; Giordani, B. ; Berent, S. and Boll, T. J. (1983). I. Q. level and performance on the Halstead-Reitan Neuropsychological test Battery for older children. *Journal of consulting and clinical psychology*, **51**, 406-13.

Sharma, M. K. and Varghese, S.T. (2005). Neuro-cognitive impairment in Euthymic ultra rapid cycling bipolar disorder, **32 (1),** 80-82.

Shukla, T. R. ; Jah, J. and Misra, M. N. (1971). The diagnostic usefulness of Benton Revised Visual Retention Test of brain damage. *Psychological studies*, **16**, 35-41.

Sidana, R. C. ; Pershad, D. and Verma, S. K. (1982). Usefulness of Psychodiagnostic test in difficult psychiatric outpatients. *Journal of Clinical Psychiatry-India*, **6(1)**, 75-80.

Siddiqui, R. S. ; Pershad, D. ; Verma, S. K. and Prabhakar, S. (1992). Cognitive dysfunction in epileptics with and without CT scan lesion. *Journal of Personality and Clinical Studies,* **8(2),** 121-24.

Siddiqui, R. S. ; Verma, S. K. and Pershad, D. (1992). Cognitive and psycho-social dysfunction in chronic schizophrenia. *Indian J. of Disability and Rehabilitation,* **6 (1),** 1-6.

Siddiqui, R. S. ; Verma, S. K. ; Pershad, D. ; Prabhakar, S. ; Suri, S. and Avasthi, A. (1994). A follow up study of cognitive functions in epileptics with CT Lesion. *Neurology India,* **42,** 19-23.

Singh, J ; Dwivedi, K. and Saxena, W. B. (1988). Psychological performance and subjective symptoms of welders in an automobile workshop. *Journal of Personality and clinical studies*, **4(2)**, 179-82.

Singh, S. (1995). A study of cognitive functioning in male alcoholics. Presented at Annual Conference of Indian Psychiatric Society, Patna, 5-8 Jan.

Singh, S. B. and Virmani, V. (1970). Perceptual disturbances in the case of primary and secondary mental retardation ; A comparative study with special reference to Bender Gestalt and Draw-a-person Test. *Indian Journal of Mental Retardation*, **3**, 40-49.

Spiers, P. A. (1982). The Luria Nebraska Neuro-Psychological Battery Revisited : A theory in Practice or just practicing. *Journal of Consulting and Clinical Psychology*, **50**, 301-306.

Srivastava, A. K. ; Kumar, S. ; Kumar, R. and Singh, B. (2004). Neuro-cognitive dysfunctions in chronic schizophrenia, **31 (2),** 132-34.

Stanley, B. and Howe, J. G. (1983). Identification of multiple sclerosis using double discrimination scales derived from the Luria-Nebraska Neuropsychological battery : An attempt at cross validation. *Journal of Consulting and Clinical Psychology*, **51**, 420-23.

Surya Gupta (1988). *Personal communication about organic test battery* (ICMR Project).

## T

Talland, G. A. (1965). *Deranged Memory*. Academic Press, New York.

Turland, D. N. and Steinhard, M. (1969). The efficiency of the memory for design test. *British Journal of Social and Clinical Psychology*, **8,** 44-49.

## V

Verma, S. K. (1974). Some perceptuo-motor disturbances in Bender-Gestalt Test as affected by Change in the orientation of paper. *Indian Journal of Clinical Psychology* **1**, 61-63.

Verma, S. K.; Pershad, D. ; Malhotra, A. and Arunima (1988). *The Revised Bhatia's Short Battery of Performance Tests of Intelligence for Adults*. National Psychological Corporation, Agra.

Verma, S. K.; Pershad, D. ; Ojha K. N. ; Misra, M. N. ; Khorana, S ; Gupta, L. N. and Shukla, T. R. (1983). Standardization of Hindi Adaptation of WAIS-R verbal scale-A multicentred Research. Report submitted to the Indian Council of Medical Research, New Delhi.

Verma, S. K. ; Pershad, D. ; Ojha, K. N. ; Misra, M. N. ; Khorona, S. ; Gupta, L. N. ; Shukla, T. R. and Singh, T. B. (1985). Hindi WAIS-R verbal scale : Interpretation of data in difficult to diagnose cases. *Indian Journal of Psychometry and Education*, **16**, 57-64.

Verma, S. K. Pershad, D. ; Singh, A. and Singh, T B. (1982). Effect of job and mass media on verbal intelligence scores. *Journal of Clinical Psychiatry*, India, **6**, 192-95.

Verma, S. K., Wig, N. N. and Shah, D. K. (1972). Validation of Bender Gestalt Test mass media on verbal intelligence scores. *Jeurnal of Clinical Psychiatry*, India, **6**, 192-95.

Verma, S. K. Wig, N. N. and Shah, D. K. (1972). Validation of Bender Gestalt Test in Indian Psychiatric Patients. *Indian Journal of Applied Psychology*, **9**, 65-69.

## W

Watson, C. G., Thomas, R., Anderson, D. and Felling, J. C. (1968). Differentiation of organics from schizophrenics at two chronicity levels by use of the Reitan Halstead Organic Test Battery. *Journal of Consulting and Clinical Psychology*, **32**, 679-84.

Wechsler, D. (1948). *Wechsler Memory Scale*. Psychological Corporation, New York.

Wechsler, D. (1958). *Wechsler Adult Intelligence Scale*. Psychological Corporation, New York.

Wechsler, D. (1958). *The Measurement and Appraisal of Adult Intelligence*. The Wilkins Co., Baltimore.

Wechsler, D. (1981). *Wechsler Adult Intelligence Scale-Revised*. Psychologica Corporation, New York.

Wells, C. E. (1977). Diagnostic evaluation and treatment in dementia. *In C. E. Wells (Ed.) Dementia*, 2nd Edition F. A. Davis, Philadelphia.

Wells, F. L. and Martin, H. A. (1923). A method of memory examination suitable fo psychotic cases. *American Journal of Psychiatry,* **8**, 243-58.

Wig, N. N., Pershad, D. and Verma, S. K. (1974). The Use of Psychometric tools ir Indian Psychiatric Research—a reappraisal. *Indian Journal of Clinica Psychology*, **1**, 8-14.

Williams, M. (1968). Measurement of memory in clinical practice. *British Journa of Social and Clinical Psychology*, **7**, 19-34.

Withers, E. and Hinton, J. (1971). Three forms of clinical tests of sensorium an their reliability. *British Journal of Psychiatry* **119**, 1-8.

Wyke, M. (1982). Editorial-interhemispheric Integration to Man. *Psychologica Medicine*, **12**, 225-30.

**Y**

Yates, A. J. (1954). The validity of some psychological tests of brain damage *Psychological Bulletin*, **42,** 359-77.

# APPENDIX

# A REVIEW OF PSYCHOMETRIC ASSESSMENT OF COGNITIVE DEFICIT IN INDIA*

Clinical psychology got its official recognition only in the year 1956, when for the first time a two year fulltime residential post-master's training course was started at the All India Institute of Mental Health at Bangalore (now known as the National Institute of Mental Health and Neurosciences). Earlier there existed a one year course in Clinical Psychology at Banaras Hindu University, which was discontinued later as it was not in a clinical setting. During this more than four decade, around 800 clinical psychologists have been trained to man the mental hospitals, child Guidance Cinics, neuropsychologial centres, pediatric departments, rehabilitation centres etc.

Now-a-days a super speciality within the clinical psychology is emerging. It has become a matter of pride to call oneself a neuropsychologist. With this background it is assumed that the tools specifi cally required in neuropsychology centres have been refined. However, this goal has not yet been satisfactorily achieved, although the departments of neuropsychology have been existing in the premier institute of Mental Health, Bangalore as well as in the All India Institute of Medical Sciences, New Delhi, for a long time.

Since no unanimously agreeable instruments to measure cognitive functioning in neuropsychological patients were available, it was considered desirable to scan the recent literature to find out the status of psychometric assessment for cognitive deficit. To fulfill this objective, Indian Psychological

---

* This Chapter is jointly contributed by R. S. Siddiqui and Dr. D. Pershad, Department of Psychiatry, P. G. I., Chandigarh.

Abstracts published under the auspicies of Indian Council of Social Sciences Research by Prof. Udai Pareek, were reviewed. All the volumes of the Abstracts starting from 1977 to 1988 were searched comprehensively. It also included all the special volumes published during this period.

The main purpose of this review was (1) to update our knowledge and to facilitate the young researchers to note the trends of psychological studies conducted so far in the area of brain pathology, (2) to furnish a list of various psychological tests that have been employed in the past to assess cognitive functioning, and (3) to provide an instant bibliographical service on the subject.

During the period of last 12 years a total number of 74 empirical and one theoretical studies could be located in the Indian Psychological Abstracts, where cognitive deficit or evidence of organic brain pathology were searched in adult human subjects. The range of types of psychiatric disorders included in these studies is shown in Table 1.

**TABLE 1**

*Samples Used in Cognitive Deficit Studies*

| Sample | Frequency | References |
|---|---|---|
| Neurology and Head Injury | 30 | 2, 9, 11, 17, 18, 19, 20, 22, 23, 24, 25, 34, 36–39, 41, 43–47, 51, 53, 55, 58, 61, 63, 69–71 |
| Epilepsy | 6 | 1, 21, 32, 33, 49, 72 |
| Schizophrenia | 13 | 5, 7, 12, 14, 19, 30, 38, 40, 45, 58, 63, 68, 70 |
| Depression | 8 | 4, 5, 14, 15, 35, 53, 60, 68 |
| Cannabis | 10 | 3, 10, 27, 28, 52, 59, 62, 66, 73, 74 |
| Ageing | 4 | 13, 16, 40, 48 |
| Inhalation of Gases | 2 | 64, 65 |
| Psychosurgery | 2 | 50, 75 |

*Contd.....*

| Sample | Frequency | References |
|---|---|---|
| Long Term Drug Treatment (Lithium) | 2 | 53, 67 |
| Yogic Exercises | 1 | 56 |
| Diabetes | 1 | 57 |
| Selection of Hockey Players | 1 | 24 |
| Mountaineering | 1 | 26 |

The studies reported in the above cited Abstracts were conducted from the year 1974 to 1988. This means nearly a 15 year span is covered. During the first five year period, i. e., from 1974 to 1978, there have been 16 studies, in the next five year period (i. e., from 1979 to 1983) 27 studies could be located and the remaining 32 studies were conducted in the last five year period, i. e., from 1984 to 1988. This shows a definite trend of psychologists concentrating more and more on cognitive deficit.

Surprisingly the cognitive deficit has been searched in varied neurological conditions, functional psychiatric conditions, Cannabis users, aged individuals, inhalation of gases, psychosurgery and different treatment procedures. Except for four studies where projective tests were used, in all the others intelligence, memory, perceptuo-motor and/or speed tests have been used. 37 studies used omnibus test batteries where tests were selected according to the specific hypothesis set for the purpose of investigation, as shown in Table 2.

**TABLE 2**

*Functions Assessed For Cognitive Deficit*

| Function or Functions | References | Total |
|---|---|---|
| Only Intelligence | 4, 16, 17, 30, 48, 50, 70, 72 | 8 |
| Only Memory | 13, 40, 64 | 3 |
| Only Perceptuo-motor Functions | 38, 42, 51, 58, 63, 66, 68, 71 | 8 |
| Intelligence + Memory | 32, 33, 39, 49, 67, 69 | 6 |

*Contd.....*

| Function or Functions | References | Total |
|---|---|---|
| Memory + Percepto-motor | 11, 15, 18, 19, 20, 21, 23, 27, 28, 34, 53, 56, 65 | 13 |
| Intelligence + Percepto-motor | 2, 7, 8, 24, 29, 31, 62 | 7 |
| Intelligence + Memory + Percepto-motor | 1, 3, 5, 9, 10, 22, 25, 26, 36, 37, 41, 43, 44, 45, 46, 47, 52, 57, 59, 60, 73, 74, 75 | 23 |
| Intelligence + Memory + Percepto-motor + Lab | 12, 61 | 2 |
| Personality and Projective Tests | 9, 14, 25, 35, 54, 55 | 6 |
| **TEST BATTERIES USED** | | |
| Nimhans Battery | 15, 18, 19, 20, 21, 23, 34 | 7 |
| P. G. I. Battery | 41, 43, 44, 45, 46, 47 | 6 |
| Luria – Nebraska Battery | 61 | 1 |
| Halsted Battery | 12 | 1 |
| Omnibus Batteries | 1, 3, 5, 9, 10, 11, 17, 22, 24, 25, 26, 27, 28, 29, 31, 32, 33, 35, 36, 37, 39, 49, 50, 52, 53, 56, 57, 59, 60, 62, 65, 67, 69, 71, 73, 74, 75 | 37 |
| One or two tests | 2, 4, 7, 8, 13, 14, 16, 30, 38, 40, 42, 48, 51, 54, 55, 58, 63, 64, 66, 68, 70, 72 | 22 |

Amongst the intelligence tests, Bhatia Battery of Performance Tests of Intelligence has been used most frequently to measure intelligence. The next in order is the Wechsler Adult Intellgence Scales used either in full or in part. Progressive Matrices by J. C. Raven, Binet Kamath Test, Cattell's Culture Fair Test and WAPIS have been used less frequently in this field.

Amongst the memory tests, Wechsler Memory Scale and P. G. I. Memory Scale have been used very frequently. In the field of perceptuo-motor functions, the Bender Visual Motor Gestalt Test has been extensively used. The Noher-

Benson Test is second in order. Benton Visual Retention Test; is used only in two studies, which migth be because of its low applicability on Indian subjects. Speed tests such as colour, word, picture or number cancellation have been tried in nearly 25% of the studies.

Projective tests like the Rorschach Test and Holtzman Inkblot Test are used only in four studies. Kapur's NIMHANS battery of Organic Brain Pathology has been used in 6 studies and Mukancan's NIMHANS Battery in one study. P. G. I. Battery of Brain Dysfunction is used in 6 studies while Halsted and Luria approaches have been used in one study each in a period of 15 years. The details of other tests along with their references are given in Tables 3 and 4.

**TABLE 3**

*Intelligence and Memory Tests Used in Cognitive Deficit Studies*

| Name of the Test | References | Total |
|---|---|---|
| **A. Intelligence** | | |
| Bhatia's Battery of Intelligence Tests, full / part | 3, 15, 17, 18, 19, 20, 21, 22, 23, 29, 30, 31, 32, 33, 34, 39, 41, 43, 44, 45, 46, 47, 48, 59, 60, 62, 67, 69, 73, 74. | 30 |
| Wechsler Adult Intelligence Scale, Full | 2, 4, 7, 12, 22, 25, 72 | 7 |
| WAIS (Verbal) | 16, 41, 45, 46, 47, 48, 70, 73 | 8 |
| Wechsler Adult Performance Intelligence Scale Indian Adaptation (WAPIS) | 36, 37, 75 | 3 |
| Raven's Progressive Matrices | 9, 17, 26, 69, 73 | 5 |
| Binet-Kamath | 36, 37, 49, 69, 72 | 5 |
| Others | 1, 5, 8, 17, 29, 50, 57, 62, 69, 75 | 10 |

*Contd.....*

| Name of the Test | References | Total |
|---|---|---|
| **B. Memory** | | |
| Wechsler Memory Scale | 3, 9, 13, 22, 25, 36, 37, 39, 53, 59, 67, 74, 75 | 13 |
| P. G. I. Memory Scale | 39, 40, 43, 44, 45, 46, 47, 52, 56, 60, 64, 65, 73 | 13 |
| Boston Memory Scale | 32, 33, 39 | 3 |
| Others | 1, 5, 10, 11, 15, 18, 19, 20, 21, 23, 26, 27, 28, 34, 41, 49, 51, 53, 57, 69 | 20 |

**TABLE 4**

*Preceptuo-motor And Other Tests Used in Cognitive Deficit Studies*

| Test | References | Total |
|---|---|---|
| Bender Visual Motor Gestalt Test | 3, 5, 9, 25, 27, 28, 31, 36, 37, 38, 41, 43, 44, 45, 46, 47, 57, 59, 60, 61, 63, 73, 74, 75 | 24 |
| Noher Benson Test | 41, 43, 44, 45, 46, 47, 73 | 7 |
| Benton Visual Retention Test | 11, 57 | 2 |
| Other Percepto-motor Tests | 1, 10, 24, 26, 27, 29, 41, 35, 42, 51, 52, 53, 57, 58, 66, 68, 71, 73 | 18 |
| Projective Tests | 9, 14, 25, 63 | 4 |
| Speed Tests like word, number, colour, cancellation Tests | 1, 8, 15, 18, 19, 20, 21, 22, 23, 26, 27, 28, 34, 49, 52, 56, 62, 65, 73 | 19 |
| Handedness Tests | 9, 36, 37 | 3 |
| Adjustment Dysfunction | 54, 65, 73 | 3 |
| Aphasia Test | 2 | 1 |
| CT-Scan, EEG and Laboratory Tests | 7, 12, 22, 37 | 4 |

The above review suggests amply that the newer approaches like that of Luria and Halsted, that have been used more popularly in western countries for the assessment of brain dysfunction have not yet had much footing in India. There might be many reasons for their not being used here in addition to the difficulties in their applicability on Indian subjects. Most frequently intelligence tests, memory and perceptuo-motor tests have been used in the batteries of cognitive dysfunction. The P. G. I. Battery of Brain Dysfunction has also used these tests.

The P. G. I. Battery of Brain Dysfunction, as described earlier in this book, seems to have a promising future. At the moment, the P. G. I. Battery of Brain Dysfunction is being used in two M. D. research works and three Ph. D. research projects at the post graduate Institute of Medical Education and Research, Chandigarh, Panjab University, Dayanand Medical College, Ludhiana and at other places.

## REFERENCES

### A

1. Abraham, A. and K. V. Mathur (1980). Usefulness of Psychological Tests in the diagnosis of Temporal Lobe Epilepsy. *Neurology India*, **28(3)**, 155-61.
2. Abrol, B. M. et al (1975). Cognitive Dysfunction in Aphasic patients Using Wechsler Adult Intelligence Scale as a measure. *Indian Journal of Mental Retardation*, **8**, 33-41.
3. Agarwal A. K. ; B. B. Sethi and S. C. Gupta (1975). Physical and cognitive effects of chronic Bhang (Cannabis) intake. *Indian Journal of Psychiatry*, **17(1)**, 1-7.
4. Ahuja, Veena (1975). Comparison between deterioration in depression as obtained from WAIS index (Wechsler) and revised WAIS Index by J. Kraus. *Journal of Psychological Researches*, **21(3)**, 98-101.

## B

5. Bagadia, U. N. ; L. P. Shah, P. V. Pradhan, Jyothi Doshi and R. Abhyankar (1981). Evaluation of Cognitive effects of E. C. T. Preliminary observations. *Indian Journal of Psychiatry*, **23(4)**, 324-29.
6. Barodawalla, S. F. (1978). Psychological Evaluation of Brain Damaged patients. *Community Psychiatry Journal*, **4(3)**, 21-24.
7. Bilder, R. M. ; G. Dagreef ; A. K. Mandurangi, R. O. Reider, H. A. Sachaim and S. Mukherjee (1988). Neuropsychological deterioration and CT-Scan findings in chronic schizophrenia. *Schizophrenia Research* **1(1)**, 37-45.

## C

8. Chatterjee, R. G., S. R. Royghatak and B. Paul (1984). A study of parental loss in relation to cognitive style and intelligence. *Personality Study and Group Behaviour*, **4(1)**, 58-65.
9. Chatterjee, S. B. and R. Kishore (1979). Psychiatric disability in brain trauma (a longitudinal study of 37 cases). *Indian Journal of Psychiatry*, **21**, 127-32.
10. Chawla, P. L.; N. K. Binde and J. S. Neki (1976). Effects of oral administration of Cannabis indica leaf powder (bhang) on some aspects of cognition. *Journal of All India Institute of Medical Sciences*, **2(2)**, 127-31.

## D

11. Dev, B. (1983). Closed Head Injury and Cognitive Recovery. *Neurology India* **31(4)**, 41-44.
12. Dewan, M. J.; A. K. Pandurangi, B. F. Levy, M. L. Boucher and L. F. Major (1985). Are schizophrenics with abnormal dexamethazone suppression test results a distinct subgroup ? *Acta sychiatrica Scandinavia*, **72(3)**, 274-77.
13. Dubey, B. L. ; S. Singh and T. S. Boparai (1978). Can we control ageing and memory deterioration ? *Indian Journal of Clinical Psychology*, **5**, 85-88.
14. Dwived, K. D. and T. R. Shukla (1982). Differentiating variables of Holtzman Inkblot technique for depression and schizophrenia. *Journal of Clinical Psychiatry India*, **6(2)**, 130-32.

## G

15. Gangadhar, B. N. ; R. L. Kapur, and S. Kalyansundram (1982). Comparison of electroconvulsive therapy with imipramine in and ogenous depression -A double blind study. *British Journal of Psychiatry*, **141**, 367-71.

16. Gupta, L. N., S. K. Verma, D. Pershad and A. K. Mathur, (1982). Hindi WAIS-R the Bikaner experience. (I and II) *Journal of Rajasthan Psychiatric Society*, **5**, 25-29. (II 30-33).

## J

17. Janaki, S. S., Kothari, M. S. Gulati, S. R. Jayaram, S. B. Mathur and S. Gupta (1970). Trial of encephabol in neurological disorders. *The Indian Practitioners*.. **23(11)**, 689-96.

## K

18. Kapoor, Malavika, (1978). Measurement of organic brain dysfunction. *Indian Journal of Clin ical Psychology*, **5**, 1-9.

19. Kapoor, M. (1978). Validity of Tests of Organic Brain Dysfunction in Psychotic population. *Indian Journal of Clinical Psychology*, **5**, 10-15.

20. Kapoor, M. (1978). A short screening battery of tests to detect organic brain dysfunction. *Journal of Clinical Psychology*, **34(1)** 104-11.

21. Kapoor, M. (1981). A comparison of patients with functional fits and epileps on tests of organic brain dysfunction. *Indian Journal of Clinical Psychology*, **8**, 81-84.

22. Kapoor, N. and A. K. Coughlan (1980). Confabulation and Frontal Lobe Dysfunction. *Journal of Neurology, Neurosurgery and Psychiatry*, **43(5)**, 461-63.

23. Keshavan, M. S. S., M. Channabasvanna and G. N. Narayan Reddy (1981). Post- Traumatic Psychiatric Disturbances, Patterns and Predictors of Outcome. *British Journal of Psychiatry*, **138** (Feb) 157-60.

24. Khan, H. A. and M. Khan (1987). Contribution of neuropsychological tests in selection of elite Indian hockey players. *Snipes Journal*, **10 (1 & 2)**, 53-59.

25. Kishore, R. and K. Dutt. (1988). Psychiatric assessment in head injury cases. *Indian Journal of Clinical Psychology*, **15(1)**, 34-37.

**M**

26. Machoda, T. and Chitranjan Andrage (1985). The Psyche at high altitude. *Indian Journal of Clinical Psychology*, **12(2)**, 41-45.

27. Mehndiratta, S. S. ; N. N. Wig, and S. K. Verma. (1978). Some psychological correlates of long term heavy Cannabis users. *British Journal of Psychiatry*, **132**, 482-86.

28. Mehndiratta S. S. ; S. K. Verma, Ravinder Dang, Anil K. Malhotra, Karobi Das and Ritu Nehra. (1988). Cannabis and cognitive function, are evaluation study. *British Journal of Addiction*, **83**, 749-53.

29. Misra, G. and L. B. Tripathi (1980). *Psychological consequences of prolonged deprivation*, Agra : National Psychological Corporation, 251 p.

30. Mishra, J. P., M. Verma and Bhagat, R. (1983). A comparative study of functioning intelligence of paranoid schizophrenics and unspecified chronic schizophrenic patients. *Indian Journal of Clinical Psychology*, **10(2)**, 283-85.

31. Misra S. L. (1986). Efficacy of psychometry in late sequeiae of head injury. *Indian Journal of Clinical Psychology*, **13(1)**, 21-24.

32. Mohan, V. ; V. K. Verma, B. B. Sawhney and D. K. Menon (1976). Cognitive functions in the epileptics. *Indian Journal of Clinical Psychology*, **3(1)**, 85-86.

33. Mohan, V. ; V. K. Verma, B. B. Sawhney and D. K. Menon (1976). Relationship of cognitive functions of epileptics with clinical variables. *Neurology India*, **24(3)**, 134-40.

34. Mukandan, C. R. ; G. Reddy, N. Narayan, A. S. Hegde, J. Shankar and V. G. Kaliaperumal. (Jan, 1987). Neuropsychological and clinical recovery in patients with head trauma. *NIMHANS Journal*, **5(1)**, 23-31.

35. Murgai Neena and K. Sathyawati (1987). Cognitive distortions in neurotic depression. *Journal of Personality and Clinical Studies*, **35(1)**, 5-38.

**N**

36. Nagaraja D. and M. Rao. (1983). Comparative neuropsychological evaluation of dominent and nondominent hemispheric lesions in occlusive cerebrovascular disease. *Indian Journal of Clinical Psychology*, **10(2)**, 381-36.

37. Nagaraja D. and P. Medhu Rao (1986). Comparative neuropsychological evaluation of dominent and nondominent hemispheric lesions in occlusive vascular disease. *Indian Journal of Clinical Psychology*, **13(1)**, 71-76.

**P**

38. Panda J. K. R. Gupta, and R. L. Narang (1987). A comparative study of Bender Gestalt Test Test on Organic, schizophrenic patients and normals. *Indian Journal of Psychological Medicine*, **10(1)**, 6-8.

39. Pershad D. (1977). *Construction and standardization of clinical test of memory in simple Hindi*, Agra : National Psychological Corporation, 124 p.

40. Pershad D. (1979). Forgetting and New learning deficit in elderly subjects. *Journal of Psychological Researches*, **23(2)**, 72-75.

41. Pershad, D. and S. K. Verma (1978). Cross validation of a screening test for organic brain disease. *Neurology, India*, **26**, 110-17.

42. Pershad, D. and S. K. Verma (1980). Clinical Utility of Hooper's Visual Organization test, a preliminary investigation. *Indian Journal of Clinical Psychology*, **7**, 67-70.

43. Pershad, D. and S. K. Verma (1987). Cross Validation of Psychiatric evaluation of organic brain pathology. *Journal of Personality and Clinical Studies*, **3(1)**, 23-26.

44. Pershad, D. ; S. K. Verma ; S. Malhotra, S. Prabhakar and H. A. Khan (1984). Utility of Psychological tests in the diagnosis of brain dysfunction cases. *Indian Journal of Applied Psychology*, **21(1)**, 25-32.

45. Persahd, D. ; S. K. Verma ; S. Malhotra and S. Prabhakar (1984). Corss validity of the battery of Cognitive Tests for the Assessment of brain damage cases. *Indian Journal of Psychology,* **59(3&4),** 36-44.

46. Pershad, D. ; S. K. Verma ; S. Malhotra and S. Prabhakar (1984). Cognitive rating in the identification of organic brain dysfunction ; a psychometric approach. *Bulletin P. G. I.,* **18(3),** 129-36.

47. Pershad, D. ; S. K. Verma ; S. Malhotra and S. Prabhakar (1984). Screening of Organic Brain Dysfunction. *Indian Journal of Psychiatry,* **26(4),** 349-55.

48. Pershad, D. ; S. K. Verma ; T. R. Shukla ; S. Khorana, M. N. Misra ; K. N. Ojha and L. N. Gupta (1994). Some experience with WAIS-R Hindi verbal section with aged subjects. *Indian Journal of Applied Psychology,* **21(2),** 47-54.

**R**

49. Rangaswamy, K. (1985). Cognitive functions of adolescent epileptics. *Child Psychiatry Quarterly,* **18(2),** 53-56.

50. Rao, A. V. and R. R. Chinnian (1974). Psychosurgery with special reference to Obscessive Compulsive Neurosis. *Indian Journal of Psychiatry,* **16(4),** 294-98.

51. Rao, S. L. ; B. N. Gangadhar and A. S. Hegde (1986). Information processing deficits as a diagnostic tool in post conclusion syndrome. *Neurology India,* **34,** 271-74.

52. Ray, R. ; G. G. Prabhu ; D. Mohan ; L. M. Nath and J. S. Neki (1979). Chronic cannabis use and cognitive functions. *Indian Journal of Medical Research,* June, **69,** 996-1000.

**S**

53. Sugumas, A. ; N. Nammalvar and A. Venkobarao (1980). Memory changes in patients on long term Lithium therapy (*in A. V. Rao and S. P. Devi Editors' Depressive illness,*Maduri. A. V. Rao, pp. 317-23.

54. Sabhesan, S. ; R. Arumunghan ; P. Ramaswamy and M. Natrajan (1987). Persistent alcohol abuse and late outcome in head injury. *Indian Journal of Psychological Medicine,* **10(2),** 61-65.

55. Sebhesan, S. ; M. S. Jothi ; Packia Premila and M. Natrajan (1988). Frontal lobe injuries-psychosocial outcome and experienced burden. *Indian Journal of Psychological Medicine,* **11(2),** 101-06.

56. Sahasi, G. (1984). A replicated study of the effects of Yoga on cognitive functions. *Indian Psychological Review,* **27(1-4),** 33-35.

57. Sarda Menon, M. and S. Balakrishnan (1982). Intellectual deficits in insulin dependent diadetics. *Indian Journal of Psychological Medicine,* **5(1),** 13-21.

58. Sen Mazumdar, D.P. and T.K. Mazumdar (1983). Abstract and Concrete behaviour of organic, schizophrenic and normal subjects on Goldestein-Scheerer Cube Test, *Indian Journal of Clinical Psychology,* **10(1),** 5-10.

59. Sethi, B.B. ; J. K. Trivedi and H. Singh (1981). Long term effects of cannabis. *Indian Journal of Psychiatry,* **26(1),** 51-54.

60. Sharma, I. ; P. Singh and S. S. Agnihotri (1984). Cognitive Dysfunction in Depression, *Indian Journal of Psychiatry,* **26(1),** 51-54.

61. Sharma, S. ; V. K. Nandkumar and M. Mazumdar (1981). Luria neuropsychological investigation in organic brain dysfunction. *Acta Psychiatrica Scandanavia,* **63(5),** 425-30.

62. Shuklam, T. R. ; M. Naik ; D. K. Kenswar and J. P. Mishra (1981). A study of some cognitive functions of ganja addicts. *Indian Journal of Clinical Psychiatry,* **5(1),** 60-62.

63. Shukla, V. ; R. R. Tripathi and N. K. Dhar (1987). Validation of pitrowskis Rorschach signs of organicity against Bender Visual Motor Gestalt Test. *Indian Journal of Clinical Psychology,* **14,** 84-86.

64. Singh, Jaya ; Kamal Divedi and V. B. Sexena (1987). Memory and subjective symptoms in automobile painters. *Indian Journal of Clinical Psychology,* **14,** 43-45.

65. Singh, R. ; B. Singh ; S. Gautam and S. N. Sinha (1988). Perceptual distortion among cannabis users. *Journal of Personality and Clincial Studies,* **4(1),** 97-99.

66. Singh, R.K. ; B. B. Sethi and J. K. Trivedi (1982). Lithium and Memory. *Indian Journal of Psychiatry,* **24(1),** 15-21.

67. Srinivas, K. V. ; A. Abraham and A. Verghese (1987). Right temporal lobe function in psychiatric disorders. *Indian Journal of Clinical Psychology,* **14,** 40-92.

68. Srinivas, K. (1983). Recovery of intellectual functions after injury of young brain – a preliminary report based on a study of patients with infantile hemiplegia. *Neurology India,* **31(2),** 47-49.

69. Srivastava, A. and T. R. Shukla (1988). WAIS-R and WAIPS (Form PR) in differential diagnosis of schizophrenia and organicity. *Indian Journal of Clinical Psychology,* **15(1),** 42-44.

**U**

70. Uma, H. and S. L. Rao (1987). Information processing in patients with closed head injury. *Journal of Psychological Researches,* **31(2),** 70-76.

**V**

71. Vaidyalingam, M. (1979). The intellectual deterioration in epileptic psychosis. *Indian Journal of Psychological Medicine,* **2(1),** 10-6.

72. Varma, V. K. ; A. K. Malhotra ; R. Dang ; K. Das and R. Nehra (1988). Cannabis and cognitive functions : A prospective study. *Drug and Alcohol Dependence,* **21,** 147-52.

73. Vaya, S. L. and B. H. Buch (1986). Assessment of cognitive functions in cannabis abusers of mental hospital patients. *Indian Journal of Clinical Psychology,* **13(2),** 199-202.

74. Virudhagirinathan, B. S. ; V. Balasubramaniyam and T. S. Kanaka (1983). Psychological appraisal in stereotoxic cingulumotomy for drug addiction. *Neurology India,* **31(3),** 37-40.

T M. Regd. No. 564838


**Dwarka Pershad** *(Chandigarh)*
**S. K. Verma** *(Chandigarh)*

# Response Sheet of PGIBBD-PV
*(English Version)*

***Please fill up the following Informations :* Date** [ | | | | | | | ]

**Name** ______________________________

**Age** ____________ **Sex** ____________ **Education** ____________

| Sr. No. | Test Variables → Sub-tests ↓ | Dysfunction Rating Score | | |
|---|---|---|---|---|
| | | 3 | 2 | 0 |
| 1. | Remote Memory | 0-2 | 3-4 | 5+ |
| 2. | Recent Memory | 0-2 | 3-4 | 5+ |
| 3. | Mental Balance | 0-2 | 3-4 | 5+ |
| 4. | Attention and Concentration | 0-2 | 3-4 | 5+ |
| 5. | Delayed Recall | 0-2 | 3-4 | 5+ |
| 6. | Immediate Recall | 0-2 | 3-4 | 5+ |
| 7. | Retention for Similar Pairs | 0-2 | 3-4 | 5+ |
| 8. | Retention for Dissimilar Pairs | 0-2 | 3-4 | 5+ |
| 9. | Visual Retention | 0-2 | 3-4 | 5+ |
| 10. | Recognition | 0-2 | 3-4 | 5+ |
| 11. | *P/K ×100 | 251+ | 200-250 | upto 199 |
| 12. | Performance Quotient | 60 | 61-80 | 81+ |
| 13. | T.Q. On information | 60 | 61-80 | 81+ |
| 14. | T.Q. on Digit Span | 60 | 61-80 | 81+ |
| 15. | T.Q. on Arithmetic | 60 | 61-80 | 81+ |
| 16. | T.Q. on Comprehension | 60 | 61-80 | 81+ |
| 17. | Difference in P. Q. and V. Q. | 24+ | 15-23 | upto 14 |
| 18. | Nahor and Benson Test | 6+ | 4-5 | 0-3 |
| 19. | Bender-Gestalt Test | 9+ | 5-8 | 0-4 |
| | **Total** | | | |

Remarks Dysfunction Rating Score =

**Estd. 1971** **www.npcindia.com** ☎:(0562) 2464926
**NATIONAL PSYCHOLOGICAL CORPORATION**
**4/230, KACHERI GHAT, AGRA-282 004 (INDIA)**

## Sub-Test I. Memory Scale

### I. Remote Memory

1. How old are you ?
2. Where were you born ?
3. (a) When were you married ?
   (b) When did you start earning ?
   (c) When did you left study/pass high school ?
4. How old is your youngest child/brother/sister ?
5. When did you come first time in this clinic/department (Hospital) ?
6. When did you come here last time ?

### II. Recent Memory

1. What did you eat in your last dinner ?
2. What did you eat this morning ?
3. What is the name of this month ?
4. What day is today ?
5. Who came to visit you or to whom you visited yesterday ?

### III. Mental Balance

1. Recite A to Z (Alphabet of any language).
2. Count backward from 20 to 1.
3. Count backward by minusing 3s starting from 40.

## IV. Attention and Concentration

1. I am going to say some numbers. Listen them carefully, when I read them, you will repeat them in the same order.

| | | | |
|---|---|---|---|
| 5—7—3 | ☐ | 4—1—7 | ☐ |
| 5—3—8—7 | ☐ | 6—1—5—8 | ☐ |
| 1—6—4—9—5 | ☐ | 2—9—7—6—3 | ☐ |
| 3—4—1—7—9—6 | ☐ | 6—1—5—8—3—9 | ☐ |
| 7—2—5—9—4—8—3 | ☐ | 4—7—1—5—3—8—6 | ☐ |
| 4—7—2—9—1—6—8—5 | ☐ | 9—2—8—8—3—1—7—4 | ☐ |

2. I am going to read some more numbers but you will be required to repeat them backward, for example, I say 2, 5 you will say 5, 2.

| | | | |
|---|---|---|---|
| 8—5 | ☐ | 2—8 | ☐ |
| 4—3—7 | ☐ | 8—5—1 | ☐ |
| 8—5—6—3 | ☐ | 3—7—5—9 | ☐ |
| 4—7—2—9—1 | ☐ | 9—2—5—8—4 | ☐ |
| 2—5—9—4—8—3 | ☐ | 7—1—5—3—9—6 | ☐ |
| 3—5—8—6—1—9—2 | ☐ | 6—3—7—1—4—8—5 | ☐ |
| 8—5—2—3—6—1—9—4 | ☐ | 2—8—4—5—9—7—1—3 | ☐ |

## V. Delayed Recall

I am going to read name of some objects. Listen carefully and when I ask you to repeat, you will do so (read at the rate of one word per second and ask the subject to repeat it after an interval of one minute).

| 1 | | 2 | |
|---|---|---|---|
| Umbrella | ☐ | Fish | ☐ |
| Flower | ☐ | Lamp | ☐ |
| Clock | ☐ | Rupee | ☐ |
| Picture | ☐ | Taj | ☐ |
| Pencil | ☐ | Toy | ☐ |

**VI. Immediate Recall**

I am going to read a few small sentences one by one. Listen them carefully because when I am through I would like you to tell me the whole sentence as precisely as you can.

1. Ram got up from the chair, opened the door and went to market.
2. Patient was asked to lie down on the table, he was seen, medicine was prescribed and was told to come next day.
3. Mohan did not have water in his house, he picked-up the bucket, went to street well, filled it up and returned back.

**VII. Verbal Retention for Similar Pairs**

Now, I am going to read to you a list of pairs, i.e., two words at a time. Listen carefully. When I name one word of the pair you will tell the second word of the pair (read the pairs at the rate of 2 seconds per pair, with a pause of 5 seconds between pairs; give an interval of 10 seconds after reading last pair and then present a stimulus word of the pair and ask the subject to recall the 2nd word of the pair).

| | | |
|---|---|---|
| 1. | Tree ☐ | Flower ☐ |
| 2. | Sweet ☐ | Sour ☐ |
| 3. | Man ☐ | Woman ☐ |
| 4. | Day ☐ | Night ☐ |
| 5. | Black ☐ | White ☐ |

### VIII. Verbal Retention for Dissimilar Pairs

Instruction and administration procedure is the same as above for subtest VII, with a difference that stimulus words are to be presented in the order as mentioned for each of the trial. If subject fails to give correct answer, correct it and proceed to next stimulus word. If all his answers are correct in first trial even then other two trials are to be completed but in no case, pairs be repeated, only incorrect answers are to be corrected.

| | | | |
|---|---|---|---|
| **Table—Black** | 4 | 2 | 1 |
| **Tree—High** | 2 | 1 | 5 |
| **Lamp—Uneven** | 1 | 5 | 3 |
| **Child—Bitter** | 3 | 4 | 2 |
| **Dream—Deep** | 5 | 3 | 4 |

### IX. Visual Retention

I am going to show you a card, see it carefully. After some time (15 seconds) I will take it away and when I ask you (after 30 seconds) to draw them, draw the things you saw in the card from your memory on this paper (give a paper, a pencil and an erasure to the subject but do not instruct whether he can use the erasure or not).

**X. Recognition**

I am showing you a card containing pictures of many objects, see the whole card attentively (expose for 30 seconds). After some time (120 seconds interval) I will place before you another card. From this you will be required to identify and name the objects you saw in earlier card (Do not tell the subject, the exact number of objects seen in first card and how many things he is yet to identify).

**Summary of Memory Scale**

| Sub-Tests | I | II | III | IV | V | VI | VII | VIII | IX | X | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Raw Scores | | | | | | | | | | | |
| Converted-Scores | | | | | | | | | | | |

## Sub-Test II. Battery of Performance Tests of Intelligence

| KOHS' BLOCK | | | PASS-A-LONG | | | Remarks |
|---|---|---|---|---|---|---|
| Item No. | Time Taken | Score | Item No. | Time taken | Score | |
| $B_1$ | | | | | | |
| $B_2$ | | | $B_1$ | | | |
| $B_3$ | | | $B_2$ | | | |
| $B_4$ | | | $B_3$ | | | |
| $B_5$ | | | $B_4$ | | | |
| $B_6$ | | | $B_5$ | | | |
| $B_7$ | | | $B_6$ | | | |
| $B_8$ | | | $B_7$ | | | |
| $B_9$ | | | $B_8$ | | | |
| $B_{10}$ | | | | | | |
| Total Score | | | Total Score | | | |
| T. Q. | | | T.Q. | | | |

**Performance Quotient =**

## Sub-Test III. Verbal Adult Intelligence Scale (VAIS)

**1. INFORMATIONS**

Discontinue after 7 consective failures response

1. Ball
2. Year
3. Thermometer
4. Sun
5. Prime Minister
6. Guru Nanak
7. Flag
8. Lohri
9. Ramayana
10. Kuran/Bible
11. Women Ht
12. Independence
13. Birbal
14. Gandhi
15. Rubber
16. Taj Mahal
17. Population
18. Triveni
19. Bombay
20. Direction-Delhi-Madras
21. Curd
22. Dark coloured clothes
23. Year-Weeks
24. Temperature
25. Ceylone
26. England
27. Geeta
28. Meghdoot
29. Mahabharata
30. Vedas
31. Delhi-Bombay Distance
32. Lok Sabha
33. Veins

TOTAL SCORE

**2. DIGIT SPAN**

Discontinue after failures on Both TRIALS of any item.

DIGITS FORWARD SCORE =
DIGITS BACKWARD SCORE =

TOTAL SCORE =

**3. ARITHMETIC**

**Discontinue after 5 consecutive failures**

| **Problem** | **Score 1 or 0** |
|---|---|
| 1. | |
| 2. | |
| 3. | |
| 4. | |
| 5. | |
| 6. | |
| 7. | |
| 8. | |
| 9. | |
| 10. | |
| 11. | |
| 12. | |
| 13. | |
| 14. | |
| 15. | |

**TOTAL SCORE =**

| 4. COMPREHENSION<br>Discontinue after 6 failures | Response | Score |
|---|---|---|
| 1. Engine | | |
| 2. Clothes | | |
| *3. Health | | |
| 4. Bad habbits | | |
| 5. Movie (Cinema Hall) | | |
| 6. City land | | |
| *7. Eatables | | |
| 8. Credit | | |
| 9. Envelope | | |
| *10. Cheque | | |
| 11. Tax | | |
| 12. Unity | | |
| 13. Empty Vessel | | |
| 14. Forest | | |
| *15. Child law | | |
| 16. Driving Licence | | |
| 17. Deaf man | | |
| 18. Strike-iron | | |
| **Total Score** | **(Max. 36)** | |

*If the subject replies with only one idea, ask for a second response. Rephrase the test item appropriately saying. Tell me another reason why (another thing people can do.)

**SUMMARY OF V A I S**

| **Sub-tests** | **Information** | **Digit Span** | **Arithmetic** | **Comprehension** |
|---|---|---|---|---|
| **Raw Scores** | | | | |
| **Test Quotients** | | | | |

***Verbal I. Q.—***

## RESPONSE FOR NAHOR BENSON TEST

RESPONSE FOR BENDER GESTALT TEST

## SUB-TEST IV. NAHOR BENSON TEST

| Total No. | I | II | III | IV | V | VI | VII | VIII | Total Score |
|---|---|---|---|---|---|---|---|---|---|
| Figure wrongly reproduced | | | | | | | | | |

## SUB-TEST V. BENDER GESTALT TEST

| | Score |
|---|---|
| Perseveration | 4 |
| Rotation or Reversal | 4 |
| Concretism | 4 |
| Added angles | 3 |
| Separation of lines | 3 |
| Overlap | 3 |
| Distortion | 3 |
| Embolishments | 2 |
| Partial Rotation | 2 |
| Omission | 1 |
| Abbreviation | 1 |
| Separation | 1 |
| Absense of erasure | 1 |
| Closure | 1 |
| Point of contact | 1 |
| **TOTAL SCORE** | |